

CLOUD COMPUTING

Methodology, Systems,
and Applications

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and Applications

Edited by
Lizhe Wang • Rajiv Ranjan
Jinjun Chen • Boualem Benatallah



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To my beloved mother, to whom I owe too much.

– Lizhe

To my parents and beloved wife, who are the pillars of my life.

– Rajiv

To my parents, to whom I owe too much.

– Jinjun

To my parents, without whom I would not be where I am today.

– Boualem

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Foreword

Clouds have emerged as an important computing paradigm and are quickly dominating all aspects of the IT landscape. In fact, most organizations today are considering Cloud services as part of their IT roadmap, as users or providers or both.

Clouds are characterized by on-demand access to different levels of computing utilities, an abstraction of unlimited resources, and a usage-based payment model where users essentially “rent” virtual resources (or capabilities) and pay for what they use. Underlying these cloud services are, typically, consolidated and virtualized data centers that provide virtual machine (VM) containers hosting applications from large numbers of distributed users.

Clouds and the notion of computing as a service provides opportunities for exploring potentially interesting cost-benefit trade-off (for example, between capital and operational costs) models for operational flexibility and agility, approaches for reducing environmental impact, etc., and are leading to interesting economic and business models. Furthermore, integrating these public cloud platforms with more traditional data centers and Grids provides opportunities for on-demand scale-up, scale-out and scale-down, and allows service offering and applications capabilities to no longer be constrained by local infrastructure limitations.

Clearly, such a seductive paradigm can have a significant impact on a range of application in academia, industry and government. However, transitioning to the Cloud mind-set, integrating clouds into the currently computing infrastructure, and moving current applications and their operations to Clouds present a non-trivial challenge and often require new paradigms and practices at all levels. Furthermore, these challenges often go far beyond just the technical aspects. For example, cultural, legal, regulatory and social challenges are quickly overshadowing technical challenges. While certain usage modes are natural to Clouds, others are not and, as a result, the Cloud paradigm must coexist and complement other usage modes, such as, for example, high-performance computing. It is critical that these challenges be fundamentally understood and addressed.

I would like to congratulate the editors of this volume for putting together such a comprehensive and timely collection. This book brings together chapters authored by internationally recognized researchers, describing leading research efforts that are trying to get to the heart of many of the above listed challenges, with an overall goal to provide a comprehensive overview of the state-of-the-art of this emerging research area, from fundamental con-

cepts to more specific technologies and application use cases. As a result, I do believe that this book will provide a tremendous resource to students, researchers and practitioners, and have a significant impact on this important and growing field.

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Preface

Cloud computing is the latest evolution of computing, where IT resources are offered as services. The hardware and software systems that manage these services are referred to as Infrastructure as a Service (IaaS) and Platform as a Service (PaaS), while actual applications managed and delivered by IaaS and PaaS are referred to as Software as a Service (SaaS). The combination of virtualization, IaaS, and PaaS hold the potential to revolutionize software application (SaaS) life-cycle management. If Cloud computing is properly applied within an overall IT strategy, it can help small and medium business enterprises (SMEs) and governments lower their IT costs, by taking advantage of economies of scale and automated IT operations, while at the same time optimizing investment in in-house computing infrastructure. The benefit of such an environment is efficiency and flexibility, through creation of a more dynamic computing enterprise, where the supported functionalities are no longer fixed or locked to the underlying infrastructure. This offers tremendous automation opportunities in a variety of computing domains including, but not limited to, e-Government, e-Research, high-performance computing, web hosting, social networking, multi-media, and e-Business.

The goal of this book is to give a comprehensive overview of the state-of-the-art of this emerging research area, which many believe to be the next platform for provisioning and delivering SaaS applications in various computing domains. The book also envisions future research topics and directions. This book is expected to serve as the most important reference and a milestone for research on Cloud computing since 2007, when the term “Cloud computing” was coined.

The book is organized into three parts. Part I focuses on the fundamentals of cloud computing and includes chapters that present the insight, concepts, methodologies, taxonomies, and architectures related to existing and future software systems and architectures. In Chapter 1, A. Kalapatapu and M. Sarkar provide an insight into evolution of Cloud computing and its origins, and direction. In Chapter 2, A. Edlund and I. Livenson present a comprehensive study on how Cloud computing can be leveraged for accelerating the innovation and business values for IT startups. In Chapter 3, R. Teckelmann, A. Sulistio, and C. Reich present a detailed study on interoperability issues of Cloud infrastructures (IaaS). Chapter 4, written by C. Baun and M. Kunze, gives an overview about the most popular Cloud services and categorizes them according to their organizational and technical realization. L. Atzori, F. Granelli, and A. Pescapé in Chapter 5 study the networking solutions for

engineering Cloud computing architectures. In Chapter 6, A. Dastjerdi and R. Buyya provide a detailed taxonomy and survey of QoS management and service selection methodologies in Cloud computing environments. A detailed study on open-source software frameworks that are aimed at managing Cloud infrastructures (IaaS) is presented in Chapter 7 by P. Sempolinski and D. Thain. In Chapter 8 H. Khazaei, J. Mišić, and V. B. Mišić present an analytical model for performance evaluation of Cloud server farms, and demonstrate the manner in which important performance indicators such as request waiting time and server utilization may be assessed with sufficient accuracy. Chapter 9, written by A. Celesti, F. Tusa, M. Villari, and A. Puliafito investigates the new business advantages of a futuristic worldwide InterCloud ecosystem, considering evolutionary degree of the current infrastructures and the possible scenarios on which the InterCloud federation could be built.

Part II compiles the contributions on the research theme of Cloud computing functionality and service provisioning. In Chapter 10, S. Nepal, S. Chen, and J. Yao discuss the issue of building elastic and secure Cloud storage services—TrustScore, which allows end-users to scale their storage space to meet the ever-increasing need while improving utilization and manageability. A. Goscinski, M. Brock, and P. Church in Chapter 11 investigate what challenges exist when one attempts to use Clouds for high performance computing (HPC) and they further demonstrate the effectiveness of HPC in Clouds through broad benchmarking and through a new prototype system used to satisfy the needs of HPC clients. Chapter 12, written by E. Jiménez-Domingo, Á. Lagares-Lemos, and J. M. Gómez-Berbís, addresses the issue of architecting multitenancy application stacks in Clouds. V. Emeakaroha et al. in Chapter 13 present an innovative framework – LoM2HiS, for the mapping low level resource metric to high-level SLA parameters. In Chapter 14, E. Caron et al. examine Cloud computing IaaS and PaaS providers from three important resource management issues including load balancing, auto-scaling, and monitoring. Monitoring of IaaS and SaaS components is fundamental to ensure QoS guarantees in Cloud computing environments. To this end, G. Katsaros et al. in Chapter 15 analyze the requirements of an efficient monitoring mechanism for Cloud environments. To handle peak loads, M. Mattess et al. present an approach in Chapter 16 that leases Cloud infrastructure services in an opportunistic way. Energy efficient management of Cloud computing infrastructures that consists of many power-hungry components such as processors, memory modules, and network devices is an important issue. In Chapter 17, Y. Choon et al. present energy-efficiency models for provisioning and migrating applications in Cloud environments. Lack of security, privacy, and trust of data are the major hindrances in the path to moving sensitive applications to public Cloud Computing environments. S. Hamouda and J. Glauert in Chapter 18 present a detailed taxonomy on how these issues are currently handled in context of Clouds.

Part III focuses on specific case studies and applications that are deployed and provisioned over Cloud infrastructures. This part of the book also includes

discussion on possible research directions. In Chapter 19, J. Shi discusses the fundamentals of architecting applications on Clouds with a focus on three objectives: maximal application survivability, unlimited application performance scalability and zero data losses. To develop a software framework that can provide a common access to all existing Cloud-based services (IaaS, PaaS, and SaaS) is challenging. F. Moscato et al. in Chapter 20 address this challenge by defining a common ontology for Cloud service negotiation and establishment. Cloud environments offer promising platforms for deploying and managing elastic RDBMS systems. L. Zhao, S. Sakr, and A. Liu in Chapter 21 explore the recent advancements in web scale data management on Clouds. In order to allow cost-effective scaling of videoconference systems, the use of Cloud computing appears to be a viable choice, mainly due to its elasticity and infinite resource capacity. To this end, J. Cervino et al. in Chapter 22 describe a new service that delivers a videoconferencing system through Clouds; further, they also analyze the cost and resource usage related to this case study. L. Carril et al. in Chapter 23 present a case study on designing a computing and storage service for planning radiotherapy treatment using Cloud infrastructures. In Chapter 24 K. Fletcher and X. Liu present a methodology to analyze Cloud security requirements and develop policies to deal with both internal and external security challenges. Integration of traditional HPC Grids with Cloud infrastructures is going to play a pivotal role for future scientific studies and experiments. H. Kim et al., in Chapter 25, experimentally investigate, from an applications perspective, interesting usage of nodes and scenarios for integrating HPC Grids and Clouds, and how they can be effectively enabled using an autonomic scheduling system. In Chapter 26, J. Kaylor, K. Läufer, and G. Thiruvathukal propose a file system, called RestFS, as a connector abstraction for flexible resource and service composition. Finally Y. Wei et al. present the Aneka Cloud computing application platform, in Chapter 27.

The compilation of this book has been possible because of the efforts and contributions of many individuals. Firstly, the editors would like to thank all contributors for their immense hard work in putting together excellent chapters that are informative, comprehensive, rigorous and, above all, timely. The editors would like to thank the team led by Nora Konopka, Amy Blalock and Brittany Gilbert, at Taylor & Francis Group/CRC Press for patiently helping us put this book together. Lastly, we would like to thank our families for their support throughout this journey and would like to dedicate this book to them.

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Part I

Fundamentals of Cloud Computing: Concept, Methodology, and Overview

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Cloud Computing: An Overview

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CONTENTS

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1.1 Introduction

Cloud computing, with the revolutionary promise of turning computing into a 5th utility, after water, electricity, gas, and telephony, has the potential to transform the face of Information Technology (IT), especially the aspects of service-renderition and service management. Though there are myriad ways of defining the phenomenon of Cloud Computing, we put forth the one coined by NIST (National Institute of Standards and Technology). According to them, Cloud Computing is defined as “*A model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction*” [84]. Loosely speaking, Cloud computing represents a new way to deploy computing technology to give users the ability to access, work on, share, and store information using the Internet [762]. The cloud itself is a network of data centers, each composed of many thousands of computers working together that can perform the functions of software on a personal or business computer by providing users access to powerful applications, platforms, and services delivered over the Internet. It is in essence a set of network enabled

services that is capable of providing scalable, customized and inexpensive computing infrastructures on demand, which could be accessed in a simple and pervasive way by a wide range of geographically dispersed users. The Cloud also assures application based Quality-of-Service (QoS) guarantees to its users. Thus, Cloud Computing provides the users with large pools of resources in a transparent way along with a mechanism for managing the resources so that a user can access it ubiquitously and without incurring unnecessary performance overhead. The ideal way to describe Cloud Computing then would be to term it as “Everything as a Service” abbreviated as XaaS [100]. Below, we sum up the key features of Cloud Computing:

- Agility – helps in rapid and inexpensive re-provisioning of resources.
- Location Independence – resources can be accessed from anywhere and everywhere.
- Multi-Tenancy – resources are shared amongst a large pool of users.
- Reliability – dependable accessibility of resources and computation.
- Scalability – dynamic provisioning of data helps in avoiding various bottleneck scenarios.
- Maintenance – users (companies/organizations) have less work in terms of resource upgrades and management, which in the new paradigm will be handled by service providers of Cloud Computing.

However, Cloud Computing doesn’t imply that it consists of only one cloud. The term “Cloud” symbolizes the Internet, which in itself is a network of networks. Also, not all forms of remote computing are Cloud Computing. On the contrary, Cloud Computing is nothing but services offered by providers who might have their own systems in place. [18]

1.2 Cloud Computing: Past, Present, and Future

A hundred years ago, companies stopped generating their own power with steam engines and dynamos and plugged into the newly built electric grid. The cheap power pumped out by such electric utilities did not just change how businesses operated — it set off a chain reaction of economic and social transformations that brought the modern world into existence. Today, a similar revolution is under way. Hooked up to the Internet’s global computing grid, massive information-processing plants have begun pumping data and software code into our homes and businesses. This time, it’s computing (instead of electricity) that’s turning into a utility. Nicholas Carr in his book *The Big*

Switch: Rewiring the world from Edison to Google [164] has finely portrayed the transition of Cloud from its past till its present form. Cloud Computing has integrated various positive aspects of different computing paradigms, resulting in a hybrid model that has evolved gradually over the years beginning in 1960 when *John McCarthy* rightfully stated that “*computation may someday be organized as a public utility*” [608].

“The Cloud” is based on the idea of generating computing facility on demand. Just the way one turns on a faucet to get water or plug into an electric socket on the wall to get electricity, similarly Cloud Computing intends to create a paradigm where most of the features and functions of stand-alone computers today can be streamed for a user over the Internet [764]. Further probing into the philosophy of cloud computing will reveal that the concept dates back to the era of “Mainframes,” where resources (like memory, computational capabilities) of centralized powerful computers owned by large organizations were used/shared by several users over a small geographical local area. Today Cloud Computing boasts of an architecture where the powerful computers are replaced by supercomputers and perhaps even a network of supercomputers and the users are dispersed over vast geographic areas, accessing these computing resources via the Internet (network of networks). In the past, issues like dearth of bandwidth, perception, loss of control, trust and feasibility proved to be major road blocks in realizing the Cloud concept of service rendition. Today most of these challenges have been overcome, or countermeasures are in place to resolve the challenges. Faster bandwidth, virtualization and more particular skills around cloud type technologies help in realizing the Cloud Computing paradigm.

The concept of Cloud Computing — as nascent as it might appear — itself, has undergone significant evolution. The first generation of Cloud Computing which evolved along with the “Internet Era” was mainly intended for “e-business services.” The current generation of Cloud services has progressed several steps to now include “IT as a Service” which can be thought of as “consumerized internet services.” Using standardized, highly virtualized infrastructure and applications, IT can drive higher degrees of automation and consolidation, thus reducing the cost of maintaining existing solutions and delivering new ones. In addition, externally supplied infrastructure, software, and platform services are delivering capacity augmentation and a means of using operating expense funding instead of a heavy duty capital. [388]

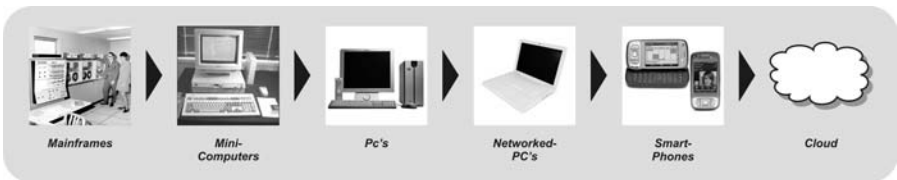


FIGURE 1.1
History of Computing.

Today, “the network is the (very big, very powerful) computer.” The capabilities of the Cloud as a centralized resource can match to industrial scales. This implies that processing power involving thousands of machines embedded in a network has surpassed even the capabilities of the very high-performance supercomputers. By making this technology available through the network on an on-demand and as-needed basis, the Cloud holds the promise of giving individuals, businesses, organizations and governments around the world access to extraordinary computing power from any location and any device. It is a fact that data and information is growing at a break-neck pace. HP predicted that within three more years (by 2014) there will be more information produced and consumed than in the history of mankind. Thus, the next version of Cloud will enable access to information through services that are set in the context of the consumer experience. This is significantly different — it means that data will be separated from the applications — a paradigm where processes can be broken into smaller pieces and automated through a collection of services, woven together with access to massive amounts of data. It will eliminate the need for large scale, complex applications that are built around monolithic processes. Changes can be accomplished by refactoring service models, and integration achieved by subscribing to new data feeds. This will create new connections, new capabilities, and new innovations surpassing those that exist today. [388]. Thus it is envisioned that by 2020 most people will access

	Cloud 1 E-business Services			Cloud 2 IT as a Service		Cloud 3 Everything as a Service	
	1990	1995	2000	2005	2010	2015	2020
Primary forcing function	■ Internet based supply chain integration and e-commerce			■ Consumerized internet services ■ Low cost IT		■ Pervasive business and consumer services	
technology orientation	■ Web based app design ■ EAI & message bus Integration ■ Internal protocols ■ 3-tier architecture			■ Web 2.0 & SCA app design ■ Virtualization ■ Cloudbased technology platforms		■ Data oriented, context aware services ■ Vertical and horizontal cloud ecosystems	
IT organization design	■ Organized around technology domains ■ Technology-centric			■ Organized around service supply chain ■ Service-centric		■ Organized around value networks ■ Service-centric	

FIGURE 1.2
Clouds Past, Present, and Future.

software applications online and share and access information through the use of remote server networks, rather than depending primarily on tools and information housed on their individual, personal computers. It is predicted

that cloud computing will become more dominant than the desktop in the next decade [84]. In other words, most users will perform the majority of their computing and communicating activities through connections to servers that will be operated and owned by service-providing organizations. Most technologists and industrialists believe that cloud computing will continue to expand and come to dominate information transactions because it offers many advantages, allowing users to have easy, instant, and individualized access to tools and information they need wherever they are, locatable from any networked device. To validate this claim, the PEW INTERNET & AMERICAN LIFE PROJECT carried out a survey with a highly diverse population set. *71% of the survey takers believed that by 2020, most people won't do their work with software running on a general-purpose PC. Instead, they will work in Internet-based applications such as Google Docs, and in applications run from smart phones* [630]. However, quality of service guarantees, interoperability between existing working platforms and security concerns are some of the issues that still continue to plague the growing popularity of Cloud computing.

1.3 Cloud Computing Methodologies

Cloud Computing is based on two main techniques — (i) Service Oriented Architecture and (ii) Virtualization.

(i) *Service Oriented Architecture (SOA)*: Since the paradigm of Cloud computing perceives of all tasks accomplished as a “Service” rendered to users, it is said to follow the *Service Oriented Architecture*. This architecture comprises a flexible set of design principles used during the phases of system development and integration. The deployment of a SOA-based architecture will provide a loosely-integrated suite of services that can be used within multiple business domains. The enabling technologies in SOA allow services to be discovered, composed, and executed. For instance, when an end-user wishes to accomplish a certain task, a service can be employed to discover the required resources for the task. This will be followed by a composition service which will plan the road-map to provide the desired functionality and quality of service to the end-user. [579, 761]

(ii) *Virtualization*: The concept of virtualization is to relieve the user from the burden of resource purchases and installations. The Cloud brings the resources to the users. Virtualization may refer to *Hardware* (execution of software in an environment separated from the underlying hardware resources), *Memory* (giving an application program the impression that it has contiguous working memory, isolating it from the underlying physical memory implementation), *Storage* (the process of completely abstracting logical storage from physical storage), *Software* (hosting of multiple virtualized environments within a sin-

gle Operating System (OS) instance), *Data* (the presentation of data as an abstract layer, independent of underlying database systems, structures and storage) and *Network* (creation of a virtualized network addressing space within or across network subnets) [486]. Virtualization has become an indispensable ingredient for almost every Cloud; the most obvious reasons being the ease of abstraction and encapsulation. Amongst the other important reasons for which the Clouds tend to adopt virtualization are:

- (i) Server and application consolidation – as multiple applications can be run on the same server resources can be utilized more efficiently.
- (ii) Configurability – as the resource requirements for various applications could differ significantly, (some require large storage, some require higher computation capability) virtualization is the only solution for customized configuration and aggregation of resources which are not achievable at the hardware level.
- (iii) Increased application availability – virtualization allows quick recovery from unplanned outages as virtual environments can be backed up and migrated with no interruption in services.
- (iv) Improved responsiveness – resource provisioning, monitoring and maintenance can be automated, and common resources can be cached and reused. [784]

In addition, these benefits of virtualization tend to facilitate the Cloud to meet stringent SLA (Service Level Agreement) requirements in a business setting which otherwise cannot be easily achieved in a cost-effective manner. Without virtualization, systems have to be over provisioned to handle peak load and hence waste valuable resources during idle periods.

1.4 The Cloud Architecture and Cloud Deployment Techniques

Geared with the knowledge of SOA and Virtualization, we now take a look at the overall Cloud architecture. From the end user's perspective, Figure 4.1 depicts a basic Cloud Computing architecture involving multiple components. Cloud architecture closely resembles the UNIX philosophy of involving multiple components which work together over universal interfaces [762]. Recall that the Cloud computing paradigm represents a Service oriented mechanism of managing and dispatching resources. Before we delve into studying the actual architecture of Cloud computing it will be beneficial to examine the possible characteristics that will be required to realize such a system. It is common knowledge that the architectural requirements of the Cloud will vary depending on the application for which the Cloud is being used. For instance, social networking applications like Facebook and Orkut will have a very differ-

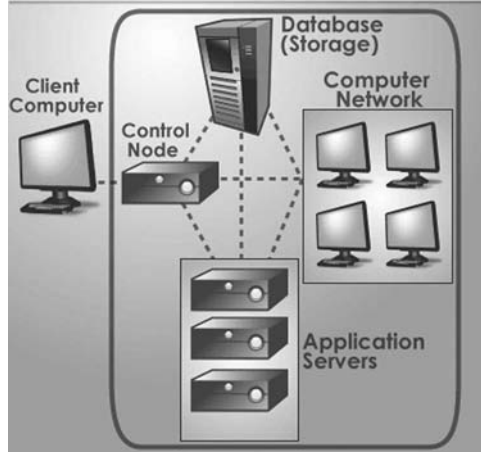


FIGURE 1.3
Basic Cloud Computing Architecture [86].

ent set of requirements, constraints and deliverables from the architecture in comparison to, say, a remote patient health monitoring application. However, some common architectural characteristics can still be identified. For instance, (i) the system should be scalable with the ability to include thousands to perhaps tens of thousands of members. (ii) It should be able to interoperate between various service requirements and effectively share resources amongst its users. (iii) The system should be easy to maintain and upgrade, maintaining user transparency during these processes. (iv) As outlined earlier, managing resources like servers and storage devices virtually, thereby creating a virtual organization, is absolutely crucial.

To mitigate the problem of designing customized Cloud architecture for each and every application and also to streamline the architecture design process of the Cloud, scientists resorted to the age old concept of a generalized “Layered approach” [163]. As with the classical 7-layer OSI model of data networks [759], the layered model in Cloud computing serves the same general purpose. Depending on the service requirement of an application, these layers are shuffled to create a customized architecture. The layered architecture adheres to the principle of Service Oriented Architecture (SOA) which forms the core of the Cloud computing paradigm. The components of a basic layered architecture are shown in the Figure 1.4, below [630], namely the Client, its required Services, the Applications that the Client runs, the Platform on which these applications run, the Storage requirement and finally the Infrastructure required to support the Client’s computing needs.

We devote a few sentences on each component below.

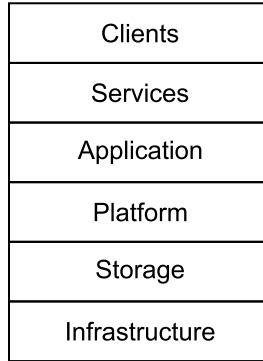


FIGURE 1.4

Layered Architecture for a Customized Cloud Service [760].

Clients: the Clients of a Cloud comprise computer hardware and/or computer software that relies on the computational capability of the Cloud for application or service delivery. Examples include computers, mobile devices, operating systems, and browsers.

Services: this refers to the different service models made available by the Cloud like SaaS (Software-as-a-Service), IaaS (Infrastructure-as-a-Service) and PaaS (Platform-as-a-Service). This layer acts as a middleman between the user and the vast amount of resources accessible to the user. A resource includes “products, services and solutions that are delivered and consumed in real time over the Internet” [630]. Examples include location services and Search Engines among others.

Application: the Cloud enables resource management and user activity tracking from central locations rather than at each customer’s site, enabling customers to access applications remotely via the Internet. Cloud application services deliver software as a service over the Internet, eliminating the need to install and run the application on the customer’s own computer, thereby simplifying (almost eliminating) maintenance and support at the customer’s end. Examples include Web Application and Peer-to-Peer computing.

Platform: it facilitates deployment of applications without the cost and complexity of buying and managing the underlying hardware and software layers. This layer delivers a computing platform and/or solution stack as a service, often consuming Cloud infrastructure and sustaining Cloud applications. Examples include Web Application Frameworks like Ruby on Rails and Web Hosting.

Storage: the storage layer consists of computer hardware and/or computer software products that are specifically designed for the storage of Cloud services. Computer hardware comprises huge data centers that are used for resource sharing. Examples include Amazon SimpleDB, and Nirvanix SDN

(Storage Delivery Network).

Infrastructure: this layer delivers computer infrastructure, typically a platform virtualization environment as a service. It includes management of virtual resources too. Rather than purchasing servers, software, data center space or network equipment, clients instead buy those resources as a fully outsourced service. Examples include Network Attached Storage and Database services.

The main advantage of such a layered architecture is the ease with which they can be modified to suit a particular service. The way in which these components interact leads to various architectural styles. There are two basic architectural styles on which most of the services are based. They are:

- **Outside-In:** This architectural style is inherently a top-down design emphasizing the functionality of the components. Implementing this style leads to a better architectural layering with various functionalities. It infuses more feasibility enabling better integration and interoperation of components.
- **Inside-Out:** This architectural style, on the other hand, is inherently a bottom-up design which takes an infrastructural point of view of the components. This style is more application oriented than service oriented. [631]

It is to be noted that incorporating new functionalities in a pre-existing architectural scheme is done in an incremental fashion. The ease of transforming an existing architecture into another, depends on the complexity of the architecture, the functionalities of the components and their integration. The vast landscape of the services and their growing complexity has lead to implementation of innovative architectural styles and several hybrid architectures. [221,631]

Cloud Deployment Techniques

Cloud deployment is the manner in which a Cloud is designed to provide a particular service. Obviously these deployment methods will vary according to the way in which a Cloud provides service to the users. Thus, their deployment techniques are user specific [494]. For instance, a deployment technique might depend on the level of security commissioned for a particular user.

Figure 1.5 depicts the various Cloud deployment techniques which predominantly comprise (i) the Public deployment, (ii) the Private deployment or (iii) the Hybrid deployment. We discuss each of these deployment strategies briefly below.

(i) **Public Cloud:** it is the traditional mainstream Cloud deployment technique whereby resources are dynamically provisioned by third party providers who share them with the users and bill the users on a fine grained utility computing basis. It offers easy resource management, scalability and flexibility

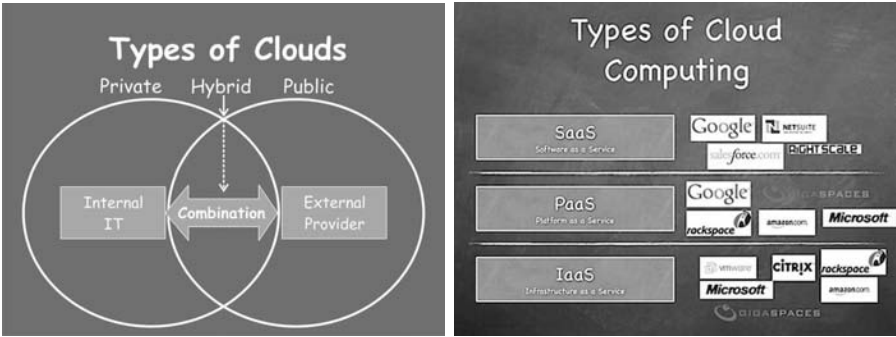


FIGURE 1.5
Cloud Deployment Techniques [325].

with an economical pay-as-you-go model which is extremely viable especially for small businesses. On the negative side, the user lacks visibility and control over the computing infrastructure. Since computing infrastructures are shared between various organizations, these Clouds face various security and compliance issues. Amazon's Web Services and Google's AppEngine are few examples of Public Clouds, also known as external Clouds. [630]

(ii) **Private Cloud:** in this Cloud deployment technique, the computing infrastructure is solely dedicated to a particular organization or business. These Clouds are more secure because they belong exclusively to a particular organization [494]. These Clouds are more expensive because one needs in-house expertise for their maintenance. Private Clouds are further classified based on their location as:

(a) *On-Premise Clouds* – these refer to Clouds that are for a particular organization hosted by the organization itself. Examples of such Clouds would include Clouds related to military services which have a considerable amount of confidential data.

(b) *Externally hosted Clouds* – these refer to Clouds that are also dedicated for a particular organization but are hosted by a third party specializing in Cloud infrastructure. These are cheaper than On-premise Clouds. Examples of such Clouds would be small businesses using services from VMware, Amazon etc. Such Clouds are also known as Internal Clouds. [631]

(iii) **Hybrid Cloud:** this deployment technique integrates the positive attributes of both the Public Cloud and Private Cloud paradigm. For instance, in a Hybrid Cloud deployment, critical services with stringent security requirements may be hosted on Private Clouds while less critical services can be

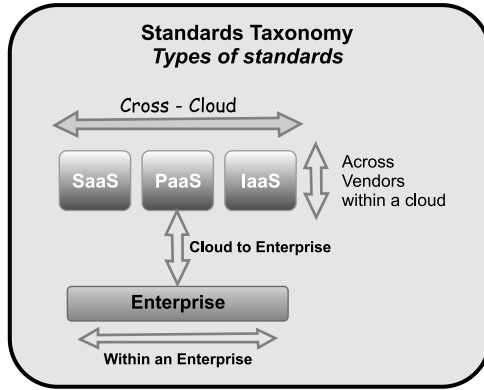


FIGURE 1.6
Types of Cloud Services [569].

hosted on the Public Clouds. The criticality, flexibility and scalability requirement of a service governs its classification into either the Public or Private Cloud domain. Each Cloud in the Hybrid domain retains its unique entity. However, they function synchronously to gracefully accommodate any sudden rise in computing requirements. Hybrid Cloud deployment is definitely the current trend amongst the major leading Cloud providers currently [630].

(iv) **Community Cloud:** this deployment technique is similar to Public Clouds with the only difference being the distribution of the sharing rights on the computing resources. In a Community Cloud, the computing resources are shared amongst organizations of the same community. So this Cloud covers a particular group of organizations, which have the same functionalities. For example, all Government organizations within the state of California may share the computing infrastructure on the Cloud to manage data related to citizens residing in California [608].

1.5 Cloud Services

The Cloud can provide us with a myriad service models and services. They include SaaS (Software as a Service), PaaS (Platform as a Service), HaaS (Hardware as a Service), DaaS ([Development, Database, Desktop] as a Service), IaaS (Infrastructure as a Service), BaaS (Business as a Service), FaaS (Framework as a Service), OaaS (Organization as a Service) amongst oth-

ers [100]. However, Cloud Computing products can be broadly classified into three main Services (SaaS, PaaS and IaaS) which are showcased in Figure 1.6 along with their relationship to a user (Enterprise). The following section is an attempt to familiarize the reader with the several different Cloud Services that are currently rendered:

Infrastructure-as-a-Service (IaaS): This service provisions for hardware related services like storage, and virtual servers on a pay-as-you-go basis. The main advantage of IaaS is the usage of latest technology at all times with regard to computer infrastructure which allows users to achieve faster service. Organizations can use IaaS to quickly build new versions of applications or environments without incurring unnecessary purchase and configuration delay. On-demand scaling via resource virtualization and use-based billing makes IaaS competent enough for any kind of businesses. The major companies already providing IaaS are Amazon [57, 58], Rackspace, GoGrid, AT&T and IBM. [367]

Platform-as-a-Service (PaaS): PaaS offerings may include facilities for application design, application development, testing, deployment and hosting as well as application services such as team collaboration, web service integration and marshalling, database integration, security, scalability, storage, persistence, state management, application versioning, application instrumentation and developer community facilitation. These services may be provisioned as an integrated solution over the web, providing an existent managed higher-level software infrastructure for building particular classes of applications and services. The platform includes the use of underlying computing resources, typically billed similar to IaaS products, although the infrastructure is abstracted away below the platform. Major companies providing PaaS are Google's AppEngine [92], Microsoft Azure, and Force.com etc. [608, 630, 631]

Software-as-a-Service (SaaS): Provides specific already created applications as fully or partially remote services. Sometimes it is in the form of web-based applications and other times it consists of standard non-remote applications with Internet-based storage or other network interactions. It allows a user to use the provider's application using a thin client interface. Users can access a software application hosted by the Cloud vendor on pay-per-use basis [527]. It is a multi-tenant platform. The pioneer in this field has been Salesforce.com offering online Customer Relationship Management (CRM) space. Other examples are online email providers like Google's Gmail and Microsoft's hotmail, Google docs and Microsoft's online version of office called BPOS (Business Productivity Online Standard Suite). [16, 608, 630, 631]

Other than the above services David Linthicum has described a more granular classification of services which includes:

Storage-as-a-Service (SaaS): Storage as a Service is a business model that helps a smaller company or individual in renting storage spaces from a large company. Storage as a Service is generally seen as a good alternative for a

small or mid-sized business that lacks the capital budget and/or technical personnel to implement and maintain their own storage infrastructure. SaaS is also being promoted as a way for all businesses to mitigate risks in disaster recovery, provide long-term retention for records and enhance both business continuity and availability. Examples include Nirvanix, Cleversafe's dsNET etc.

Database-as-a-Service (DbaaS): It constitutes delivery of database software and related physical database storage as a service. A managed service, offered on a pay-per-usage basis that provides on-demand access to a database for the storage of application data is what constitutes DbaaS. Examples include Amazon, Force.com etc.

Information-as-a-Service (IfaaS): Information as a service accepts the idea that data resides within many systems and repositories. Its main function is to standardize the access of data by applying a standard set of transformations to the various sources of data thus enabling service requestors to access the data regardless of vendor or system. Examples include IBM, Microsoft etc.

Process-as-a-Service (PaaS): Refers to a remote resource that's able to bind many resources together, either hosted within the same Cloud computing resource or remote, to create business processes. These processes are typically easier to change than applications, and thus provide agility to those who leverage these process engines that are delivered on-demand. Process-as-a-service providers include Appian Anywhere, Akemma, and Intensil.

Integration-as-a-Service (IaaS): Integration-as-a-Service includes most of the features and functions found within traditional Enterprise Application Integration technology, but delivered as a service. Integration-as-a-Service takes the functionality of system integration and puts it into the Cloud, providing for data transport between the enterprise and SaaS applications or third parties. Examples include Amazon SQS, OpSource Connect, Boomi, and Mule OnDemand.

Security-as-a-Service (SaaS): Delivers core security services remotely over the Internet like anti-virus, log management etc. While typically the security services provided are rudimentary, more sophisticated services are becoming available such as identity management. Security-as-a-Service providers include Cisco, McAfee, Panda Software, Symantec, Trend Micro and VeriSign.

Management/Governance-as-a-Service (MaaS): Provides the ability to manage one or more Cloud services, typically simple things such as topology, resource utilization, virtualization, and uptime management. Governance systems are becoming available as well, such the ability to enforce defined policies on data and services. Management/governance-as-a-service providers include RightScale, rPath, Xen, and Elastra.

Testing-as-a-Service (TaaS): These systems have the ability to test other Cloud applications, Web sites, and internal enterprise systems, and do not require a hardware or software footprint within the enterprise. They also have the ability to test services that are remotely hosted. SOASTA is one of the

many Testing-as-a-Service providers [608, 631].

1.6 Cloud Applications

Cloud Applications are not only developed with a business perspective but also take into account activities oriented towards socializing and sharing information. This information may be as basic as checking news headlines or more sensitive in nature, such as searches for health or medical information. Thus Cloud Computing is often a better alternative than local servers handling such applications.

Virtualization is the main basis of Cloud Computing, thus it is fully enabled with virtual appliances. A virtual appliance is an application which has all its components bundled and streamlined with the operating system [762]. The main advantage of a Cloud Computing application is that the provider can run various instances of an application with minimum labor and expense. A Cloud service provider needs to anticipate a few issues before launching its application in the Cloud computing environment. Keeping in mind the issues an application must be designed to scale easily, tolerate failures and include management tools [388]. We discuss these issues in the following section.

- *Scale*: Application in a Cloud environment needs to have maximum scalabilities and to ensure this, one should start building the application in the simplest manners avoiding complex design patterns and enhancements. The next step would be to split the functions of an application and integrate them loosely. The most important step in ensuring on demand scalability of an application is sharding, which can be described as splitting up the system into many smaller clusters instead of scaling the single system up so as to serve all users.
- *Failures*: Any application due to one or the other reason is bound to face failure at some point of time. To tolerate failures, an application must operate in an asynchronous fashion and one should spread the load across multiple clusters so that the impact of failure gets distributed. The best way to tolerate failures would be testing the application for all kinds of failure scenarios, and also users should be aware of the real cost incurred if an application faces any kind of failure.
- *Management Tools*: Having a proper management tool helps in automating the application configuration and updates, thereby reducing management overhead. The management system helps in not just minimizing economic expenditures but also leading to optimized usage of resources. The most

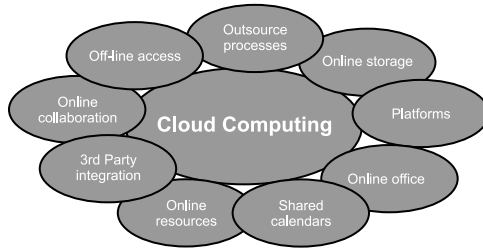


FIGURE 1.7
Cloud Applications [569].

difficult and expensive problem that can be handled with a proper management tool is variability [795]. This helps in delivering an application that can boast of consistent performance.

Applications in Cloud Computing have been designed keeping in mind the various types of services offered by it. Cloud Computing has impacted people's perception of applications over the Internet. The applications of Cloud computing are streamlined with each and every field of sciences. In Figure 1.7 we have highlighted a few areas in which Cloud Computing has shown great potential for developing various applications which have fostered the growth of businesses and enterprises.

These applications can be categorized into broad areas like datacenters and storage, security, military applications, the health industry, platforms and software usages, applications with high performance computing resources, and last but not the least, the growing need for virtual environments. [221, 494]

1.7 Issues with Cloud Computing

So far we have focused on the promises that the Cloud holds for users and applications alike. However, there are issues that need to be resolved before this technology can be exploited to its maximum potential [238]. In the section below we enumerate and discuss the burning issues which might deter the phenomenal growth of Cloud Computing technology.

(i) Security Issues: Security is as much of a concern in Cloud computing as it would be in any other computing paradigms. Cloud Computing can be vaguely defined as outsourcing of services, which in turn causes users to lose significant control over their data. There is always a risk of seizure associated with the public Clouds. For instance, an organization sharing data in an environment

where other organizations are doing the same is always under the threat of compromising the security and privacy of its data if any other organization in the shared scheme happens to violate the security protocols [390]. Moreover, in a virtualized environment one needs to consider the security of not just the physical host but also the virtual machine. This is because if the security of a physical host is compromised, then automatically all virtual machines face security threat and vice versa. Since the majority of services in Cloud computing are provided using web browsers, there are many security issues related with it as well [392]. Flooding is also a major issue where an attacker sends huge amounts of illegitimate service requests which cause the system to run slow thereby hampering the performance of the overall system. Cloud networks stand the potential threat of both Indirect Denial of Service attacks and Distributed Denial of Service attacks.

(ii) Legal and Compliance Issues: Clouds are sometimes bounded by geographical boundaries. Provision of various services is not location dependent but because of this flexibility Clouds face Legal & Compliance issues. These issues are related mainly to the vendors though they still affect the end users. These issues are broadly classified as functional (services in the Clouds that have legal implications for both service providers and end users), jurisdictional (where governments administer laws to follow) and contractual (terms and conditions). Issues include (a) **Physical Location of the data** referring to where the data is physically located and if a dispute occurs, which jurisdiction will help in resolving it (b) **Responsibilities of the data** where if a vendor is hit by a disaster will the businesses using its services be covered under insurance (c) **Intellectual Property Rights** which deals with the way trade secrets are maintained [732].

(iii) Performance and QoS Related Issues: For any computing paradigm performance is of utmost importance. Quality of Service (QoS) varies as the user requirements vary. One of the critical QoS related issues is the optimized way in which commercial success can be achieved using Cloud computing. If a provider is not able to deliver the promised QoS it may tarnish its reputation [238]. Since Software-as-a-Service (SaaS) deals with provision of softwares on virtualized resources, one faces the issue of Memory and Licensing constraints which directly hamper the performance of a system.

(iv) Data Management Issues: The main purpose of Cloud Computing is to put the entire data on the Cloud with minimum infrastructure requirements for the end users. The main issues related to data management are scalability of data, storage of data, data migration from one Cloud to another and also different architectures for resource access [127]. Since data in Cloud computing even includes high confidential information it is of utmost importance to manage these data effectively. There has been an instance where an online storage service called The Linkup got shut down after losing access to as much as 45% of its customers. While transferring data, i.e., data migration, in a Cloud has to be done very carefully as it could lead to bottlenecks at each

and every layer of the network model, as huge chunks of data are associated with the Cloud [762].

(v) Interoperability Issues: The Cloud computing interoperability idea was conceived by Reuven Cohen. Reuven Cohen is founder and Chief Technologist for Toronto based Enomaly Inc. Vint Cerf, who is a co-designer of the Internet's TCP/IP standards and widely considered a father of the Internet, spoke about the need for data portability standards for Cloud computing. Companies such as Microsoft, Amazon, IBM, and Google all own their independent Clouds but they lack interoperability amongst them. Each service provider has its own architecture, which caters to a specific application. To make such uniquely distinct Clouds interoperate is a non-trivial problem. The lack of standardized protocols in the domain of Cloud computing further makes interoperability a challenge. The key issues hampering implementation of interoperable Clouds are the large scale access and computational capabilities of the Clouds, resource contention and the dynamic nature of the Cloud. However, interoperability amongst the various Clouds would only add to the value of this technology, making it more widely accessible, fault tolerant, and thereby robust.

1.8 Cloud Computing and Grid Computing: A Comparative Study

Cloud computing should not be confused with Grid Computing, Utility Computing and Autonomic Computing. Grid computing consists of clusters of loosely coupled, networked computers acting in concert to perform very large tasks. Utility computing packages computing resources as a metered service. Autonomic computing stresses self management of resources. Grids require many computers, typically in the thousands, and commonly use servers, desktops, and laptops. Clouds also support non-grid environments [284]. The differences between Utility computing and Cloud computing are crucial. Utility computing relates to the business model in which application, infrastructure, resources, hardware and/or software are delivered. In contrast, Cloud computing relates to the way we design, build, deploy and run applications that operate in a virtualized, shared-resource environment accompanied by the coveted ability to dynamically grow, shrink and self-heal. Thus the major aspects which separate Cloud Computing from other computing paradigms are user centric interfaces, on-demand services, QoS guarantees, autonomous system organization, scalability and flexibility of services [369].

In the mid 1990s, the term "Grid" was coined to describe technologies that would allow consumers to obtain computing power on demand. Ian Foster and others posited that by standardizing the protocols used to request com-

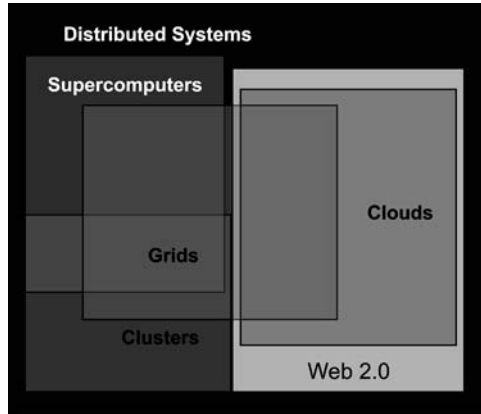


FIGURE 1.8
Cloud and Grid Computing [283].

puting power, we could spur the creation of a Computing Grid, analogous in form and utility to the electric power grid. Figure 1.8 showcases the overlap and distinctions amongst several cutting edge computing technologies [369]. Web 2.0 covers almost the whole spectrum of service-oriented applications, whereas Cloud Computing lies mostly on the large-scale computing domain. Supercomputing and Cluster computing have been more focused on traditional non-service applications. Grid Computing overlaps with all these fields and is generally considered to be catering to smaller scaled computing requirements than the Supercomputers and the Clouds [284].

Half a decade ago, Ian Foster gave a three point checklist to help define what is, and what is not a Grid. We present his checklist below:

- 1.) Coordinates resources that are not subject to centralized control,
- 2.) Uses standard, open, general-purpose protocols and interfaces, and
- 3.) Delivers non-trivial qualities of service.

Although the third point holds true for Cloud Computing, neither point one nor two is applicable for today's Clouds. The vision for Clouds and Grids are similar but the implementation details and technologies used differ considerably. In the following section, we discuss the differences in these two technologies based on their architecture, business model, and resource management techniques.

Architecture

Grids provide protocols and services at five different layers as identified in the Grid protocol architecture shown in Figure 1.9.

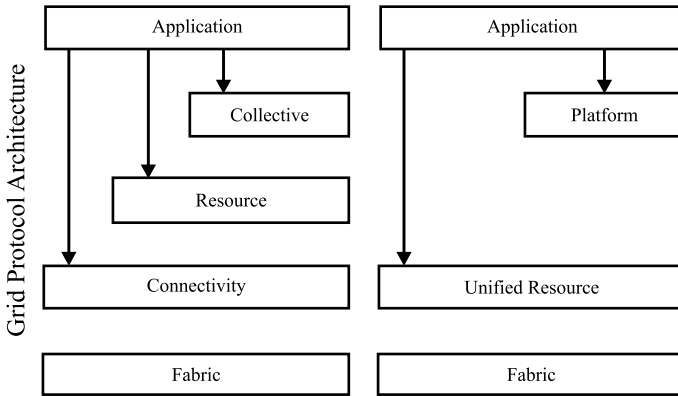


FIGURE 1.9
Cloud Architecture vs. Grid Architecture [283].

At the *fabric layer*, Grids provide access to different resource types such as computation, storage and network resource, code repository, etc. The *connectivity layer* defines core communication and authentication protocols for easy and secure network transactions. The *resource layer* defines protocols for the publication, discovery, negotiation, monitoring, accounting and payment of sharing operations on individual resources. The *collective layer* captures interactions across collections of resources and directory services. The *application layer* comprises whatever user applications are built on top of the above protocols and APIs environments. [284, 370]

In contrast, the Cloud architecture presents a four-layer architecture composed of (i) Fabric (ii) Unified resource (iii) Platform and (iv) Application layers. (It is to be noted that there are other paradigms of cloud architecture as well. We however, choose to focus on the 4-layer architecture for our discussion).

The Fabric layer contains the raw hardware level resources, such as computational resources, storage resources, and network resources. The Unified Resource Layer contains resources that have been abstracted/encapsulated (usually by virtualization) so that they can be exposed to upper layer and end users as integrated resources, for instance, a virtual computer/cluster, a logical file system, a database system, etc. The Platform layer adds a collection of specialized tools, middleware and services on top of the unified resources to provide a development and/or deployment platform. Finally, the Application layer contains the applications that run in the Clouds. Clouds in general provide services at three different levels, namely IaaS, PaaS, and SaaS, although some providers can choose to expose services at more than one level. [284]

Resource Management

With regard to resource management we compare Clouds and Grids using the following models:

(i) Computational Model: Most Grids use a batch-scheduled computational model, in which a local resource manager (LRM), such as PBS, Condor and SGE manages the computational resources for a Grid site. Users submit batch jobs via GRAM that occupies a certain amount of resources for a stipulated period of time [787]. In contrast to the Grid's paradigm of allocating dedicated resources to specific "jobs" for a stipulated amount of time with a scheduling algorithm governing the process, the Cloud computing model takes a resource sharing approach by all users at all times [371]. The goal here (in Cloud computing) is to achieve low queuing latency for resource access. Moreover, this design should allow latency sensitive applications to operate natively on Clouds, although ensuring the required level of QoS is likely to be one of the major challenges for Cloud Computing, especially as the Cloud scales to incorporate more users. [284]

(ii) Data model: The importance of data has caught the attention of the Grid community for the past decade; Data Grids have been specifically designed to tackle data intensive applications in Grid environments, with the concept of virtual data playing a crucial role. Virtual data captures the relationship between data, programs, and computations and prescribes various abstractions that a data grid can provide. For instance, (a) location transparency where data can be requested without regard to data location, a distributed metadata catalog is engaged to keep track of the locations of each piece of data (along with its replicas) across grid sites, and privacy and access control are enforced; (b) materialization transparency where data can be either recomputed on the fly or transferred upon request, depending on the availability of the data and the cost to re-compute [378]. There is also representation transparency where data can be consumed and produced no matter what their actual physical formats and storages are, data are mapped into some abstract structural representation and manipulated in that way. In contrast, the Cloud computing paradigm is centralized around Data. Cloud Computing and Client Computing will coexist and evolve hand in hand, while data management (mapping, partitioning, querying, movement, caching, replication, etc.) will become more and more important for both Cloud Computing and Client Computing with the increase of data-intensive applications. [163]

(iii) Virtualization: Grids do not rely on virtualization as much as Clouds do, but that might be more due to policy issues and having each individual organization maintain full control of their resources (i.e., not by virtualization). However, there are efforts in Grids to use virtualization as well, such as Nimbus, which provides the same abstraction and dynamic deployment capabilities. A virtual workspace is an execution environment that can be deployed dynamically and securely in the Grid. In addition, Nimbus can also provision

a virtual cluster for Grid applications (e.g., a batch scheduler, or a workflow system), which is also dynamically configurable — a growing trend in Grid Computing. [284]

(iv)Monitoring: Another challenge that virtualization brings to the Clouds is the potential difficulty in retaining fine grained control over the monitoring of resources. Although many Grids (such as TeraGrid) also enforce restrictions on what kind of sensors or long-running services a user can launch, Cloud monitoring is not as straightforward as in Grids, because Grids in general have a different trust model in which users via their identity delegation can access and browse resources at different Grid sites, and Grid resources are not highly abstracted and virtualized as in Clouds. Monitoring can be argued to be less important in Clouds, as users interact with a more abstract layer that is potentially more sophisticated. This abstract layer could respond to failures and quality of service (QoS) requirements automatically in a general-purpose way irrespective of application logic. In the near future, user-end monitoring might be a significant challenge for Clouds, but it will become less important as Clouds become more sophisticated and more self-maintained and self-healing.

(v)Programming Model: Grids primarily target large-scale scientific computations, so they must scale to leverage large number/amount of resources, and we would also naturally want to make programs run fast and efficiently in Grid environments. Programs must also run correctly, so reliability and fault tolerance must be considered. Clouds (such as Amazon Web Services, Microsoft's Azure Services Platform) have generally adopted Web Services APIs where users access, configure and program Cloud services using predefined APIs exposed as Web services, and HTTP and SOAP are the common protocols chosen for such services. Although Clouds adopted some common communication protocols such as HTTP and SOAP, the integration and interoperability of all the services and applications remain the biggest challenge as users need to tap into a federation of Clouds instead of a single Cloud provider. [519]

(v)Application model: Grids generally support many different kinds of applications, ranging from high performance computing (HPC) to high throughput computing (HTC). HPC applications are efficient at executing tightly coupled parallel jobs within a particular machine with low-latency interconnects and are generally not executed across a wide area network Grid. On the other hand, Cloud computing could in principle cater to a similar set of applications. The one exception that will likely be hard to achieve in Cloud computing (but has had much success in Grids) are HPC applications that require fast and low latency network interconnects for efficient scaling to many processors. As Cloud computing is still in its infancy, the applications that will run on Clouds are not well defined, but we can certainly characterize them to be loosely coupled, transaction oriented (small tasks in the order of millisec-

onds to seconds), and likely to be interactive (as opposed to batch scheduled as they are currently in Grids).

(vi)Security Model: Clouds mostly comprise dedicated data centers belonging to the same organization, and within each data center, hardware and software configurations and supporting platforms are in general more homogeneous as compared with those in the Grid environments. Interoperability can become a serious issue for cross-data center and cross-administration domain interactions. Imagine running your accounting service in Amazon EC2 while your other business operations are on Google infrastructure. Grids, however, are built on the assumption that resources are heterogeneous and dynamic, and each Grid site may have its own administration domain and operation autonomy. Thus security has been engineered in the fundamental Grid infrastructure. The key issues considered are: single sign-on, so that users can log on only once and have access to multiple Grid sites; this also facilitates accounting and auditing, delegation (so that a program can be authorized to access resources on a user's behalf and it can further delegate to other programs), privacy, integrity and segregation. Moreover, resources belonging to one user cannot be accessed by unauthorized users and cannot be tampered with during transfer. In contrast, currently, the security model for Clouds seems to be relatively simpler and less secure than the security model adopted by the Grids. Cloud infrastructure typically relies on Web forms (over SSL) to create and manage account information for end-users, and allows users to reset their passwords and receive new passwords via emails in an unsafe and unencrypted communication. Note that new users could use Clouds relatively easily and almost instantly, with a credit card and/or email address. To contrast this, Grids are stricter about security.

Business models:

In a Cloud-based business model, a consumer pays the provider on the basis of resource consumption, akin to the utility companies charging for basic utilities such as electricity, gas, and water. The model relies on economics of scale in order to drive prices down for users and profits up for providers. Today, Amazon essentially provides a centralized Cloud consisting of Compute Cloud EC2 and Data Cloud S3. The former is charged based on per instance-hour consumed for each instance type and the latter is charged by per GB-Month of storage used. In addition, data transfer is charged by TB /month data transfer, depending on the source and target of such transfer. The business model for Grids (at least that found in academia or government labs) is project-oriented, in which the users or community represented have a certain number of service units (i.e., CPU hours) they can spend.

Thus Clouds and Grids share a lot of commonality in their vision, architecture and technology, but they also differ in various aspects such as security, programming model, business model, computational model, data model, ap-

plications, and abstractions. We believe a close comparison such as this can help the two communities understand, share, and evolve infrastructure and technology within and across, and accelerate Cloud computing to leap from early prototypes to production systems. [360]

1.9 Conclusion

Cloud Computing plays a significant role in varied areas like e-business, search engines, data mining, virtual machines, batch oriented scientific computing, online TV amongst many others. Cloud computing has the potential to become an integral part of our lives. Examples include, (i) the Cloud operating system which provides users with all the basic features of an operating system like data storage and applications; (ii) mapping services which help users in finding routes to various places; (iii) Telemedicine applications for collecting data of a patient and calling emergency services in dire need.

As an increasing number of businesses move toward Cloud based services, issues like interoperability, security, portability, migration and standardized protocols are proving to be critical concerns. For instance, the need for higher transparency in scheduling tasks with guaranteed QoS is proving to be a challenging issue in the area of data management over the Clouds. In addition, the service driven model of Cloud Computing often leads to concerns and queries regarding the Service Level Agreement (SLA) of enterprises.

The research and business community alike are coming up with new and innovative solutions to tackle the numerous issues that are making its way as Cloud Computing is shaping itself as the Future of Internet. The potentials are endless and the sky is the limit as we watch “a computing environment to elastically provide virtualized resources as a service over the Internet in a pay-as-you-go manner” [631] bloom and blossom to its maximum potential.

APPENDIX: Comparisons of different Cloud computing technologies

In this section, we present a comparative data-set that showcases the various solution and service strategies of some of the major Cloud service providers today, like Amazon Web Services, GoGrid, FlexiScale, Google, Nimbus and a few others. Detailed description of this data set is available at [608].

TABLE 1.1

Outages in Different Cloud Services

Vendor	Service and outage	Outage Duration
Microsoft	Malfunction in Windows Azure	22 hours
Google	Gmail and Google Apps	2.5 hours
	Google search outage due to programming error	40 Mins
	Gmail site unavailable due to outage in contacts system	1.5 hours
	Google App engine partial outage	5 hours
Amazon	Authentication overload	2 hours
	single bit error leading to protocol blowup	6–8 hours
Flexiscale	Core network failure	18 hours

TABLE 1.2

Comparison of Different Cloud Computing Technologies and Solution Provider

Feature	Amazon Web Services	GoGrid	Flexiscale
Computing Architecture	Provides IaaS.Gives client API's to manage their Infrastructure	Provides IaaS.Designed to deliver a guaranteed QoS level and reconfigures itself depending on demand.	Provides IaaS.Functions similar to GoGrid's architecture but allows multi -tier architectures.
Load Balancing	Round-Robin Load Balancing, HAproxy.	F5 Load Balancing. Algorithms used are round-robin, sticky session, SSL least connect,source address.	Automatic Equalization of server load within clusters.
Fault Tolerance	System alerts automatically, does failover and resyncs to last known state.	Instantly scalable and reliable file level backup service.	Full Self-Service.
Interoperability	Supports horizontal interoperability	Working towards interoperability.	Working towards interoperability.
Storage	S3 and SimpleDB	First connects each server to private network and then uses transfer protocols.	Persistent storage based on a fully virtualized high end SAN/NAS back end.
Security	Type-I, Firewall, SSL, Access Control list.	No guarantee of security.	Customers have their own VLAN.
Programming Framework	Amazon Machine Image, Amazon MapReduce.	Uses REST-like Query interface and supports Java, Python and Ruby.	Supports C, C++, Java, PHP, Pearl and Ruby.

TABLE 1.3

Features of Different PaaS and SaaS Providers

Feature	Google App Engine	Azure	Force.com
Computing Architecture	Google's geo-distributed architecture	Platform is hosted on Microsoft datacenters. Provides an OS and a set of developer's cloud services.	Facilitates multi-tenant architecture allowing a single application to serve multiple customers.
Load Balancing	Automatic scaling and Load Balancing	Built in hardware Load Balancing. Containers are used as load balancer.	Load Balancing among tenants.
Fault Tolerance	Automatically pushed to a number of fault tolerant servers.	On Failure SQL data services will automatically begin using another replica of container.	Self-Management and Self-Learning.
Interoperability	Supports interoperability between platforms of different vendors and programming languages.	Supports interoperability between platforms.	Application level integration between different clouds.
Storage	Proprietary database (Big Table distributed storage)	SQL Server Data Services (SSDS).	Database deals in terms of relationship fields.
Security	Google's Secure Data Connector (SDC).SDC uses TLS based server authentication and uses RSA/128-bit or higher AES CBC/SHA.	Security Token Service (STS) creates a security assertion markup language token according to rule.	SysTest SAS, 70 Type II.
Programming Framework	MapReduce framework supporting Python and Java	Microsoft NET.	Apex for database service and supports NET, Apache Axis (Java, C++).

TABLE 1.4

Comparisons of Different Open Source-Based Cloud Computing Services

Feature	Eucalyptus	OpenNebula	Nimbus
Computing Architecture	Can configure multiple clusters in a Private Cloud	It is based on Haizea scheduling. Focuses on efficient, dynamic and scalable management of VM's within private clouds.	It has a client side cloud computing interface to Globus enabled TeraPort cluster. Nimbus Context Broker combines several deployed VM's into "turnkey" virtual cluster.
Load Balancing	Simple Load Balancing cloud controller	Nginx server is used as Load Balancer with round-robin or weighted selection mechanism.	Triggers self configuring virtual clusters.
Fault Tolerance	Separate clusters within a cloud reduce the risk of correlated failure.	Daemon is restarted and a persistent database backend is used to store host and VM info.	Checks worker nodes periodically and performs recovery operation.
Interoperability	Multiple private clouds use the same backend infrastructure.	Interoperable between intra cloud services.	Standards "rough consensus and working code."
Storage	Walrus is used.	SQLite3 is used as a backend component. Persistent storage for ONE data structures.	GridFTP and SCP.
Security	WS-security for authentication, cloud controller generates the public/private key.	Firewall, virtual private network tunnel.	PKI credential required, works with GRID proxies VOMS, shibboleth custom PDP's.
Programming Framework	Supports Hibernate, Axis2 and Java.	Supports Java and Ruby.	Supports Python and Java.

Bibliography

- [1] eXtensible Access Control Markup Language (XACML): http://www.oasisopen.org/committees/tc_home.php?wg_abbrev=xacml.
- [2] SAML V2.0 Technical Overview, OASIS, <http://www.oasis-open.org/specs/index.php#saml>.
- [3] Forum of Federations: <http://www.forumfed.org/en/index.php>.
- [4] Amazon Elastic Compute Cloud (Amazon EC2): <http://aws.amazon.com/ec2/>.
- [5] OpenQRM, “the next generation, open-source Data-center management platform,” <http://www.openqrm.com/>.
- [6] Resources and Services Virtualisation without Barriers (RESERVOIR) European Project, <http://www.reservoir-fp7.eu/>.
- [7] OASIS Web Services Resource Framework (WSRF): http://www.oasis-open.org/committees/tc_home.php?wg_abbrev=wsrf.
- [8] GridShib: Bridging SAML/Shibboleth and X.509 PKI for campus and grid interoperability, <http://gridshib.globus.org/>.
- [9] Extensible Messaging and Presence Protocol (XMPP): Core, <http://xmpp.org/rfcs/rfc3920.html>.
- [10] Extensible Messaging and Presence Protocol (XMPP): Instant Messaging and Presence, <http://xmpp.org/rfcs/rfc3921.html>.
- [11] StartSSL The Swiss Army Knife of Digital Certificates & PKI: <http://www.startssl.com/>.
- [12] inContext project: Unleash Team Power, <http://www.in-context.eu>, 2007.
- [13] inContext project WP2 Deliverable 2.1: Analysis and specification and context modeling techniques, <http://www.in-context.eu/page.asp?PageRef=10>. 2007.
- [14] W3C. Web services activity.
- [15] Xml-rpc specification.

- [16] Software as a service:strategic backgrounder. *Software and Information Industry Association*, 2001.
- [17] Worldwide server power and cooling expense 2006-2010. IDC Document #203598, 2006.
- [18] What is cloud computing. *Jrnl of Parallel and Distributed Computing*, 2008.
- [19] Amazon web services cloud watch Web Site, November 2010.
- [20] Cloudkick Web Site, November 2010.
- [21] Enstratus Web Site, December 2010.
- [22] Eucalyptus monitoring Web Site, November 2010.
- [23] Gogrid exchange Web Site, November 2010.
- [24] Gogrid load balancing service Web Site, November 2010.
- [25] Makara Web Site, December 2010.
- [26] J. Seltzer, D. Read, and D.D. Clark. End-to-end arguments in system design. *Second International Conference on Distributed Computing Systems*, pages 509–512, 1981.
- [27] Microsoft azure appfabric Web Site, November 2010.
- [28] Msdn forum - microsoft azure load balancing service Web Site, November 2010.
- [29] Msdn library - azure monitoring and diagnostics Web Site, November 2010.
- [30] Multi-scale modeling and simulation, 2010. [Online], <http://www.math.princeton.edu/multiscale/>.
- [31] Nagios project Web Site, November 2010.
- [32] Opennebula information manager Web Site, November 2010.
- [33] Opennebula scheduling policies Web Site, November 2010.
- [34] OpenNebula Service Management Project, December 2010.
- [35] Paraleap windows azure dynamic scaling Web Site, November 2010.
- [36] Rackspace cloud tools Web Site, November 2010.
- [37] RightScale Web Site, December 2010.
- [38] Scalr Web Site, December 2010.

- [39] Rackspace cloud sites load balancing service Web Site, January 2011.
- [40] Grid Computer Operating System For Web Applications - AppLogic, 3tera.: <http://www.3tera.com/AppLogic/>.
- [41] J.C. Mouriño, J. López, F.G. G. Castaño, D. Rodríguez, L. Domínguez, D. González, J. Pena, F. Gómez, D. G. Castaño, M. Pombar, A. Gómez, C. Fernández. Monte Carlo Verification of IMRT treatment plans on Grid. In *HealthGrid*, pages 105–114, 2007.
- [42] Aalto Venture Garage seed accelerator <http://www.aaltovg.com>.
- [43] Agnar Aamodt and Enric Plaza. Case-based reasoning: Foundational issues, methodological variations, and system approaches. *AI Communications*, 7:39–59, 1994.
- [44] Daniel J. Abadi. Data Management in the Cloud: Limitations and Opportunities. *IEEE Data Eng. Bull.*, 32(1):3–12, 2009.
- [45] Karl Aberer and Zoran Despotovic. Managing trust in a peer-2-peer information system. In *CIKM*, pages 310–317, 2001.
- [46] A. Abouzeid, K. Bajda-Pawlikowski, D. Abadi, A. Rasin, and A. Silberschatz. Hadoopdb: An architectural hybrid of mapreduce and dbms technologies for analytical workloads. *PVLDB*, 2(1):922–933, 2009.
- [47] ActiveMQ. Messaging and integration pattern provider. <http://activemq.apache.org/>.
- [48] Atul Adya, William J. Bolosky, Miguel Castro, Gerald Cermak, Ronnie Chaiken, John R. Douceur, Jon Howell, Jacob R. Lorch, Marvin Theimer, and Roger P. Wattenhofer. Farsite: federated, available, and reliable storage for an incompletely trusted environment. In *Proceedings of the 5th Symposium on Operating Systems Design and Implementation (OSDI)*, OSDI '02, pages 1–14, New York, NY, USA, 2002. ACM.
- [49] R. Agarwal and M. Gort. First Mover Advantage and the Speed of Competitive Entry: 1887-1986. *Journal of Law and Economics*, 44(1):161–178, 2002.
- [50] International Atomic Energy Agency. IAEA (2009). Directory of Radiotherapy Centres (DIRAC). <http://www-naweb.iaea.org/nahu/dirac/query3.asp>.
- [51] Gagan Aggarwal, Mayank Bawa, Prasanna Ganesan, Hector Garcia-Molina, Krishnaram Kenthapadi, Rajeev Motwani, Utkarsh Srivastava, Dilys Thomas, and Ying Xu 0002. Two can keep a secret: A distributed architecture for secure database services. In *CIDR*, pages 186–199, 2005.

- [52] Gagan Aggarwal, Mayank Bawa, Prasanna Ganesan, Hector Garcia-Molina, Krishnaram Kenthapadi, Rajeev Motwani, Utkarsh Srivastava, Dilys Thomas, and Ying Xu. Two Can Keep A Secret: A Distributed Architecture for Secure Database Services. In *CIDR*, pages 186–199, 2005.
- [53] Divyakant Agrawal, Amr El Abbadi, Fatih Emekçi, and Ahmed Metwally. Database Management as a Service: Challenges and Opportunities. In *ICDE*, pages 1709–1716, 2009.
- [54] Ahronovitz, M. et al. Cloud Computing Use Cases White Paper, Version 4.0. http://opencloudmanifesto.org/Cloud_Computing_Use_Cases_Whitepaper-4_0.pdf.
- [55] S. Ajuja, N. Carriero, and D. Gelernter. Linda and friends. *Computer*, 19(8):26–34, 1986.
- [56] S. F. Altschul, W. Gish, W. Miller, E. W. Myers, and D. J. Lipman. Basic Local Alignment Search Tool. *Journal of Molecular Biology*, 215(3):403–410, October 1990.
- [57] Amazon. Amazon elastic compute cloud (amazon ec2). *Jrnl of Parallel and Distributed Computing*, 2008.
- [58] Amazon. Amazon simple storage service (amazon s3). *Jrnl of Parallel and Distributed Computing*, 2008.
- [59] Amazon. Amazon elastic compute cloud: <http://aws.amazon.com/ec2/>, August 2009.
- [60] Amazon. Amazon ec2 high performance computing: <http://aws.amazon.com/ec2/hpc-applications/>, September 2010.
- [61] Amazon. Amazon ec2 instance types: <http://aws.amazon.com/ec2/instance-types/>, September 2010.
- [62] Amazon. Amazon ec2 service level agreement: <http://aws.amazon.com/ec2-sla/>, November 2010.
- [63] Amazon Web Services. <http://aws.amazon.com>, Dec. 2010.
- [64] amazon.com. Amazon Elastic Compute Cloud. <http://aws.amazon.com/ec2/>.
- [65] Amazon Elastic Compute Cloud (Amazon EC2): <http://aws.amazon.com/ec2/>.
- [66] AMD Virtualization (AMD-V) Technology. <http://sites.amd.com/us/business/it-solutions/virtualization/Pages/amd-v.aspx>.

- [67] D. W. O. Rogers B. A. Faddegon G. X. Ding C. M. Ma W. J. Mackie. TR. BEAM: a Monte Carlo code to simulate radiotherapy treatment units. In *Med Phys.*, pages 503–524, 1995.
- [68] Novell BEA Systems. *Complete Transition of TUXEDO to BEA*. http://www.bea.com/framework.jsp?CNT=pr00002.htm&FP=/content/news_events/press_releases/1996, 1996.
- [69] J. Pena, D. M. González-Castaño, F. Gómez, A. Gago-Arias, F. J. González-Castaño, D. Rodríguez-Silva, A. Gómez, C. Mouriño, M. Pombar, M. Sánchez. eIMRT: a web platform for the verification and optimization of radiation treatment plans. In *J Appl Clin Med Phys*, volume 10, 2009.
- [70] J. Pena, D. M. González-Castaño, F. Gómez, F. F. Sánchez-Doblado, G. H. Hartmann. Automatic determination of primary electron beam parameters in Monte Carlo simulation. In *Med Phys.*, pages 1076–1084, 2007.
- [71] R. R. Patel and D. W. Arthur. The emergence of advanced brachytherapy techniques for common malignancies. In *Hematology/Oncology Clinics of North America*, pages 97–118, 2006.
- [72] B. R. B. Walters and D. W. O. Rogers. *DOSXYZnrc Users Manual*, 2002.
- [73] M. P. Coleman et. al. Responding to the challenge of cancer in Europe. *Institute of Public Health of the Republic of Slovenia*, 2008.
- [74] U. Amaldi G. Kraft. Particle accelerators take up the fight against cancer. *CERN Courier*, 46(10):17–20, 2006.
- [75] L. E. Taylor and M. Ding. A review of intensity-modulated radiation therapy. In *Current Oncology Reports*, pages 294–299, 2008.
- [76] A. Taylor and M. E. Powell. Intensity-modulated radiotherapy—what is it? In *Cancer Imaging*, pages 68–73, 2004.
- [77] M. Brada, M. Pijls-Johannesma, D. De Ruysscher. Proton therapy in clinical practice: Current clinical evidence. In *Journal of Clinical Oncology*, pages 965–970, 2007.
- [78] D. Schulz-Ertner, O. Jäkel, W. Schlegel. Radiation therapy with charged particles. In *Seminars in Radiation Oncology*, pages 249–259, 2006.
- [79] B. D. Kavanagh, R. D. Timmerman. Stereotactic radiosurgery and stereotactic body radiation therapy: An overview of technical considerations and clinical applications. In *Hematology/Oncology Clinics of North America*, pages 87–95, 2006.

- [80] T. S. Lawrence, R. K. Ten Haken, A. Giaccia. *Cancer: Principles and Practice of Oncology*, chapter Principles of Radiation Oncology. Philadelphia: Lippincott Williams and Wilkins, 2008.
- [81] B. Sotomayor, R. S. Montero, I. M. Llorente, I. Foster. Virtual Infrastructure Management in Private and Hybrid Clouds, *IEEE Internet Computing*, vol. 13, no. 5, pp. 14–22, Sep./Oct. 2009.
- [82] S. E. Noda, T. Lautenschlaeger, M. R. Siedow. Technological advances in radiation oncology for central nervous system tumors. In *Seminars in Radiation Oncology*, pages 179–186, 2008.
- [83] David G. Andersen, Jason Franklin, Michael Kaminsky, Amar Phanishayee, Lawrence Tan, and Vijay Vasudevan. Fawn: a fast array of wimpy nodes. In *Proceedings of the 22nd ACM Symposium on Operating Systems Principles 2009, SOSP 2009, Big Sky, Montana, USA, October 11-14, 2009*, pages 1–14, 2009.
- [84] Janna Anderson, Elon University, and Lee Rainie. Technical seminar report on cloud computing. *Intel Executive Summary*, 2005.
- [85] Juan M. Andrade. *The TUXEDO System: Software for Constructing and Managing Distributed Business Applications*, 1996. Addison-Wesley Professional.
- [86] Andrew. Wordpress. *3rd International Conference on Grid and Pervasive Computing-gpc-workshops*, 2008.
- [87] A. Andrieux, K. Czajkowski, A. Dan, K. Keahey, H. Ludwig, T. Nakata, J. Pruyne, J. Rofrano, S. Tuecke, and M. Xu. *Web Services Agreement Specification (WS-Agreement)*. Open Grid Forum, March 14, 2007.
- [88] A. Andrieux, K. Czajkowski, A. Dan, K. Keahey, H. Ludwig, J. Pruyne, J. Rofrano, S. Tuecke, and M. Xu. Web services agreement specification (WS-Agreement). In *Global Grid Forum*, 2004.
- [89] S. Androulidakis, T. Doukoglou, G. Patikis, and D. Kagklis. Service differentiation and traffic engineering in ip over wdm networks. *Communications Magazine, IEEE*, 46(5):52–59, May 2008.
- [90] Animoto: automatic production of video pieces from photos, video clips and music. based on AWS and rightscale <http://www.animoto.com>.
- [91] J. Anselmi, D. Ardagna, and P. Cremonesi. A QoS-based selection approach of autonomic grid services. In *Proceedings of the 2007 workshop on Service-oriented computing performance: aspects, issues, and approaches*, pages 1–8. ACM, 2007.
- [92] Google App. Multitenancy, 2010.

- [93] Appistry. Cloud Taxonomy: Applications, Platform, Infrastructure : <http://www.appistry.com/blogs/sam/cloud-taxonomyapplications-platform-infrastructure>, 2008.
- [94] Sam Charrington Appistry. Cloud taxonomy: Applications, platform, infrastructure, December 2008.
- [95] M. Armbrust, A. Fox, R. Griffith, A.D. Joseph, R.H. Katz, A. Konwinski, G. Lee, D.A. Patterson, A. Rabkin, I. Stoica, et al. Above the clouds: A Berkeley view of cloud computing. *EECS Department, University of California, Berkeley, Tech. Rep. UCB/EECS-2009-28*, 2009.
- [96] Michael Armbrust, Armando Fox, Rean Griffith, Anthony D. Joseph, Randy Katz, Andy Konwinski, Gunho Lee, David Patterson, Ariel Rabkin, Ion Stoica, and Matei Zaharia. A view of cloud computing. *Communications of the ACM*, 53:50–58, April 2010.
- [97] Michael Armbrust, Armando Fox, Rean Griffith, Anthony D. Joseph, Randy H. Katz, Andrew Konwinski, Gunho Lee, David A. Patterson, Ariel Rabkin, Ion Stoica, and Matei Zaharia. Above the clouds: A Berkeley view of cloud computing. Technical Report UCB/EECS-2009-28, EECS Department, University of California, Berkeley, Feb 2009.
- [98] Michael Armbrust, Anthony D Joseph, Randy H Katz, and David A Patterson. Above the Clouds : A Berkeley View of Cloud Computing. *University of California, Berkeley, Technical Report*, 2009.
- [99] Arvind. Decomposing a program for multiple processor system. In *Proceedings of the 1980 International Conference on Parallel Processing*, pages 7–14, 1980.
- [100] Everything as a Service. Gathering clouds of xaas. *Second International Symposium on Information Science and Engineering*, 2010.
- [101] G. Atanassov et al. Interval valued intuitionistic fuzzy sets. *Fuzzy sets and Systems*, 31(3):343–349, 1989.
- [102] L. Atzori and T. Onali. Operators challenges toward bandwidth management in diffserv-aware traffic engineering networks. *Communications Magazine, IEEE*, 46(5):154–160, May 2008.
- [103] Dariusz Rafal Augustyn and Lukasz Warchal. Cloud service solving n-body problem based on windows azure platform. In Andrzej Kwiecień, Piotr Gaj, and Piotr Stera, editors, *Computer Networks*, volume 79 of *Communications in Computer and Information Science*, pages 84–95. Springer Berlin Heidelberg, 2010.

- [104] Stefan Aulbach, Torsten Grust, Dean Jacobs, Alfons Kemper, and Jan Rittinger. Multi-tenant databases for software as a service: schema-mapping techniques. In *SIGMOD '08: Proceedings of the 2008 ACM SIGMOD international conference on Management of data*, pages 1195–1206, New York, NY, USA, 2008. ACM.
- [105] T. Auld, A. W. Moore, and S. F. Gull. Bayesian neural networks for internet traffic classification. *Neural Networks, IEEE Transactions on*, 18(1):223–239, Jan 2007.
- [106] AutoI project. Project Website: <http://ist-autoi.eu/autoi/index.php>.
- [107] AuverGrid. Project website. <http://www.auvergrid.fr>.
- [108] R. Aversa, B. Di Martino, N. Mazzocca, and S. Venticinque. A skeleton based programming paradigm for mobile multi-agents on distributed systems and its realization within the magda mobile agents platform. *Mob. Inf. Syst.*, 4:131–146, April 2008.
- [109] Azure service platform. <http://www.microsoft.com/azure/>.
- [110] J. S. Bagla and T Padmanabhan. Cosmological n-body simulations. *Pramana - Journal of Physics*, 49 (2), pages 161–192, 2008.
- [111] Satish Balay, Kris Buschelman, William D. Gropp, Dinesh Kaushik, Matthew G. Knepley, Lois Curfman McInnes, Barry F. Smith, and Hong Zhang. PETSc Web page, 2001. <http://www.mcs.anl.gov/petsc>.
- [112] Magdalena Balazinska, Hari Balakrishnan, and Mike Stonebraker. Contract-based load management in federated distributed systems. In *1st Symposium on Networked Systems Design and Implementation (NSDI)*, pages 197–210, San Francisco, USA, March 2004. USENIX Association.
- [113] K. Ballinger, P. Brittenham, A. Malhorta, W. A. Nagy, and S. Pharies. *Web Services Inspection Language (WS-Inspection)*, Nov. 2001.
- [114] Paul Baran. On distributed communications, rm-3420. Technical report, <http://www.rand.org/about/history/baran.list.html>, 1964.
- [115] Paul Barham, Boris Dragovic, Keir Fraser, Steven Hand, Tim Harris, Alex Ho, Rolf Neugebauer, Ian Pratt, and Andrew Warfield. Xen and the art of virtualization. In *SOSP '03: Proceedings of the nineteenth ACM symposium on Operating systems principles*, pages 164–177, New York, NY, USA, 2003. ACM.
- [116] Paul Barham, Boris Dragovic, Keir Fraser, Steven Hand, Tim Harris, Alex Ho, Rolf Neugebauer, Ian Pratt, and Andrew Warfield. Xen and the art of virtualization. *SIGOPS Oper. Syst. Rev.*, 37:164–177, October 2003.

- [117] Paul Barham, Boris Dragovic, Keir Fraser, Steven Hand, Tim Harris, Alex Ho, Rolf Neugebauer, Ian Pratt, and Andrew Warfield. Xen and the Art of Virtualization. In *Proceedings of the nineteenth ACM symposium on Operating systems principles*, pages 164 – 177, 2003.
- [118] Paul Barham, Boris Dragovic, Keir Fraser, Steven Hand, Tim Harris, Alex Ho, Rolf Neugebauer, Ian Pratt, and Andrew Warfield. Xen and the art of virtualization. In *Proceedings of the nineteenth ACM symposium on Operating systems principles*, SOSP '03, pages 164–177, New York, NY, USA, 2003. ACM.
- [119] Luiz André Barroso and Urs Hölzle. The case for energy-proportional computing. *IEEE Computer*, 40(12):33–37, 2007.
- [120] J. Basney. *A Distributed Implementation of the C-Linda Programming Language*. PhD thesis, Oberlin College, June 1995.
- [121] Christian Baun, Marcel Kunze, Jens Nimis, and Stefan Tai. *Cloud Computing: Web-basierte dynamische IT-Services*. Springer, 2010.
- [122] B. Benatallah, M. Dumas, M.C. Fauvet, and F.A. Rabhi. *Towards patterns of web services composition*. Springer-Verlag, 2003.
- [123] Hal Berenson, Phil Bernstein, Gray Jim, Jim Melton, Elizabeth O’Neil, and Patrick O’Neil. A critique of ANSI SQL isolation levels. In *Proceedings of the 1995 ACM SIGMOD international Conference on Management of Data*, pages 1–10. ACM, 1995.
- [124] Berkeley Millenium Project. Project Website: <https://www.millennium.berkeley.edu>.
- [125] Laurent Bernaille, Renata Teixeira, and Kave Salamatian. Early application identification. In *Proceedings of the 2006 ACM CoNEXT conference*, CoNEXT '06, pages 6:1–6:12, New York, NY, USA, 2006. ACM.
- [126] D. Bernstein, E. Ludvigson, K. Sankar, S. Diamond, and M. Morrow. Blueprint for the Intercloud – Protocols and Formats for Cloud Computing Interoperability. In *Proceedings of the 4th International Conference on Internet and Web Applications and Services (ICIW)*, Venice/Mestre, Italy, May 24–28 2009.
- [127] David Bernstein and Erik Ludvigson. Networking challenges and resultant approaches for large scale cloud construction. *Workshops at the Grid and Pervasive Computing Conference*, 2009.
- [128] WoShun Luk Bin Zhou, Jian Pei. A brief survey on anonymization techniques for privacy preserving publishing of social network data. *ACM SIGKDD Explorations Newsletter*, 10(Issue 2), 2008.

- [129] Andrew D. Birrell and Bruce Jay Nelson. Implementing remote procedure calls. *ACM Transactions on Computer Systems*, 2:39–59, 1984.
- [130] Tomas Bittman. The evolution of the cloud computing market. *Gartner Blog Network*, http://blogs.gartner.com/thomas_bittman/2008/11/03/the-evolution-of-the-cloud-computing-market/, November 2008.
- [131] Blobstore API from Google's App Engine: <http://code.google.com/intl/it-IT/appengine/docs/java/blobstore/>.
- [132] Harold Boley. The ruleml family of web rule languages. In José Júlio Alferes, James Bailey, Wolfgang May, and Uta Schwertel, editors, *PP-SWR*, volume 4187 of *Lecture Notes in Computer Science*, pages 1–17. Springer, 2006.
- [133] Matt Bone, Peter F. Nabicht, Konstantin Läuffer, and George K. Thiruvathukal. Taming XML: Objects first, then markup. In *Proc. IEEE Intl. Conf. on Electro/Information Technology (EIT)*, May 2008.
- [134] M. Boniface, S. C. Phillips, A. Sanchez-Macian, and M. Surridge. Dynamic service provisioning using GRIA SLAs. In *International Workshops on Service-Oriented Computing (ICSOC'07)*, 2007.
- [135] O. J. Boxma, J. W. Cohen, and N. Huffel. Approximations of the mean waiting time in an $M/G/s$ queueing system. *Operations Research*, 27:1115–1127, 1979.
- [136] Ivona Brandic. Towards self-manageable cloud services. In *33rd Annual IEEE International Computer Software and Applications Conference (COMPSAC'09)*, 2009.
- [137] Ivona Brandic, Dejan Music, Philipp Leitner, and Schahram Dustdar. Vieslaf framework: Enabling adaptive and versatile sla-management. In *Proceedings of the 6th International Workshop on Grid Economics and Business Models*, GECON '09, pages 60–73, 2009.
- [138] J. P. Brans, P. Vincke, and B. Mareschal. How to select and how to rank projects: The PROMETHEE method. *European Journal of Operational Research*, 24(2):228–238, 1986.
- [139] Matthias Brantner, Daniela Florescu, David A. Graf, Donald Kossmann, and Tim Kraska. Building a database on S3. In *SIGMOD Conference*, pages 251–264, 2008.
- [140] Brein. Business objective driven reliable and intelligent grids for real business. <http://www.eu-brein.com/>.
- [141] G. Breiter and M. Behrendt. Life cycle and characteristics of services in the world of cloud computing. *IBM Journal of Research and Development*, 53(4):3:1–3:8, July 2009.

- [142] John Brevik, Daniel Nurmi, and Rich Wolski. Predicting bounds on queuing delay for batch-scheduled parallel machines. In *PPoPP '06: Proceedings of the eleventh ACM SIGPLAN symposium on Principles and practice of parallel programming*, pages 110–118, New York, NY, USA, 2006. ACM.
- [143] Eric A. Brewer. Towards robust distributed systems (abstract). In *PODC*, page 7, 2000.
- [144] James Broberg, Rajkumar Buyya, and Zahir Tari. Metacdn: Harnessing ‘storage clouds’ for high performance content delivery. *J. Network and Computer Applications*, 32(5):1012–1022, 2009.
- [145] M. Brock and A. Goscinski. Attributed publication and selection for web service-based distributed systems. In *Proceedings of the 3rd International Workshop on Service Intelligence and Computing (SIC 2009) in conjunction with the 7th IEEE International Conference on Web Services (ICWS 2009)*, pages 732–739. IEEE Los Angeles, CA, USA, 2009.
- [146] M. Brock and A. Goscinski. Toward ease of discovery, selection and use of clusters within a cloud. *IEEE 3rd International Conference on Cloud Computing (CLOUD 2010)*, July 5–10 2010.
- [147] Chris Bunch, Navraj Chohan, Chandra Krintz, Jovan Chohan, Jonathan Kupferman, Puneet Lakhina, Yiming Li, and Yoshihide Nomura. An Evaluation of Distributed Datastores Using the AppScale Cloud Platform. In *IEEE Cloud10: International Conference on Cloud Computing*, July 2010.
- [148] Michael Burrows. The Chubby Lock Service for Loosely-Coupled Distributed Systems. In *OSDI*, pages 335–350, 2006.
- [149] R. Buyya. *Economic-based distributed resource management and scheduling for grid computing*. PhD thesis, Citeseer, 2002.
- [150] R. Buyya, C. S. Yeo, S. Venugopal, J. Broberg, and I. Brandic. Cloud computing and emerging IT platforms: Vision, hype, and reality for delivering computing as the 5th utility. *Future Generation Computer Systems*, 25(6):599–616, 2009.
- [151] R. Buyya, Chee Shin Yeo, and S. Venugopal. Market-oriented cloud computing: Vision, hype, and reality for delivering it services as computing utilities. In *High Performance Computing and Communications, 2008. HPCC '08. 10th IEEE International Conference on*, pages 5–13, Sep 2008.
- [152] R. Buyya, C.S. Yeo, S. Venugopal, J. Broberg, and I. Brandic. Cloud computing and emerging IT platforms: Vision, hype, and reality for delivering computing as the 5th utility. *Future Generation Computer Systems*, 25(6):599–616, 2009.

- [153] Rajkumar Buyya. Cloud computing: The next revolution in information technology. In *Parallel Distributed and Grid Computing (PDGC), 2010 1st International Conference on*, pages 2–3, October 2010.
- [154] Rajkumar Buyya, Chee S. Yeo, and Srikumar Venugopal. Market-Oriented Cloud Computing: Vision, Hype, and Reality for Delivering IT Services as Computing Utilities. In *HPCC '08: Proceedings of the 2008 10th IEEE International Conference on High Performance Computing and Communications*, pages 5–13. IEEE Computer Society, September 2008.
- [155] Rajkumar Buyya, Chee Shin Yeo, Srikumar Venugopal, James Broberg, and Ivona Brandic. Cloud computing and emerging it platforms: Vision, hype, and reality for delivering computing as the 5th utility. *Future Gener. Comput. Syst.*, 25:599–616, June 2009.
- [156] Rajkumar Buyya, Chee Shin Yeo, Srikumar Venugopal, James Broberg, and Ivona Brandic. Cloud computing and emerging it platforms: Vision, hype, and reality for delivering computing as the 5th utility. *Future Generation Computer Systems*, 25:599–616, June 2009.
- [157] Cactus Framework. <http://www.cactuscode.org>.
- [158] R. N. Calheiros, R. Buyya, and C. A. F. De Rose. Building an automated and self-configurable emulation testbed for grid applications. *Software: Practice and Experience*, 40(5):405–429, 2010.
- [159] Andrea Cali, Georg Gottlob, and Thomas Lukasiewicz. A general datalog-based framework for tractable query answering over ontologies. In *Proceedings of the twenty-eighth ACM SIGMOD-SIGACT-SIGART symposium on Principles of database systems*, PODS '09, pages 77–86, New York, NY, USA, 2009. ACM.
- [160] Roy Campbell, Indranil Gupta, Michael Heath, Steve Ko, Michael Kozuch, Marcel Kunze, Thomas Kwan, Kevin Lai, Hing Yan Lee, Martha Lyons, Dejan Milojicic, David O'Hallaron, and Yeng Chai Soh. Open CirrusTM Cloud Computing Testbed: Federated Data Centers for Open Source Systems and Services Research. In *HotCloud'09: Workshop in Hot Topics in Cloud Computing*, 2009.
- [161] K. Selcuk Candan, Wen-Syan Li, Thomas Phan, and Minqi Zhou. Frontiers in information and software as services. *Data Engineering, International Conference on*, 0:1761–1768, 2009.
- [162] C. Carlsson and R. Fullér. OWA operators for decision support. *Proceedings of EUFIT*, 97:8–11, 1997.
- [163] Jason Carolan and Steve Gaede. Introduction to cloud computing architecture. *Sun Microsystems Inc white paper*, 2009.

- [164] Nicholas Carr and Yan Yu. It is no longer important: the internet great change of the high ground - cloud computing. *The Big Switch: Rewiring the World, from Edison to Google*, CITIC Publishing House, October, 2008.
- [165] N. Carriero and D. Gelernter. *How to Write Parallel Programs - A First Course*. The MIT Press, Cambridge, MA, 1990.
- [166] Nicholas Carriero and David Gelernter. Linda in context. *Commun. ACM*, 32(4):444–458, 1989.
- [167] Chuck Cavaness. *Quartz Job Scheduling Framework: Building Open Source Enterprise Applications*. Prentice Hall PTR, Upper Saddle River, NJ, USA, 2006.
- [168] Nimbus Science Cloud: <http://workspace.globus.org/clouds/nimbus.html>.
- [169] Antonio Celesti, Francesco Tusa, Massimo Villari, and Antonio Puliafito. How to enhance cloud architectures to enable cross-federation. *Cloud Computing, IEEE International Conference on*, 0:337–345, 2010.
- [170] Memorial Sloan-Kettering Cancer Center and the University of Toronto. Pathway commons::sif interaction rules: http://www.pathwaycommons.org/pc/sif_interaction_rules.do, 26 December 2010.
- [171] Damien Cerbelaud, Shishir Garg, and Jeremy Huylebroeck. Opening the clouds: qualitative overview of the state-of-the-art open source vm-based cloud management platforms. In *Proceedings of the 10th ACM/IFIP/USENIX International Conference on Middleware*, Middleware '09, pages 22:1–22:8, New York, NY, USA, 2009. Springer-Verlag New York, Inc.
- [172] J. Cerviño, P. Rodríguez, J. Salvachúa, F. Escribano, and G. Huecas. Marte 3.0: Una videoconferencia 2.0. In *JITEL 2008*, pages 209–216, 2008.
- [173] Rohit Chandra. *Parallel Programming in OpenMP*. Morgan Kaufman, 2001.
- [174] M. Chandy and L. Lamport. Distributed snapshots: Determining global states of distributed systems. *ACM Transactions on Computing Systems*, 3(1):63–75, 1985.
- [175] Elizabeth Chang, Farookh Hussain, and Tharam Dillon. *Trust and Reputation for Service-Oriented Environments: Technologies For Building Business Intelligence And Consumer Confidence*. John Wiley & Sons, 2005.

- [176] Fay Chang, Jeffrey Dean, Sanjay Ghemawat, Wilson C. Hsieh, Deborah A. Wallach, Michael Burrows, Tushar Chandra, Andrew Fikes, and Robert E. Gruber. Bigtable: A Distributed Storage System for Structured Data. *ACM Trans. Comput. Syst.*, 26(2), 2008.
- [177] David Chappell. Introducing the windows Azure platform. David Chappell & Associates, October 2010.
- [178] David Chappell. Introducing Windows Aazure. David Chappell & Associates, October 2010.
- [179] S. Chen, S. Nepal and B. Yan. Sdsi: Simple distributed storage interface, csiro ict centre technical report, no. 06/322., 2006.
- [180] Shiping Chen, Surya Nepal, Jonathan Chan, David Moreland, and John Zic. Virtual storage services for dynamic collaborations. In *WETICE*, pages 186–191, 2007.
- [181] Yu Cheng, A. Leon-Garcia, and I. Foster. Toward an autonomic service management framework: A holistic vision of soa, aon, and autonomic computing. *Communications Magazine, IEEE*, 46(5):138–146, May 2008.
- [182] S. Cheshire and M. Krochmal. DNS-Based Service Discovery, Oct. 25, 2010. Internet-Draft.
- [183] R. Chinnici, J-J Moreau, A. Ryman, and S. Weerawarana. Web services description language (WSDL) version 2.0 part 1: Core language. W3C Recommendation, June 2007. Available from <http://www.w3.org/TR/wsdl20>.
- [184] C.Hoff. Cloud Taxonomy and Ontology : <http://rationalsecurity.typepad.com/blog/2009/01/cloud-computing-taxonomy-ontology.html>, 2009.
- [185] Navraj Chohan, Chris Buch, Sydney Pang, Chandra Krintz, Nagy Mostafa, Sunil Soman, and Rich Wolski. AppScale Design and Implementation. Technical report, UCSB, January 2009.
- [186] Frederick Chong, Gianpaolo Carraro, and Roger Wolter. Multi-tenant data architecture, 2006.
- [187] Online, available at <http://msdn.microsoft.com/en-us/library/aa479086.aspx>
- [188] T.C.K. Chou and J.A. Abraham. Load balancing in distributed systems. *Software Engineering, IEEE Transactions on*, SE-8(4):401–412, July 1982.

- [189] Richard Chow, Philippe Golle, Markus Jakobsson, Elaine Shi, Jessica Staddon, Ryusuke Masuoka, and Jesus Molina. Controlling data in the cloud: outsourcing computation without outsourcing control. In *Proceedings of the 2009 ACM workshop on Cloud computing security, CCSW '09*, pages 85–90, New York, NY, USA, 2009. ACM.
- [190] J. Christiansen. COPYING Y COMBINATOR – A framework for developing Seed Accelerator Programmes. *MBA Thesis*, 2009.
- [191] C. Clark, K. Fraser, S. Hand, J. G. Hanseny, E. July, C. Limpach, I. Pratt, and A. Warfield. Live Migration of Virtual Machines. In *Proceedings of the 2nd Symposium on Networked Systems Design and Implementation (NSDI)*, Boston, USA, May 2–4, 2005.
- [192] Christopher Clark, Keir Fraser, Steven Hand, Jacob Gorm Hansen, Eric Jul, Christian Limpach, Ian Pratt, and Andrew Warfield. Live migration of virtual machines. In *Proceedings of the 2nd conference on Symposium on Networked Systems Design & Implementation - Volume 2, NSDI'05*, pages 273–286, Berkeley, CA, USA, 2005. USENIX Association.
- [193] S. Clayman, A. Galis, and L. Mamatas. Monitoring virtual networks with lattice. In *Network Operations and Management Symposium Workshops (NOMS Wksp), 2010 IEEE/IFIP*, pages 239–246, April 2010.
- [194] Cloud Security Alliance. Top threats to cloud computing v 1.0. <http://www.cloudsecurityalliance.org/topthreats/csathreats.v1.0.pdf>, 2010. (accessed July 20, 2010).
- [195] Cloud4SOA. <http://www.cloud4soa.eu/>, Dec. 2010.
- [196] CloudSigma. Cloud hosting interface: <http://www.cloudsigma.com/en/platform-details/intuitive-web-interface>, October 2010.
- [197] C.A.C. Coello. A comprehensive survey of evolutionary-based multiobjective optimization techniques. *Knowledge and Information systems*, 1(3):129–156, 1999.
- [198] C.A.C. Coello and M.S. Lechuga. MOPSO: A proposal for multiple objective particle swarm optimization. *Proceedings of the Evolutionary Computation on*, pages 1051–1056, 2002.
- [199] M. Commuzzi, C. Kotsokalis, G. Spanoudakis, and R. Yahyapour. Establishing and Monitoring SLAs in Complex Service Based Systems. In *Proceedings of the International Conference on Web Services (ICWS)*, Los Angeles, USA, July 6–10 2009.
- [200] Adaptive Computing. Adaptive operating environment: <http://www.adaptivecomputing.com/solutions/solution-architecture.php>, November 2010.

- [201] Penguin Computing. Penguin computing on demand (pod): <http://www.penguincomputing.com/POD>, 16 November 2010.
- [202] Penguin Computing. Security on penguin computing on demand (pod) http://www.penguincomputing.com/files/datasheets/Security_on_POD.pdf, 10 April 2010.
- [203] Penguin Computing. Security <http://www.penguincomputing.com/POD/Security>, 16 November 2010.
- [204] M. Comuzzi, C. Kotsokalis, G. Spanoudakis, and R. Yahyapour. Establishing and monitoring SLAs in complex service based systems. In *IEEE International Conference on Web Services 2009*, 1009.
- [205] Brian F. Cooper, Raghu Ramakrishnan, Utkarsh Srivastava, Adam Silberstein, Philip Bohannon, Hans-Arno Jacobsen, Nick Puz, Daniel Weaver, and Ramana Yerneni. PNUTS: Yahoo!’s hosted data serving platform. *PVLDB*, 1(2):1277–1288, 2008.
- [206] Brian F. Cooper, Adam Silberstein, Erwin Tam, Raghu Ramakrishnan, and Russell Sears. Benchmarking cloud serving systems with YCSB. In *ACM SoCC*, pages 143–154, 2010.
- [207] Intel Corp. Intel xeon processor x5570. <http://ark.intel.com/Product.aspx?id=37111>.
- [208] R. S. Cox, J. G. Hansen, S. D. Gribble, and H. M. Levy. A safety-oriented platform for Web applications. *2006 IEEE Symposium on Security and Privacy*, 2006.
- [209] D. Crockford. The application/json Media Type for JavaScript Object Notation (JSON). RFC 4627 (Informational), July 2006.
- [210] S. Crosby, R. Doyle, M. Gering, M. Gionfriddo, S. Grarup, S. Hand, M. Hapner, D. Hiltgen, M. Johanssen, L.J. Lamers, J. Leung, F. Machida, A. Maier, E. Mellor, J. Parchem, S. Pardikar, S.J. Schmidt, R. W. Schmidt, A. Warfield, M.D. Weitzel, and J. Wilson. Open Virtualization Format Specification (OVF). Technical Report DSP0243, 2009.
- [211] Cloud security alliance: Security guidance for critical areas of focus in cloud computing <http://www.cloudsecurityalliance.org/guidance>.
- [212] Cloud security alliance: Top threats to cloud computing <http://www.cloudsecurityalliance.org/topthreats/csathreats.v1.0.pdf>.
- [213] Carlo Curino, Evan Jones, Yang Zhang, Eugene Wu, and Sam Madde. Relational Cloud: The Case for a Database Service. In *CIDR*, 2011.

- [214] CycleComputing. Cyclecloud: <http://www.cyclecomputing.com/products/cyclecloud/overview>, November 2010.
- [215] John Daintith. *Oxford Dictionary of Computing*. Oxford University Press, 4th edition, 2004.
- [216] A. D’Ambrogio and P. Bocciarelli. A model-driven approach to describe and predict the performance of composite services. In *6th International Workshop on Software and Performance (WOSP’07)*, 2007.
- [217] Ernesto Damiani, De Capitani di Vimercati, Stefano Paraboschi, Pierangela Samarati, and Fabio Violante. A reputation-based approach for choosing reliable resources in peer-to-peer networks. In *Proceedings of the 9th ACM conference on Computer and communications security, CCS ’02*, pages 207–216, New York, NY, USA, 2002. ACM.
- [218] A. Dan, D. Davis, R. Kearney, A. Keller, R. King, D. Kuebler, H. Ludwig, M. Polan, M. Spreitzer, and A. Youssef. Web services on demand: Wsla-driven automated management. *IBM Syst. J.*, 43:136–158, January 2004.
- [219] C. Darwin. *The origin of species by means of natural selection of the preservation of favoured races in the struggle for life*. Signet Classics, 2003.
- [220] A.V. Dastjerdi, S.G.H. Tabatabaei, and R. Buyya. An Effective Architecture for Automated Appliance Management System Applying Ontology-Based Cloud Discovery. In *Cluster, Cloud and Grid Computing (CCGrid), 2010 10th IEEE/ACM International Conference on*, pages 104–112. IEEE, 2010.
- [221] D. Baran. Cloud computing basics. *ICWS*, 2007.
- [222] S. K. De, R. Biswas, and A. R. Roy. An application of intuitionistic fuzzy sets in medical diagnosis. *Fuzzy Sets and Systems*, 117(2):209–213, 2001.
- [223] Marcos Dias de Assunção, Alexandre di Costanzo, and Rajkumar Buyya. Evaluating the cost-benefit of using cloud computing to extend the capacity of clusters. In *HPDC*, pages 141–150, 2009. <http://doi.acm.org/10.1145/1551609.1551635>.
- [224] J. de Bruijn, H. Lausen, R. Krummenacher, A. Polleres, L. Predoiu, M. Kifer, and D. Fensel. The web service modeling language WSM. *WSML Final Draft D*, 16, 2005.
- [225] J.C. de Oliveira, C. Scoglio, I.F. Akyildiz, and G. Uhl. New preemption policies for diffserv-aware traffic engineering to minimize rerouting in mpls networks. *Networking, IEEE/ACM Transactions on*, 12(4):733 – 745, Aug 2004.

- [226] Jeffrey Dean and Sanjay Ghemawat. Mapreduce: Simplified data processing on large clusters. In *Proceedings of the 6th Conference on Symposium on Operating Systems Design and Implementation*, pages 10–10, Berkeley, CA, USA, 2004. USENIX Association.
- [227] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE transactions on evolutionary computation*, 6(2):182–197, 2002.
- [228] Giuseppe DeCandia, Deniz Hastorun, Madan Jampani, Gunavardhan Kakulapati, Avinash Lakshman, Alex Pilchin, Swaminathan Sivasubramanian, Peter Voss hall, and Werner Vogels. Dynamo: Amazon’s highly available key-value store. In *SOSP*, pages 205–220, 2007.
- [229] Ewa Deelman, Gurmeet Singh, Miron Livny, Bruce Berriman, and John Good. The cost of doing science on the Cloud: The montage example. In *2008 ACM/IEEE Conference on Supercomputing (SC 2008)*, pages 1–12, Piscataway, NJ, USA, 2008. IEEE Press.
- [230] Chrysanthos Dellarocas. Reputation mechanism design in online trading environments with pure moral hazard. *Info. Sys. Research*, 16:209–230, June 2005.
- [231] J. B. Dennis. Data flow supercomputers. *Computer*, pages 48–56, 1980.
- [232] Department of Defense. *Trusted Computer System Evaluation Criteria*, Dec. 26, 1985. DoD 5200.28-STD.
- [233] Todd Deshane, Zachary Shepherd, Jeanna N. Matthews, Muli Ben-Yehuda, Amit Shah, and Balaji Rao. Quantitative comparison of Xen and KVM. *Xen summit*, June 2008.
- [234] N. A. Detorie. Helical tomotherapy: A new tool for radiation therapy. In *Journal of the American College of Radiology*, pages 63–66, 2008.
- [235] Amazon DevPay <http://aws.amazon.com/devpay>.
- [236] Prashant Dewan and Partha Dasgupta. Pride: peer-to-peer reputation infrastructure for decentralized environments. In *WWW (Alternate Track Papers & Posters)*, pages 480–481, 2004.
- [237] M. D. Dikaiakos, D. Katsaros, P. Mehra, G. Pallis, and A. Vakali. Cloud computing: Distributed internet computing for it and scientific research. *Internet Computing, IEEE*, 13(5):10–13, Sep 2009.
- [238] Tham Dillon, Chen Wu, and Elizabeth Chang. Cloud computing : Issues and challenges. *24th IEEE International Conference on Advanced Information Networking and Applications*, 2010.

- [239] Distributed Management Task Force. *Interoperable Clouds – A White Paper from the Open Clouds Standards Incubator*, Nov. 2009. DSP-IS0101.
- [240] Distributed Management Task Force. *Open Virtualization Format White Paper*, Feb. 2009. DSP2017 Version 1.0.0.
- [241] Distributed Management Task Force. *Open Virtualization Format Specification*, Jan. 12, 2010. DSP0243 Version 1.1.0.
- [242] G. Dobson and A. Sanchez-Macian. Towards unified QoS/SLA ontologies. In *IEEE Services Computing Workshops (SCW’06)*, 2006.
- [243] Dropbox: cloud based online backup <http://www.dropbox.com>.
- [244] J. J. Dujmovic. Mixed Averaging by Levels (MAL)–A System and Computer Evaluation Method. In *Proceedings of the Informatica Conference, Bled, Yugoslavia*, 1973.
- [245] E. Cristensen, F. Curbera, G. Meredith, S. Weerawarana. Web Services Description Language (WSDL) 1.1: <http://www.w3.org/TR/2001/NOTE-wsdl-20010315>, 2001.
- [246] Amazon elastic compute cloud. <http://aws.amazon.com/ec2/>.
- [247] C. Perkins (ed.). *IP Mobility Support for IPv4*, Aug. 2002. RFC 3344.
- [248] Vincent. C. Emeakaroha, I. Brandic, M. Maurer, and S. Dustdar. Low level metrics to high level SLAs - LoM2HiS framework: Bridging the gap between monitored metrics and SLA parameters in cloud environments. In *High Performance Computing and Simulation Conference (HPCS’10)*, 2010.
- [249] Vincent C. Emeakaroha, Rodrigo N. Calheiros, Marco A. S. Netto, Ivona Brandic, and César A. F. De Rose. DeSVi: An architecture for detecting SLA violations in cloud computing infrastructures. In *Proceedings of the 2nd International ICST Conference on Cloud Computing (Cloud-Comp’10)*, 2010.
- [250] Patrícia Takako Endo, Glaucio Estácio Gonçalves, Judith Kelner, and Djamel Sadok. A Survey on Open-Source Cloud Computing Solutions. *Brazilian Symposium on Computer Networks and Distributed Systems*, May 2010.
- [251] ENISA – cloud computing: Benefits, risks and recommendations for information security <http://www.enisa.europa.eu/act/rm/files/deliverables/cloud-computing-risk-assessment>.
- [252] U. Villano, E. P. Mancini, M. Rak. PerfCloud: GRID Services for Performance-oriented Development of Cloud Computing Applications. In *Proceedings of WETICE*. IEEE Computer Society, July 2009.

- [253] ESPER. Event stream processing. <http://esper.codehaus.org/>.
- [254] A. Goscinski et al. The Cloud Miner: Moving Data Mining to Computational Clouds. Grid and Cloud Database Management. Springer, 2010.
- [255] I. Khalil et al. The potential of biologic network models in understanding the etiopathogenesis of ovarian cancer. *Gynecol Oncol.* 116(2):282-5, 2004.
- [256] J. Shendure et al. Next-generation dna sequencing. *Nat Biotech* 26, pages 1135–1145, 2008.
- [257] P. Shannon et al. Cytoscape: a software environment for integrated models of biomolecular interaction networks. *Genome Research*, 13(11):2498-504, 2003.
- [258] Eucalyptus Documentation Page. <http://www.eucalyptus.com/>.
- [259] C. Evangelinos and C. Hill. Cloud computing for parallel scientific hpc applications: Feasibility of running coupled atmosphere-ocean climate models on amazon’s ec2. *Cloud Computing and its Applications*, 2008.
- [260] M. Factor, K. Meth, D. Naor, O. Rodeh, and J. Satran. Object Storage: the Future Building Block for Storage Systems. In *Proceedings of the 2nd International Symposium on Mass Storage Systems and Technology*, Sardinia, Italy, June 20–24, 2005.
- [261] Ming Fan, Yong Tan, and Andrew B. Whinston. Evaluation and design of online cooperative feedback mechanisms for reputation management. *IEEE Trans. on Knowl. and Data Eng.*, 17:244–254, February 2005.
- [262] Xiaobo Fan, Wolf-Dietrich Weber, and Luiz André Barroso. Power provisioning for a warehouse-sized computer. In *34th International Symposium on Computer Architecture (ISCA 2007)*, June 9-13, 2007, San Diego, California, USA, pages 13–23.
- [263] D. Farinacci, V. Fuller, D. Meyer, and D. Lewis. *Locator/ID Separation Protocol (LISP)*, March 2 2009. Internet-Draft.
- [264] D. Fensel and C. Bussler. The web service modeling framework WSMF. *Electronic Commerce Research and Applications*, 1(2):113–137, 2002.
- [265] J. M. Fernández Salido and S. Murakami. Extending Yager’s orness concept for the OWA aggregators to other mean operators. *Fuzzy Sets and Systems*, 139(3):515–542, 2003.
- [266] D. F. Ferraiolo and D. R. Kuhn. Role-Based Access Controls. In *Proceedings of the 15th National Computer Security Conference*, Baltimore, USA, Oct 13-16 1992.

- [267] Tiago C. Ferreto, Cesar A. F. de Rose, and Luiz de Rose. Rvision: An open and high configurable tool for cluster monitoring. *Cluster Computing and the Grid, IEEE International Symposium on*, 0:75, 2002.
- [268] Joe Fialli and Sekhar Vajjhala. Java architecture for XML binding (JAXB) 2.0. Java Specification Request (JSR) 222, October 2005.
- [269] R. Fielding. Representational state transfer (rest). *Architectural Styles and the Design of Network-based Software Architectures*. University of California, Irvine, 2000.
- [270] R. Fielding, H. Frystyk, Tim Berners-Lee, J. Gettys, and J. C. Mogul. Hypertext transfer protocol - HTTP/1.1, 1996.
- [271] Roy T. Fielding. *Architectural Styles and the Design of Network-based Software Architectures*. PhD thesis, University of California, 2000.
- [272] J. Figueira, S. Greco, and M. Ehrgott. *Multiple criteria decision analysis: state of the art surveys*. Springer Verlag, 2005.
- [273] Flexiscale. <http://www.flexiant.com/products/flexiscale/>, Dec. 2010.
- [274] Daniela Florescu and Donald Kossmann. Rethinking cost and performance of database systems. *SIGMOD Record*, 38(1):43–48, 2009.
- [275] Michael Flynn. Some computer organizations and their effectiveness. *IEEE Trans. Comput.*, c(21):948, 1972.
- [276] MESSAGE-PASSING INTERFACE FORUM. *MPI: A message-passing interface standard*. <http://www.mpi.org>, 1994.
- [277] FoSII. Foundations of self-governing infrastructures. <http://www.infosys.tuwien.ac.at/linksites/FOSII/index.html>.
- [278] I. Foster. What’s faster - a supercomputer or ec2?: <http://ianfoster.typepad.com>, 2009.
- [279] I. Foster, C. Kesselman, J. M. Nick, and S. Tuecke. The physiology of the grid. *Grid computing: making the global infrastructure a reality*, pages 217–250, 2003. Wiley and Sons.
- [280] I. Foster, C. Kesselman, and S. Tuecke. The anatomy of the grid: Enabling scalable virtual organizations. *International Journal of High Performance Computing Applications*, 15(3):200, 2001.
- [281] I. Foster, Y. Zhao, I. Raicu, and S. Lu. Cloud computing and grid computing 360-degree compared. In *Grid Computing Environments Workshop, 2008. GCE’08*, pages 1–10. IEEE, 2009.

- [282] I. Foster, Yong Zhao, I. Raicu, and S. Lu. Cloud computing and grid computing 360-degree compared. In *Grid Computing Environments Workshop, 2008. GCE '08*, pages 1–10, Nov 2008.
- [283] Ian Foster, Yong Zhao, Ioan Raicu, and Shiyong Lu. Cloud computing and grid computing 360-degree compared. *3rd International Conference on Grid and Pervasive Computing-gpc-workshops*, 2008.
- [284] Ian Foster, Yong Zhao, Ioan Raicu, and Shiyong Lu. Cloud computing and grid computing 360-degree compared. 2009.
- [285] J. Franks, P. Hallam-Baker, J. Hostetler, S. Lawrence, P. Leach, A. Luotonen, and L. Stewart. *HTTP Authentication Basic and Digest Access Authentication*, June 1999. RFC 2617.
- [286] FreeCBR. Online. <http://freecbr.sourceforge.net/>.
- [287] H. M. Frutos and I. Kotsiopoulos. BREIN: Business objective driven reliable and intelligent grids for real business. *International Journal of Interoperability in Business Information Systems*, 3(1):39–42, 2009.
- [288] Kevin Fu, M. Frans Kaashoek, and David Mazières. Fast and secure distributed read-only file system. *ACM Trans. Comput. Syst.*, 20:1–24, February 2002.
- [289] W. Fu and Q. Huang. GridEye: A service-oriented grid monitoring system with improved forecasting algorithm. In *International Conference on Grid and Cooperative Computing Workshops*, 2006.
- [290] Xiaodong Fu, Ping Zou, Ying Jiang, and Zhenhong Shang. Qos consistency as basis of reputation measurement of web service. In *Proceedings of the The First International Symposium on Data, Privacy, and E-Commerce*, pages 391–396, Washington, DC, USA, 2007. IEEE Computer Society.
- [291] B. Furht. Cloud computing fundamentals. In Borko Furht and Armando Escalante, editors, *Handbook of Cloud Computing*, pages 3–19. Springer US, 2010.
- [292] Galen Gruman and Eric Knorr. What cloud computing really means. InfoWorld : <http://www.infoworld.com/article/08/04/07/15FE-cloud-computing-reality-1.html>, 2008.
- [293] Ganglia Project. Project Website: <http://ganglia.sourceforge.net/>.
- [294] J. M. García, D. Ruiz, and A. Ruiz-Cortes. On User Preferences and Utility Functions in Selection: A Semantic Approach? In *Service-Oriented Computing-ICSOC 2007 Workshops: ICSOC 2007 International Workshops, Vienna, Austria, September 17, 2007, Revised Selected Papers*, page 105. Springer Verlag, 2009.

- [295] J. M. Garcia, D. Ruiz, A. Ruiz-Cortes, and J.A. Parejo. Qos-aware semantic service selection: An optimization problem. In *Services-Part I, 2008. IEEE Congress on*, pages 384–388. IEEE, 2008.
- [296] J. M. García, I. Toma, D. Ruiz, and A. Ruiz-Cortes. A service ranker based on logic rules evaluation and constraint programming. In *Proc. of 2nd Non Functional Properties and Service Level Agreements in SOC Workshop (NFPSLASOC)*, 2008.
- [297] Francisco Garcia-Sanchez, Eneko Fernandez-Breis, Rafael Valencia-Garcia, Enrique Jimenez, Juan M. Gomez, Javier Torres-Niño, and Daniel Martinez-Maqueda. Sitio: semantic business processes based on software-as-a-service and cloud computing. In *Proceedings of the 8th WSEAS International Conference on E-Activities and information security and privacy, E-ACTIVITIES’09/ISP’09*, pages 130–135, Stevens Point, Wisconsin, USA, 2009. World Scientific and Engineering Academy and Society (WSEAS).
- [298] Gartner. Gartner highlights five attributes of cloud computing: <http://www.gartner.com/it/page.jsp?id=1035013>, June 2009.
- [299] S. I. Gass and T. Rapcsák. Singular value decomposition in AHP. *European Journal of Operational Research*, 154(3):573–584, 2004.
- [300] J. Geelan. Twenty-one experts define cloud computing: <http://cloudcomputing.sys-con.com/node/612375>, 24 January 2009.
- [301] F. Gens. A key driver of new growth. idc forecast. it cloud services forecast, 2008.
- [302] Sanjay Ghemawat, Howard Gobioff, and Shun-Tak Leung. The google file system. *SIGOPS Oper. Syst. Rev.*, 37:29–43, October 2003.
- [303] Sanjay Ghemawat, Howard Gobioff, and Shun-Tak Leung. The Google file system. In *SOSP*, pages 29–43, 2003.
- [304] Seth Gilbert and Nancy A. Lynch. Brewer’s conjecture and the feasibility of consistent, available, partition-tolerant web services. *SIGACT News*, 33(2):51–59, 2002.
- [305] A. S. Glassner et al. *An introduction to ray tracing*. Academic Press London, 1989.
- [306] Rich Mogull and Glenn Brunette. Security guidance for critical areas of focus in cloud computing. *Cloud Security Alliance*, December 2009.
- [307] Gogrid. <http://www.gogrid.com>.
- [308] Eu-Jin Goh, Hovav Shacham, Nagendra Modadugu, and Dan Boneh. Sirius: Securing remote untrusted storage. In *NDSS*, 2003.

- [309] D. E. Goldberg. *Genetic algorithms in search, optimization, and machine learning*. Addison-Wesley, 1989.
- [310] R. P. Goldberg. Architecture of virtual machines. In *Proceedings of the workshop on virtual computer systems*, pages 74–112, New York, NY, USA, 1973. ACM.
- [311] R. P. Goldberg. Survey of virtual machine research. *IEEE Computer*, 7(6):34–45, June 1974.
- [312] Google App Engine: <http://code.google.com/appengine/>.
- [313] P. Goyal. Enterprise usability of cloud computing environments: Issues and challenges. In *Enabling Technologies: Infrastructures for Collaborative Enterprises (WETICE), 2010 19th IEEE International Workshop on*, pages 54–59, June 2010.
- [314] Bernardo Cuenca Grau, Ian Horrocks, Boris Motik, Bijan Parsia, Peter Patel-Schneider, and Ulrike Sattler. Owl 2: The next step for owl. *Web Semant.*, 6:309–322, November 2008.
- [315] Jim Gray. Distributed computing economics. Microsoft Research Technical Report MSRTR- 2003-24, Microsoft Research, 2003.
- [316] R. Grønmo and M. C. Jaeger. Model-driven methodology for building qos-optimised web service compositions. In *Distributed Applications and Interoperable Systems*, pages 68–82. Springer, 2005.
- [317] William Gropp and Ewing Lusk. Fault tolerance in mpi programs. *Special Issue, Journal of High Performance Computing Applications*, 1(18):363–372, 2002.
- [318] Yaqing Gu and Dean S. Oliver. The ensemble kalman filter for continuous updating of reservoir simulation models. *Journal of Engineering Resources Technology*, 128(1):79–87, 2006.
- [319] Yaqing Gu and Dean S. Oliver. An iterative ensemble kalman filter for multiphase fluid flow data assimilation. *SPE Journal*, 12(4):438–446, 2007.
- [320] Saikat Guha, Kevin Tang, and Paul Francis. Noyb: privacy in online social networks. In *Proceedings of the first workshop on Online social networks*, WOSP ’08, pages 49–54, New York, NY, USA, 2008. ACM.
- [321] T. Guha and S. A. Ludwig. Comparison of Service Selection Algorithms for Grid Services: Multiple Objective Particle Swarm Optimization and Constraint Satisfaction Based Service Selection. In *Tools with Artificial Intelligence, 2008. ICTAI’08. 20th IEEE International Conference on*, volume 1, pages 172–179. IEEE, 2008.

- [322] D. Gunter, B. Tierney, B. Crowley, M. Holding, and J. Lee. Netlogger: A toolkit for distributed system performance analysis. In *8th International Symposium on Modeling, Analysis and Simulation of Computer and Telecommunication Systems (MASCOTS'00)*, 2000.
- [323] Chang Jie Guo, Wei Sun, Ying Huang, Zhi Hu Wang, and Bo Gao. A framework for native multi-tenancy application development and management. *E-Commerce Technology, IEEE International Conference on, and Enterprise Computing, E-Commerce, and E-Services, IEEE International Conference on*, 0:551–558, 2007.
- [324] GWOS. Project Website: <http://www.groundworkopensource.com/products>.
- [325] H9Labs. Cloud computing explained - part 2. *3rd International Conference on Grid and Pervasive Computing-gpc-workshops*, 2008.
- [326] Volker Haarslev and Ralf Mäüller. Description of the racer system and its applications. In *Proc of DL2001 Workshop on Description Logics*, pages 132–141, 2001.
- [327] H. Hacigumus, B. Iyer, and S. Mehrotra. Providing database as a service. In *Data Engineering, 2002. Proceedings. 18th International Conference on*, pages 29–38, 2002.
- [328] Hakan Hacigümüs, Balakrishna R. Iyer, Chen Li, and Sharad Mehrotra. Executing SQL over encrypted data in the database-service-provider model. In *SIGMOD Conference*, pages 216–227, 2002.
- [329] Hakan Hacigümüs, Sharad Mehrotra, and Balakrishna R. Iyer. Providing Database as a Service. In *ICDE*, 2002.
- [330] Marc J. Hadley. Web application description language (WADL). Technical report, Sun Microsystems, Inc., Mountain View, CA, USA, 2006.
- [331] Yan Hai and Zhao Chong. Security policies based on security requirements of city emergency management information system multi-layer structure. In *IT in Medicine Education, 2009. ITIME '09. IEEE International Symposium on*, volume 1, pages 351–354, August 2009.
- [332] James Hamilton. On designing and deploying internet-scale services. In *LISA'07: Proceedings of the 21st conference on Large Installation System Administration Conference*, pages 1–12, Berkeley, CA, USA, 2007. USENIX Association.
- [333] James R. Hamilton. Cooperative expendable micro-slice servers (CEMS): Low cost, low power servers for internet-scale services. In *Proceedings of the 4th Biennial Conference on Innovative Data Systems Research (CIDR)*, 2009.

- [334] M. Handley, H. Schulzrinne, E. Schooler, and J. Rosenberg. SIP: Session Initiation Protocol. RFC 2543 (Proposed Standard), March 1999.
- [335] Steve Hanna. Cloud computing: Finding the silver lining. <http://www.ists.dartmouth.edu/docs/HannaCloudComputingv2.pdf>, 2009. (accessed March 12, 2009).
- [336] J. Hartigan. *Clustering Algorithms*. John Wiley and Sons, New York, 1975.
- [337] M. M. Hassan, B. Song, S. M. Han, E. N. Huh, C. Yoon, and W. Ryu. Multi-objective Optimization Model for Partner Selection in a Market-Oriented Dynamic Collaborative Cloud Service Platform. In *Tools with Artificial Intelligence, 2009. ICTAI'09. 21st International Conference on*, pages 637–644. IEEE, 2009.
- [338] Philipp C. Heckel. Hybrid Clouds: Comparing Cloud Toolkits. Technical report, University of Mannheim, 2010.
- [339] Mark Hefke. A framework for the successful introduction of KM using CBR and semantic web technologies. *Journal of Universal Computer Science*, 10(6), 2004.
- [340] Alexander Heitzmann, Bernardo Palazzi, Charalampos Papamanthou, and Roberto Tamassia. Efficient integrity checking of untrusted network storage. In *Proceedings of the 4th ACM international workshop on Storage security and survivability*, StorageSS '08, pages 43–54, New York, NY, USA, 2008. ACM.
- [341] N. Hemsoth. Hpc in the cloud: An intelligent public cloud for hpc: Computing shares news of cluster compute involvement: <http://www.hpcinthecloud.com/features/99334309.html>, 27 July 2010.
- [342] N. Hemsoth. Intel lays groundwork to fulfill 2015 cloud vision, hpc in the cloud: <http://www.hpcinthecloud.com/features/106350403.html>, 29 October 2010.
- [343] N. Hemsoth. Renting hpc - what's cloud got to do with it?, hpc in the cloud: <http://www.hpcinthecloud.com/features/97452994.html>, 29 October 2010.
- [344] N. Hemsoth. Virtualization and performance: Brocade's dr. maria iordache, hpc in the cloud: <http://www.hpcinthecloud.com/features/104296594.html>, 14 December 2010.
- [345] N. Hemsoth. Will public clouds ever be suitable for hpc?, hpc in the cloud: <http://www.hpcinthecloud.com/features/97269804.html>, 29 October 2010.

- [346] Nicole Hemsoth. Microsoft's Dan Reed on new paradigms for scientific discovery, hpc in the cloud: <http://www.hpcinthecloud.com/news/106298163.html>, October 2010.
- [347] Heroku: a ruby cloud platform <http://www.heroku.com>.
- [348] Tony Hey, Stewart Tansley, and Kristin Tolle, editors. *The Fourth Paradigm: Data-Intensive Scientific Discovery*. Microsoft Research, October 2009.
- [349] Z. Hill and M. Humphrey. A quantitative analysis of high performance computing with amazon's ec2 infrastructure: The death of the local cluster? In *10th IEEE/ACM International Conference on Grid Computing. Banff*, pages 26–33. SIAM Press, 2009.
- [350] D. Hilley. Cloud Computing: A Taxonomy of Platform & Infrastructure-level Offerings. Technical report, Georgia Institute of Technology, 2009.
- [351] H. Lohninger. *Teach/Me Data Analysis*. Springer-Verlag, Berlin-New York-Tokyo, ISBN 3-540-14743-8, 1999.
- [352] Christina Hoffa, Gaurang Mehta, Timothy Freeman, Ewa Deelman, Kate Keahey, Bruce Berriman, and John Good. On the Use of Cloud Computing for Scientific Workflows. *SWBES 2008*, December 2008.
- [353] P. Hokstad. Approximations for the $M/G/m$ queues. *Operations Research*, 26:510–523, 1978.
- [354] X. Hu and R. Eberhart. Multiobjective optimization using dynamic neighborhood particle swarm optimization. *Proceedings of the Evolutionary Computation on*, pages 1677–1681, 2002.
- [355] Mei Hui, Dawei Jiang, Guoliang Li, and Yuan Zhou. Supporting database applications as a service. *Data Engineering, International Conference on*, 0:832–843, 2009.
- [356] S. F. Hummel, E. Schonberg, and L. E. Flynn. Factoring: A method for scheduling parallel loops. *CACM*, 35(8):90–101, August 1992.
- [357] Trung Dong Huynh, Nicholas R. Jennings, and Nigel R. Shadbolt. An integrated trust and reputation model for open multi-agent systems. *Autonomous Agents and Multi-Agent Systems*, 13:119–154, September 2006.
- [358] C. L. Hwang and K. Yoon. *Multiple attribute decision making: methods and applications: a state-of-the-art survey*, volume 13. Springer-Verlag New York, 1981.
- [359] Kai Hwang. Massively distributed systems: From grids and p2p to clouds. In *Proceedings of The 3rd International Conference on Grid and Pervasive Computing - gpc-workshops*, page xxii, 2008.

- [360] Kai Hwang. Massively distributed systems:from grids and p2p to clouds. *3rd International Conference on Grid and Pervasive Computing-gpc-workshops*, 2008.
- [361] Hyperic Community. Project Website: <http://www.hyperic.com/community>.
- [362] Hyperic HQ. Project Website: <http://www.hyperic.com/>.
- [363] Hyperic HQ. Hq inventory model. Project Website: <http://www.hyperic.com/>, 2009.
- [364] I. Horrocks, P. F. Patel-Schneider, H. Boley, S. Tabet, B. Grosz, M. Dean. SWRL: A Semantic Web Rule Language Combining OWL and RuleML: <http://www.w3.org/Submission/SWRL/>.
- [365] IBM. *WebSphere: MQ V6 Fundamentals*, 2005.
- [366] IBM. *MC91: High Availability for WebSphere MQ on Unix Platforms, Vol., 7*, 2008.
- [367] IBM. IBM corp, cloud computing. *Journal of Object Technology*, 2009.
- [368] IBM. IBM point of view: Security and cloud computing. cloud computing white paper, 2009.
- [369] I. Foster, C. Kesselman, and S. Tuecke. The anatomy of the grid:enabling scalable virtual organization. *The Intl Jrnl of High Performance Computing Applications* 15(3):200-222, 2001.
- [370] I. Foster, C. Kesselman, and S. Tuecke. The physiology of the grid:an open grid services architecture for distributed systems integration. *Globus Project*, 2002.
- [371] I. Foster, J. Vockler, M. Wilde, and Y. Zhao. Chimera: a virtual data system for representing,querying and automating data derivation. *SS-DBM*, 2002.
- [372] R. Ihaka and R. Gentleman. R: A language for data analysis and graphics. *Journal of Computational and Graphical Statistics*, pages 299–314, 1996.
- [373] Silicon Graphics International. Hpc cloud computing - cyclone: http://www.sgi.com/products/hpc_cloud/cyclone/, 11 November 2010.
- [374] Silicon Graphics International. Hpc cloud computing: Cyclone applications: http://www.sgi.com/products/hpc_cloud/cyclone/applications.html, 11 November 2010.

- [375] Silicon Graphics International. Sgi announces cyclone cloud computing for technical applications: http://www.sgi.com/company_info/newsroom/press_releases/2010/february/cyclone.html, 16 November 2010.
- [376] A. Iosup, H. Li, M. Jan, S. Anoep, C. Dumitrescu, L. Wolters, and D.H.J. Epema. The grid workloads archive. *Future Generation Computer Systems*, 2008.
- [377] Alexandru Iosup, Dick H. J. Epema, Todd Tannenbaum, Matthew Farrellee, and Miron Livny. Inter-operating Grids through delegated match-making. In *2007 ACM/IEEE Conference on Supercomputing (SC 2007)*, pages 1–12, New York, USA, November 2007. ACM Press.
- [378] I. Raicu, Y. Zhao, C. Dumitrescu, I. Foster, and M. Vilde. Falkon: A fast and light-weight task execution framework. *IEEE/ACM SuperComputing*, 2007.
- [379] IT Infrastructure Library. <http://www.itil-officialsite.com>, Dec. 2010.
- [380] Jabber Software Foundation. Extensible messaging and presence protocol (xmpp): Core. 2004. Website available at <http://xmpp.org/rfc/rfc3920.html>
- [381] Keith R. Jackson, Lavanya Ramakrishnan, Karl J. Runge, and Rollin C. Thomas. Seeking supernovae in the clouds: a performance study. In *Proceedings of the 19th ACM International Symposium on High Performance Distributed Computing*, HPDC '10, pages 421–429, New York, NY, USA, 2010. ACM.
- [382] Dean Jacobs and Stefan Aulbach. Ruminations on multi-tenant databases. In Alfons Kemper, Harald Schöning, Thomas Rose, Matthias Jarke, Thomas Seidl, Christoph Quix, and Christoph Brochhaus, editors, *BTW*, volume 103 of *LNI*, pages 514–521. GI, 2007.
- [383] M. Jaeger and G. Rojec-Goldmann. SENECA—simulation of algorithms for the selection of web services for compositions. *Technologies for E-Services*, pages 84–97, 2006.
- [384] M. C. Jaeger, G. Muhl, and S. Golze. QoS-aware composition of web services: A look at selection algorithms. In *Web Services, 2005. ICWS 2005. Proceedings. 2005 IEEE International Conference on*. IEEE, 2005.
- [385] M. C. Jaeger, G. Rojec-Goldmann, and G. Muhl. Qos aggregation for web service composition using workflow patterns. In *Enterprise Distributed Object Computing Conference, 2004. EDOC 2004. Proceedings. Eighth IEEE International*, pages 149–159. IEEE, 2005.

- [386] Ravi Chandra Jammalamadaka, Roberto Gamboni, Sharad Mehrotra, Kent E. Seamons, and Nalini Venkatasubramanian. gvault: a gmail based cryptographic network file system. In *Proceedings of the 21st annual IFIP WG 11.3 working conference on Data and applications security*, pages 161–176, Berlin, Heidelberg, 2007. Springer-Verlag.
- [387] Ravi Chandra Jammalamadaka, Roberto Gamboni, Sharad Mehrotra, Kent E. Seamons, and Nalini Venkatasubramanian. . a middleware approach for building secure network drives over untrusted internet data storage. In *Proc. ACM conference on Extending database technology: Advances in database technology*, pages 710–714, 2008.
- [388] Jeffery F. Rayport and Andrew Heyward. Envisioning the cloud: The next computing paradigm. 2009. *Technical Report*, available at <http://www.hp.com/hpinfo/analystrelations/marketspace-090320-Envisioning-the-cloud.pdf>
- [389] Meiko Jensen and Nils Gruschka. Flooding attack issues of web services and service-oriented architectures. In *GI Jahrestagung (1)*, pages 117–122, 2008.
- [390] Meiko Jensen, Jorg Schwenk, Nils Gruschka, and Luigi Lo Iacono. On technical security issues in cloud computing. *IEEE International Conference on Cloud Computing*, 2009.
- [391] Jeremy Geelan. Twenty-one experts define cloud computing. Virtualization : <http://virtualization.sys-con.com/node/612375>, 2008.
- [392] J. Heiser and M. Nicolett. Assessing the security risks of cloud computing. *Gartner Report*, 2009.
- [393] Jim Gray. The dangers of replication and a solution. In *ACM SIGMOD International Conference on Management of Data Archive*, pages 173 – 182, Montreal, Quebec, Canada, 1996.
- [394] Enrique Jimenez Domingo, Javier Torres Nino, Angel Lagares Lemos, Miguel Lagares Lemos, Ricardo Colomo Palacios, and Juan Miguel Gomez Berbis. Cloudio: A cloud computing-oriented multi-tenant architecture for business information systems. *Cloud Computing, IEEE International Conference on*, 0:532–533, 2010.
- [395] Chris Dyer Jimmy Lin. *Data-IntensiveText Processing with MapReduce*. Morgan & Claypool, 2010.
- [396] Chao Jin and Rajkumar Buyya. Mapreduce programming model for .net-based cloud computing. In Henk Sips, Dick Epema, and Hai-Xiang Lin, editors, *Euro-Par 2009 Parallel Processing*, volume 5704 of *Lecture Notes in Computer Science*, pages 417–428. Springer Berlin / Heidelberg, 2009.

- [397] JMS. Java messaging service. <http://java.sun.com/products/jms/>.
- [398] D. Johnson, C. Perkins, and J. Arkko. *Mobility Support in IPv6*, June 2004. RFC 3775.
- [399] Don W Jones. *The Definitive Guide to Monitoring the Data Center, Virtual Environments, and the Cloud*. Nimsoft, 2010.
- [400] E. Jopling and D. Neill. Vno phenomenon could shake up the world’s telecom market. Gartner Research Report g00131283, 2005.
- [401] Joulemeter. Joulemeter: VM, server, client, and software energy usage. <http://research.microsoft.com/en-us/projects/joulemeter/>.
- [402] Gueyoung Jung, M.A. Hiltunen, K.R. Joshi, R.D. Schlichting, and C. Pu. Mistral: Dynamically managing power, performance, and adaptation cost in cloud infrastructures. In *Distributed Computing Systems (ICDCS), 2010 IEEE 30th International Conference on*, pages 62–73, june 2010.
- [403] Fu K. Group sharing and random access in cryptographic storage file system. master’s thesis. MIT, 1999.
- [404] Avinash W. Kadam. Information security policy development and implementation. *Inf. Sys. Sec.*, 16:246–256, September 2007.
- [405] Mahesh Kallahalla, Erik Riedel, Ram Swaminathan, Qian Wang, and Kevin Fu. Plutus: Scalable secure file sharing on untrusted storage. In *Proceedings of the 2nd USENIX Conference on File and Storage Technologies*, pages 29–42, Berkeley, CA, USA, 2003. USENIX Association.
- [406] R. E. Kalman. A new approach to linear filtering and prediction problems. *Transactions of the ASME–Journal of Basic Engineering*, 82(82 (Series D)):35–45, 1960.
- [407] Aditya Kalyanpur, Bijan Parsia, Evren Sirin, Bernardo C. Grau, and James Hendler. Swoop: A Web Ontology Editing Browser. *Web Semantics: Science, Services and Agents on the World Wide Web*, 4(2):144–153, June 2006.
- [408] Sepandar D. Kamvar, Mario T. Schlosser, and Hector Garcia-Molina. The eigentrust algorithm for reputation management in p2p networks. In *Proceedings of the 12th international conference on World Wide Web, WWW ’03*, pages 640–651, New York, NY, USA, 2003. ACM.
- [409] Aman Kansal, Feng Zhao, Jie Liu, Nupur Kothari, and Arka A. Bhattacharya. Virtual machine power metering and provisioning. In *Proceedings of the 1st ACM symposium on Cloud computing, SoCC ’10*, pages 39–50, New York, NY, USA, 2010. ACM.

- [410] Murat Kantarcioglu and Chris Clifton. Security Issues in Querying Encrypted Data. In *DBSec*, pages 325–337, 2005.
- [411] Thomas Karagiannis, Konstantina Papagiannaki, and Michalis Faloutsos. Blinc: multilevel traffic classification in the dark. *SIGCOMM Comput. Commun. Rev.*, 35:229–240, August 2005.
- [412] A. Karaman. Constraint-based routing in traffic engineering. In *Computer Networks, 2006 International Symposium on*, pages 1–6, 2006.
- [413] David R. Karger, Eric Lehman, Frank Thomson Leighton, Rina Panigrahy, Matthew S. Levine, and Daniel Lewin. Consistent Hashing and Random Trees: Distributed Caching Protocols for Relieving Hot Spots on the World Wide Web. In *STOC*, pages 654–663, 1997.
- [414] John F. Karpovich, Andrew S. Grimshaw, and James C. French. Extensible file system (ELFS): an object-oriented approach to high performance file I/O. In *OOPSLA '94: Proceedings of the ninth annual conference on Object-oriented programming systems, language, and applications*, pages 191–204, New York, NY, USA, 1994. ACM.
- [415] G. Katsaros, G. Kousiouris, S. Gogouvitis, D. Kyriazis, and T. Varvarigou. A service oriented monitoring framework for soft real-time applications. In *Service-Oriented Computing and Applications*, 2010.
- [416] Harry Katzan. Cloud software service: Concepts, technology, economics. *Service Science*, 1:256–269, 2009.
- [417] Joe Kaylor, Konstantin Läufer, and George K. Thiruvathukal. Online layered file system (OLFS): A layered and versioned filesystem and performance analysis. In *Proc. IEEE Intl. Conf. on Electro/Information Technology (EIT)*, May 2010.
- [418] Joe Kaylor, Konstantin Läufer, and George K. Thiruvathukal. RestFS: A FUSE filesystem to expose REST-ful services. <http://restfs.googlecode.com/>, 2010–2011.
- [419] Joe Kaylor, George K. Thiruvathukal, and Konstantin Läufer. Naked object file system (NOFS): A framework to expose an object-oriented domain model as a filesystem. Technical report, Loyola University Chicago, May 2010.
- [420] K. Keahey, R. Figueiredo, J. Fortes, T. Freeman, and M. Tsugawa. Science Clouds: Early Experiences in Cloud Computing for Scientific Applications, August 2008.
- [421] Katarzyna Keahey and Tim Freeman. Contextualization: Providing One-Click Virtual Clusters. *eScience 2008*, December 2008.

- [422] Chiaw K. Kee. Security policy roadmap - process for creating security policies. http://www.sans.org/reading_room/whitepapers/policyissues/security-policy-roadmap-process-creating-security-policies_494, 2001. (accessed August 13, 2010).
- [423] Bettina Kemme, Ricardo Jiménez-Peris, and Marta Patiño-Martínez. *Database Replication*. Synthesis Lectures on Data Management. Morgan & Claypool Publishers, 2010.
- [424] J. Kennedy, R.C. Eberhart, et al. Particle swarm optimization. In *Proceedings of IEEE international conference on neural networks*, volume 4, pages 1942–1948. Perth, Australia, 1995.
- [425] J. O. Kephart and D. M. Chess. The vision of autonomic computing. *IEEE Computer*, 36(1):41–50, 2003.
- [426] J.O. Kephart and W.E. Walsh. An artificial intelligence perspective on autonomic computing policies. In *Policies for Distributed Systems and Networks, 2004. POLICY 2004. Proceedings. Fifth IEEE International Workshop on*, pages 3–12. IEEE, 2004.
- [427] Brian W. Kernighan and Rob Pike. *The UNIX Programming Environment*. Prentice Hall Professional Technical Reference, 1983.
- [428] Roy W. Keyes, Christian Romano, Dorian Arnold, and Shuang Luan. Radiation therapy calculations using an on-demand virtual cluster via cloud computing. Technical Report arXiv:1009.5282, University of New Mexico, Sep 2010. Comments: 12 pages, 4 figures.
- [429] Vishal Kher and Yongdae Kim. Building trust in storage outsourcing: Secure accounting of utility storage. In *Proceedings of the 26th IEEE International Symposium on Reliable Distributed Systems, SRDS '07*, pages 55–64, Washington, DC, USA, 2007. IEEE Computer Society.
- [430] Hyunjoo Kim, Shivangi Chaudhari, Manish Parashar, and Christopher Marty. Online risk analytics on the cloud. In *Cluster Computing and the Grid, 2009. CCGRID '09. 9th IEEE/ACM International Symposium on*, pages 484–489, May 2009.
- [431] Hyunjoo Kim, M. Parashar, D.J. Foran, and Lin Yang. Investigating the use of autonomic cloudbursts for high-throughput medical image registration. In *Grid Computing, 2009 10th IEEE/ACM International Conference on*, pages 34–41, October 2009.
- [432] Ted H. Kim and Gerald J. Popek. Frigate: an object-oriented file system for ordinary users. In *COOTS'97: Proceedings of the 3rd conference on USENIX Conference on Object-Oriented Technologies (COOTS)*, pages 9–9, Berkeley, CA, USA, 1997. USENIX Association.

- [433] T. Kimura. Diffusion approximation for an $M/G/m$ queue. *Operations Research*, 31:304–321, 1983.
- [434] T. Kimura. Optimal buffer design of an $M/G/s$ queue with finite capacity. *Communications in Statistics $\tilde{U}$ Stochastic Models*, 12(6):165–180, 1996.
- [435] T. Kimura. A transform-free approximation for the finite capacity $M/G/s$ queue. *Operations Research*, 44(6):984–988, 1996.
- [436] Avi Kivity, Yaniv Kamay, Dor Laor, Uri Lublin, and Anthony Liguori. kvm: the linux virtual machine monitor. In *Proceedings of the Linux Symposium*, pages 225–230, Ottawa, Ontario, Canada, 2007.
- [437] S. R. Kleiman. Vnodes: An architecture for multiple file system types in Sun UNIX. In *Proc. Summer USENIX Technical Conf.*, pages 238–247, 1986.
- [438] L. Kleinrock. *Queueing Systems, volume 1, Theory*. Wiley-Interscience, 1975.
- [439] B. Koller and L. Schubert. Towards autonomous sla management using a proxy-like approach. *Multiagent Grid Systems*, 3(3):313–325, 2007.
- [440] J. Koomey. Estimating total power consumption by servers in the U.S. and the world. http://hightech.lbl.gov/documents/DATA_CENTERS/svrpwrusecompletefinal.pdf, 2007.
- [441] Donald Kossmann, Tim Kraska, and Simon Loesing. An evaluation of alternative architectures for transaction processing in the cloud. In *SIGMOD Conference*, pages 579–590, 2010.
- [442] D. Krafzig, K. Banke, and D. Slama. *Enterprise SOA: Service-Oriented Architecture Best Practices (The Coad Series)*. Prentice Hall PTR Upper Saddle River, NJ, USA, 2004.
- [443] Tim Kraska, Martin Hentschel, Gustavo Alonso, and Donald Kossmann. Consistency Rationing in the Cloud: Pay only when it matters. *PVLDB*, 2(1):253–264, 2009.
- [444] Sriram Krishnan. *Programming Windows Azure: Programming the Microsoft Cloud*. O'REILLY, Sebastopol, CA, USA, May 2009.
- [445] K. Kritikos and D. Plexousakis. Semantic qos metric matching. 2006.
- [446] John Kubiawicz, David Bindel, Yan Chen, Steven Czerwinski, Patrick Eaton, Dennis Geels, Ramakrishan Gummadi, Sean Rhea, Hakim Weatherspoon, Westley Weimer, Chris Wells, and Ben Zhao. Oceanstore: an architecture for global-scale persistent storage. *SIG-PLAN Not.*, 35:190–201, November 2000.

- [447] P. Lairds. Cloud Computing Taxonomy. In *Procs. Interop09*, pages 201–206. IEEE Computer Society, May 2009.
- [448] Avinash Lakshman and Prashant Malik. Cassandra: structured storage system on a p2p network. In *PODC*, page 5, 2009.
- [449] Avinash Lakshman and Prashant Malik. Cassandra: a decentralized structured storage system. *Operating Systems Review*, 44(2):35–40, 2010.
- [450] S. Lamparter, A. Ankolekar, R. Studer, and S. Grimm. Preference-based selection of highly configurable web services. In *Proceedings of the 16th international conference on World Wide Web*, pages 1013–1022. ACM, 2007.
- [451] S. Lamparter, A. Ankolekar, R. Studer, D. Oberle, and C. Weinhardt. A policy framework for trading configurable goods and services in open electronic markets. In *Proceedings of the 8th international conference on Electronic commerce: The new e-commerce: innovations for conquering current barriers, obstacles and limitations to conducting successful business on the internet*, pages 162–173. ACM, 2006.
- [452] Leslie Lamport. Time, clocks, and the ordering of events in a distributed system. *CACM*, 21(7):558–565, 1978.
- [453] Lattice Monitoring Framework. Project Website: <http://clayfour.ee.ucl.ac.uk/lattice>.
- [454] F. Le Faucheur. RFC 4124: Protocol Extensions for Support of Diffserv-aware MPLS Traffic Engineering. Technical report, IETF, June 2005.
- [455] N. Leavitt. Is cloud computing really ready for prime time? *Growth*, 27:5, 2009.
- [456] Neal Leavitt. Is Cloud Computing Really Ready for Prime Time? *Computer*, 42:15–20, 2009.
- [457] Cynthia Bailey Lee and Allan Snavely. On the user-scheduler dialogue: Studies of user-provided runtime estimates and utility functions. *Int. J. High Perform. Comput. Appl.*, 20(4):495–506, 2006.
- [458] Kevin Lee, Norman W. Paton, Rizos Sakellariou, and A. A. Fernandes Alvaro. Utility driven adaptive workflow execution. In *CCGRID '09: Proceedings of the 2009 9th IEEE/ACM International Symposium on Cluster Computing and the Grid*, pages 220–227, Washington, DC, USA, 2009. IEEE Computer Society.
- [459] Tsang-yeen Lee, Huey-ming Lee, Homer Wu, and Jin-shieh Su. Data Transmission Encryption and Decryption Algorithm in Network Security. *Proceedings of the 6th WSEAS International Conference on Simulation, Modelling and Optimization*, pages 417–422, 2006.

- [460] Y. C. Lee and A. Y. Zomaya. Energy conscious scheduling for distributed computing systems under different operating conditions. *IEEE Trans. Parallel Distrib. Syst.*, in press.
- [461] Young Choon Lee, Chen Wang, Albert Y. Zomaya, and Bing Bing Zhou. Profit-driven service request scheduling in clouds. In *Proceedings of the 2010 10th IEEE/ACM International Conference on Cluster, Cloud and Grid Computing*, CCGRID '10, pages 15–24, Washington, DC, USA, 2010. IEEE Computer Society.
- [462] Zhou Lei, Bofeng Zhang, Wu Zhang, Qing Li, Xuejun Zhang, and Junjie Peng. Comparison of Several Cloud Computing Platforms. *Second International Symposium on Information Science and Engineering*, pages 23–27, 2009.
- [463] Alexander Lenk, Markus Klems, Jens Nimis, Stefan Tai, and Thomas Sandholm. What's inside the cloud? an architectural map of the cloud landscape. In *Proceedings of the 2009 ICSE Workshop on Software Engineering Challenges of Cloud Computing*, CLOUD '09, pages 23–31, Washington, DC, USA, 2009. IEEE Computer Society.
- [464] Alexander Lenk, Markus Klems, Jens Nimis, Stefan Tai, and Thomas Sandholm. What's inside the cloud? an architectural map of the cloud landscape. In *Proceedings of the 2009 ICSE Workshop on Software Engineering Challenges of Cloud Computing*, CLOUD '09, pages 23–31, Washington, DC, USA, 2009. IEEE Computer Society.
- [465] Ang Li, Xiaowei Yang, Srikanth Kandula, and Ming Zhang. Cloudcmp: comparing public cloud providers. In *Proceedings of the 10th annual conference on Internet measurement*, IMC '10, pages 1–14, New York, NY, USA, 2010. ACM.
- [466] Feifei Li, Marios Hadjieleftheriou, George Kollios, and Leonid Reyzin. Dynamic authenticated index structures for outsourced databases. In *In SIGMOD*, pages 121–132. SIGMOD, 2006.
- [467] Henry Li. *Introducing Windows Azure*. Apress, Berkely, CA, USA, 2009.
- [468] Jie Li, Marty Humphrey, Deb Agarwal, Keith Jackson, Catharine van Ingen, and Youngryel Ryu. escience in the cloud: A modis satellite data reprojection and reduction pipeline in the windows azure platform. In *Parallel Distributed Processing (IPDPS), 2010 IEEE International Symposium on*, pages 1–10, 2010.
- [469] W. Li, L. Ping, and X. Pan. Use Trust Management Module to Achieve Effective Security Mechanisms in Cloud Environment. In *Proceedings of the International Conference On Electronics and Information Engineering (ICEIE)*, Kyoto, Japan, Aug. 1–3, 2010.

- [470] Xin Li, Christopher White, Zhou Lei, and Gabrielle Allen. Reservoir model updating by ensemble kalman filter-practical approaches using grid computing technology. In *Petroleum Geostatistics 2007*, Cascais, Portugal, August 2007.
- [471] Zhen Li and Manish Parashar. A computational infrastructure for grid-based asynchronous parallel applications. In *HPDC '07: Proceedings of the 16th international symposium on High performance distributed computing*, pages 229–230, New York, NY, USA, 2007. ACM.
- [472] Libvirt. Project Website: <http://www.libvirt.org>.
- [473] M. Lieberman and D. Montgomery. First-Mover Advantages. *Strategic Management Journal*, 9(Special Issue: Strategy Content Research):41–58, 1988.
- [474] John Little. A proof for the queueing formula: $l = \lambda w$. *Operations Research: A Journal of the Institute for Operations Research and the Management Sciences*, pages 383–387, 1961.
- [475] Xiaoqing (Frank) Liu, Lijun Dong, and Hungwen Lin. High order object-oriented modeling technique for structured object-oriented analysis. *ACIS Int. J Comp. Inf. Sci.*, 2(2):74–96, 2001.
- [476] Yan Liu, Mingguang Zhuang, Qingling Wang, and Guannan Zhang. A new approach to web services characterization. In *Asia-Pacific Services Computing Conference, 2008. APSCC '08. IEEE*, pages 404–409, Dec 2008.
- [477] Yan Liu, Mingguang Zhuang, Biao Yu, Guannan Zhang, and Xiaojing Meng. Services characterization with statistical study on existing web services. In *Web Services, 2008. ICWS '08. IEEE International Conference on*, pages 803–804, Sep 2008.
- [478] Z. Liu, T. Liu, L. Cai, and G. Yang. Quality Evaluation and Selection Framework of Service Composition Based on Distributed Agents. In *2009 Fifth International Conference on Next Generation Web Services Practices*, pages 68–75. IEEE, 2009.
- [479] Lokad.Cloud Project. <http://code.google.com/p/lokad-cloud/>.
- [480] D. A. Low, W. B. Harms, S. Mutic, and J. A. Purdy. A technique for the quantitative evaluation of dose distributions. *Medical Physics*, 25(5):656–661, 1998.
- [481] Wei Lu, Jared Jackson, and Roger Barga. Azureblast: a case study of developing science applications on the cloud. In *Proceedings of the 19th ACM International Symposium on High Performance Distributed Computing*, HPDC '10, pages 413–420, New York, NY, USA, 2010. ACM.

- [482] H. Ludwig, A. Keller, A. Dan, R. P. King, and R. Franck. Web service level agreement (WSLA) language specification. *IBM Corporation*, 2003.
- [483] H. Ludwig, A. Keller, A. Dan, R. P. King, and R. Franck. Web service level agreement (WSLA) language specification. *IBM Corporation*, 2003.
- [484] S. A. Ludwig and SMS Reyhani. Selection algorithm for grid services based on a quality of service metric. In *High Performance Computing Systems and Applications, 2007. HPCS 2007. 21st International Symposium on*, page 13. IEEE, 2007.
- [485] S.A. Ludwig and SMS Reyhani. Selection algorithm for grid services based on a quality of service metric. In *High Performance Computing Systems and Applications, 2007. HPCS 2007. 21st International Symposium on*, page 13. IEEE, 2007.
- [486] Ian Lumb, Eunmi Choi, and Bhaskar Prasad Rimal. Virtualization for dummies. *Second International Symposium on Information Science and Engineering*, 2010.
- [487] Lutz Schubert. The Future of Cloud Computing - Opportunities for European Cloud Computing Beyond 2010. <http://cordis.europa.eu/fp7/ict/ssai/docs/cloud-report-final.pdf>.
- [488] B. N. W. Ma and J. W. Mark. Approximation of the mean queue length of an $M/G/c$ queueing system. *Operations Research*, 43:158–165, 1998.
- [489] Dan Ma. The Business Model of “Software-As-A-Service.” In *Proceedings of the International Conference on Service Computing*, July 2007.
- [490] G. S. Machado, D. Hausheer, and B. Stiller. Considerations on the Interoperability of and between Cloud Computing Standards. In *Proceedings of the 27th Open Grid Forum (OGF27)*, Banff, Canada, Oct. 12–15, 2009.
- [491] Huajian Mao, Nong Xiao, Weisong Shi, and Yutong Lu. Wukong: Toward a cloud-oriented file service for mobile devices. In *Proceedings of the 2010 IEEE International Conference on Services Computing, SCC '10*, pages 498–505, Washington, DC, USA, 2010. IEEE Computer Society.
- [492] Maplesoft, Inc. *Maple 13*. CRC Press, Waterloo, ON, Canada, 2009.
- [493] Sojan Markose, Xiaoqing (Frank) Liu, and Bruce McMillin. A systematic framework for structured object-oriented security requirements analysis in embedded systems. In *EUC '08: Proceedings of the 2008 IEEE/IFIP International Conference on Embedded and Ubiquitous Computing*, pages 75–81, Washington, DC, USA, 2008. IEEE Computer Society.

- [494] M. Armbrust, A. Fox, R. Griffith, A. Joseph, R. Katz, A. Konwinski, G. Lee, D. Patterson, A. Rabkin, and I. Stoica. Above the clouds: a Berkeley view of cloud computing. *EECS Department, University of California, Berkeley, Tech Rep*, 2009.
- [495] Paul Marshall, Kate Keahey, and Tim Freeman. Elastic Site: Using Clouds to Elastically Extend Site Resources. *IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid 2010)*, May 2010.
- [496] Paul Marshall, Kate Keahey, and Tim Freeman. Elastic site: Using clouds to elastically extend site resources. *Cluster Computing and the Grid, IEEE International Symposium on*, 0:43–52, 2010.
- [497] Paul Marshall, Kate Keahey, and Tim Freeman. Elastic site: Using clouds to elastically extend site resources. *Cluster Computing and the Grid, IEEE International Symposium on*, 0:43–52, 2010.
- [498] D. Martin, M. Burstein, J. Hobbs, O. Lassila, D. McDermott, S. McIlraith, S. Narayanan, M. Paolucci, B. Parsia, T. Payne, et al. OWL-S: Semantic markup for web services. *W3C Member Submission*, 22:2007–04, 2004.
- [499] David Martin, Massimo Paolucci, Sheila Mcilraith, Mark Burstein, Drew McDermott, Deborah McGuinness, Bijan Parsia, Terry Payne, Marta Sabou, Monika Solanki, Naveen Srinivasan, and Katia Sycara. Bringing Semantics to Web Services: The OWL-S Approach. In J. Cardoso and A. Sheth, editors, *SWSWPC 2004*, volume 3387 of *LNCS*, pages 26–42. Springer, 2004.
- [500] M. L. Massie, B. N. Chun, and D. E. Culler. The Ganglia distributed monitoring system: Design, implementation and experience. *Parallel Computing*, 30(7):817–840, 2004.
- [501] M. Maurer, I. Brandic, V. C. Emeakaro, and S. Dustdar. Towards knowledge management in self-adaptable clouds. In *4th International Workshop of Software Engineering for Adaptive Service-Oriented Systems (SEASS’10)*, 2010.
- [502] E. M. Maximilien and M. P. Singh. A framework and ontology for dynamic web services selection. *Internet Computing, IEEE*, 8(5):84–93, 2004.
- [503] D. L. McGuinness, F. Van Harmelen, et al. OWL web ontology language overview. *W3C recommendation*, 10:2004–03, 2004.
- [504] McGuinness, D. L., van Harmelen, F. OWL Web Ontology Language Overview. W3C Recommendation: <http://www.w3.org/TR/2004/REC-owl-features-20040210/>, 2004.

- [505] Michael McIntosh, Paula Austel, and Yorktown Heights. XML Signature Element Wrapping Attacks and countermeasures. *IBM research division*, 23691, 2005.
- [506] Marvin McNett, Diwaker Gupta, Amin Vahdat, and Geoffrey M. Voelker. Usher: An Extensible Framework for Managing Clusters of Virtual Machines. In *Proceedings of the 21st Large Installation System Administration Conference (LISA)*, November 2007.
- [507] David Meisner, Brian T. Gold, and Thomas F. Wenisch. Powernap: eliminating server idle power. In *Proceeding of the 14th international conference on Architectural support for programming languages and operating systems*, ASPLOS '09, pages 205–216, New York, NY, USA, 2009. ACM.
- [508] P. Mell and T. Grance. The nist definition of cloud computing; <http://csrc.nist.gov/groups/SNS/cloud-computing/>, 18 December 2009.
- [509] P. Mell and T. Grance. The NIST Definition of Cloud Computing. Technical report, National Institute of Standards & Technology, Oct. 7, 2009.
- [510] Members of EGEE-II. An EGEE comparative study: Grids and clouds - evolution or revolution. Technical report, Enabling Grids for E-science Project : <https://edms.cern.ch/document/925013/>, 2008.
- [511] D. A. Menasc . QoS issues in Web services. *Internet Computing, IEEE*, 6(6):72–75, 2002.
- [512] D. A. Menasc , E. Casalicchio, and V. Dubey. On optimal service selection in Service Oriented Architectures. *Performance Evaluation*, 2009.
- [513] T. Metsch, A. Edmonds, and V. Bayon. Using Cloud Standards for Interoperability of Cloud Frameworks. Technical report, RESERVOIR and SLA@SOI, Apr. 2010.
- [514] A. Michael, F. Armando, G. Rean, D. J. Anthony, K. Randy, K. Andy, L. Gunho, P. David, R. Ariel, S. Ion, et al. Above the clouds: A berkeley view of cloud computing. *EECS Department, University of California, Berkeley, Tech. Rep. UCB/EECS-2009-28*, 2009.
- [515] Microsoft. Azure: <http://www.microsoft.com/azure/default.aspx>, 5 May 2009.
- [516] Microsoft. Microsoft pushes its technical computing initiative forward with windows hpc server 2008 r2: <http://www.microsoft.com/Presspass/press/2010/sep10/09-17WindowsHPCServerPR.aspx>, 22 October 2010.

- [517] Microsoft. *SQL Server Online Book*. <http://msdn.microsoft.com/en-us/library/ms187956.aspx>, 2010.
- [518] Dejan Milojevic. Cloud computing: Interview with russ daniels and franco travostino. *IEEE Internet Computing*, 12(5):7–9, 2008.
- [519] M. Isard, M. Budiu, Y. Yu, A. Birrell, and D. Fetterly. Dryad: distributed data-parallel programs from sequential building blocks. *Euro. Conf on computer systems (EuroSys)*, 2007.
- [520] M. Miyazawa. Approximation of the queue-length distribution of an $M/GI/s$ queue by the basic equations. *Journal of Applied Probability*, 23:443–458, 1986.
- [521] MONIT. Project Website: <http://monit.com/monit>.
- [522] Andrew W. Moore and Denis Zuev. Internet traffic classification using bayesian analysis techniques. *SIGMETRICS Perform. Eval. Rev.*, 33:50–60, June 2005.
- [523] J. N. Morse. Reducing the size of the nondominated set: Pruning by clustering. *Computers & Operations Research*, 7(1-2):55–66, 1980.
- [524] Angela Moscaritolo. Most organizations falling short on cloud security policies. <http://www.scmagazineus.com/most-organizations-falling-short-on-cloud-security-policies/article/167415/>, 2010. (accessed September 17, 2010).
- [525] Francesco Moscato, Beniamino Di Martino, Salvatore Ventcinque, and Angelo Martone. Overfa: a collaborative framework for the semantic annotation of documents and websites. *Int. J. Web Grid Serv.*, 5:30–45, March 2009.
- [526] Hamid R. Motahari-Nezhad, Bryan Stephenson, and Sharad Singhal. Outsourcing business to cloud computing services: Opportunities and challenges. *Development*, 10(4):1–17, 2009.
- [527] M. Turner, D. Budgen, and P. Brereton. Turning software into a service. *IEEE Computer*, Vol 36, 2008.
- [528] Ahuva W. Mu’alem and Dror G. Feitelson. Utilization, predictability, workloads, and user runtime estimates in scheduling the IBM SP2 with backfilling. *IEEE Transactions on Parallel and Distributed Systems*, 12(6):529–543, 2001.
- [529] MySQL. mysqlslap - load emulation client. <http://dev.mysql.com/doc/refman/5.1/en/mysqlslap.html>.
- [530] Nagios. Project Website: <http://www.nagios.org/>.

- [531] Nagios Enterprises, LLC. Nagios XI - Features. <http://assets.nagios.com/datasheets/nagiosxi/Nagios%20XI%20-%20Features.pdf>.
- [532] Nagios Plugins Development Team. Nagios plug-in development guidelines. <http://nagiosplug.sourceforge.net/developer-guidelines.html>.
- [533] Ripal Nathuji and Karsten Schwan. Virtualpower: coordinated power management in virtualized enterprise systems. In *Proceedings of twenty-first ACM SIGOPS symposium on Operating systems principles*, SOSP '07, pages 265–278, New York, NY, USA, 2007. ACM.
- [534] National Institute of Standards and Technology. Cloud computing. <http://csrc.nist.gov/groups/SNS/cloud-computing/index.html>, 2009. (accessed October 03, 2009).
- [535] Alon Naveh, Efraim Rotem, Avi Mendelson, Simcha Gochman, Rajshree Chabukswar, Karthik Krishnan, and Arun Kumar. Power and thermal management in the intel core duo processor. *Intel Technology Journal*, 10(2):109–122, 2006.
- [536] Surya Nepal, Zaki Malik, and Athman Bouguettaya. Reputation propagation in composite services. In *Proceedings of the 2009 IEEE International Conference on Web Services*, ICWS '09, pages 295–302, Washington, DC, USA, 2009. IEEE Computer Society.
- [537] SuitCloud NetSuite. Suitecloud platform, 2010.
- [538] Patrick Nicolas. Introduction to multi-tenant web applications, 2006.
- [539] Nimbus Home Page. <http://www.nimbusproject.org/>.
- [540] National Institute of Standards and Technology (NIST), Cloud Standards: <http://csrc.nist.gov/groups/SNS/cloud-computing/>.
- [541] NIST definition of cloud computing <http://www.nist.gov/itl/cloud>.
- [542] Natalya F. Noy, Michael Sintek, Stefan Decker, Monica Crubézy, Ray W. Ferguson, and Mark A. Musen. Creating semantic web contents with protégé-2000. *IEEE Intelligent Systems*, 16:60–71, March 2001.
- [543] S. A. Nozaki and S. M. Ross. Approximations in finite-capacity multi-server queues with poisson arrivals. *Journal of Applied Probability*, 15:826–834, 1978.
- [544] D. Nurmi, R. Wolski, C. Grzegorzcyk, G. Obertelli, S. Soman, L. Yousseff, and D. Zagorodnov. The Eucalyptus open-source cloud-computing system. In *9th International Symposium on Cluster Computing and the Grid (CCGRID'09)*, 2009.

- [545] Daniel Nurmi, Rich Wolski, Chris Grzegorzczak, Graziano Obertelli, Sunil Soman, Lamia Youseff, and Dmitrii Zagorodnov. Eucalyptus : A Technical Report on an Elastic Utility Computing Architecture Linking Your Programs to Useful Systems. Technical report, Computer Science Department, University of California, Santa Barbara, 2008.
- [546] Daniel Nurmi, Rich Wolski, Chris Grzegorzczak, Graziano Obertelli, Sunil Soman, Lamia Youseff, and Dmitrii Zagorodnov. Eucalyptus: A Technical Report on an Elastic Utility Computing Architecture Linking Your Programs to Useful Systems. In *UCSB Computer Science Technical Report Number 2008-10*, August 2008.
- [547] Daniel Nurmi, Rich Wolski, Chris Grzegorzczak, Graziano Obertelli, Sunil Soman, Lamia Youseff, and Dmitrii Zagorodnov. The Eucalyptus Open-source Cloud-computing System. In *CCA'08: Proceedings of Cloud Computing and Its Applications workshop*, October 2008.
- [548] Daniel Nurmi, Rich Wolski, Chris Grzegorzczak, Graziano Obertelli, Sunil Soman, Lamia Youseff, and Dmitrii Zagorodnov. The Eucalyptus Open-source Cloud-computing System. *9th IEEE/ACM International Symposium on Cluster Computing and the Grid*, pages 124–131, 2009.
- [549] O. Lassila and R. Swick. Resource Description Framework (RDF) Model and Syntax Specification.: <http://www.w3.org/TR/REC-rdf-syntax/>, 1998.
- [550] University of Manchester. Dataflow research project, 1997.
- [551] Open Grid Forum: Open Cloud Computing Interface (OCCI): <http://forge.ogf.org/sf/projects/occi-wg>.
- [552] OpenNebula Home Page. <http://www.opennebula.org/>.
- [553] Open Cloud Manifesto, Spring 2009 : <http://www.opencloudmanifesto.org>.
- [554] Oracle. *Oracle 11g Online Library*. http://download.oracle.com/docs/cd/B28359_01/server.111/b28313/usingpe.htm#i1007101, 2010.
- [555] Oracle Corporation. [http://www.virtualbox.org/manual/ UserManual.html](http://www.virtualbox.org/manual/UserManual.html).
- [556] Organization for the Advancement of Structured Information Standards (OASIS). *Web Services Dynamic Discovery (WS-Discovery)*, July 1, 2009. Version 1.1.
- [557] S. Ostermann, R. Prodan, and T. Fahringer. Extending grids with cloud resource management for scientific computing. In *Grid Computing, 2009 10th IEEE/ACM International Conference on*, pages 42 –49, oct. 2009.

- [558] Dustin Owens. Securing elasticity in the cloud. *Queue*, 8(5):10–16, 2010.
- [559] M. Tamer Özsu and Patrick Valduriez. *Principles of Distributed Database Systems, Second Edition*. Prentice-Hall, 1999.
- [560] E. Page. Tables of waiting times for $M/M/n$, $M/D/n$ and $D/M/n$ and their use to give approximate waiting times in more general queues. *J. Operational Research Society*, 33:453–473, 1982.
- [561] Mayur R. Palankar, Adriana Iamnitchi, Matei Ripeanu, and Simson Garfinkel. Amazon s3 for science grids: a viable solution? In *Proceedings of the 2008 international workshop on Data-aware distributed computing*, DADC '08, pages 55–64, New York, NY, USA, 2008. ACM.
- [562] Mayur R. Palankar, Adriana Iamnitchi, Matei Ripeanu, and Simson Garfinkel. Amazon S3 for science Grids: a viable solution? In *International Workshop on Data-aware Distributed Computing (DADC'08) in conjunction with HPDC 2008*, pages 55–64, New York, NY, USA, 2008. ACM.
- [563] G. Pallis. Cloud computing: The new frontier of internet computing. *Internet Computing, IEEE*, 14(5):70–73, No.v 2010.
- [564] I.V. Papaioannou, D.T. Tsesmetzis, I.G. Roussaki, and M.E. Anagnostou. A QoS ontology language for Web-services. In *Advanced Information Networking and Applications, 2006. AINA 2006. 20th International Conference on*, volume 1, page 6. IEEE, 2006.
- [565] P. Papakos, L. Capra, and D.S. Rosenblum. VOLARE: context-aware adaptive cloud service discovery for mobile systems. In *Proceedings of the 9th International Workshop on Adaptive and Reflective Middleware*, pages 32–38. ACM, 2010.
- [566] Michael P. Papazoglou, Paolo Traverso, Schahram Dustdar, and Frank Leymann. Service-oriented computing: State of the art and research challenges. *Computer*, 40:38–45, November 2007.
- [567] A. V. Parameswaran and A. Chaddha. Cloud Interoperability and Standardization. *SETLabs Briefings*, 7(7):19–27, 2009.
- [568] K. E. Parsopoulos and M. N. Vrahatis. Particle swarm optimization method in multiobjective problems. In *Proceedings of the 2002 ACM symposium on applied computing*, pages 603–607. ACM, 2002.
- [569] Gopi Kannan Parthiban. Cloud computing use cases white paper version 1.0. *3rd International Conference on Grid and Pervasive Computing-gpc-workshops*, 2008.

- [570] Mukaddim Pathan, James Broberg, and Rajkumar Buyya. Maximizing utility for content delivery clouds. In *Proceedings of the 10th International Conference on Web Information Systems Engineering, WISE '09*, pages 13–28, Berlin, Heidelberg, 2009. Springer-Verlag.
- [571] Paul McFedries. The cloud is the computer. IEEE Spectrum Online, : <http://www.spectrum.ieee.org/aug08/6490>, 2008.
- [572] R. Pawson. *Naked Objects*. PhD thesis, Trinity College, Dublin, Ireland, 2004.
- [573] Vern Paxson. Bro: a system for detecting network intruders in real-time. *Comput. Netw.*, 31:2435–2463, December 1999.
- [574] Jan-Simon Pendry and Marshall Kirk McKusick. Union mounts in 4.4BSD-lite. In *TCON'95: Proc. of the USENIX 1995 Technical Conf.*, pages 3–3, Berkeley, CA, USA, 1995. USENIX Association.
- [575] Somesh Jha Peng Ning, Paul Syverson. Robust Defenses for Cross-Site Request Forgery . *Proceedings of the 15th ACM conference on Computer and communications security*, october 2008.
- [576] Donald H. Perkins. *Introduction to High Energy Particle Physics*. Cambridge University Press, ISBN 978-0521621960, 2000.
- [577] R. Perlman. The ephemerizer: Making data disappear, sun microsystems whitepaper, 2005.
- [578] Gunnar Peterson. Don't trust. and verify: A security architecture stack for the cloud. *IEEE Security and Privacy*, 8:83–86, September 2010.
- [579] P. Goyal, R. Mikkilineni, and M. Ganti. The fcaps in the business services fabric model. in *proceedings of WETICE 2009:18th IEEE International Workshops on Enabling Technologies:Infrastructure for Collaborative Enterprises*, 2009.
- [580] Rob Pike, Dave Presotto, Sean Dorward, Bob Flandrena, Ken Thompson, Howard Trickey, and Phil Winterbottom. Plan 9 from Bell Labs. *Computing Systems*, 8(3):221–254, Summer 1995.
- [581] Rob Pike, Dave Presotto, Ken Thompson, Howard Trickey, and Phil Winterbottom. The use of name spaces in Plan 9. *SIGOPS Oper. Syst. Rev.*, 27(2):72–76, 1993.
- [582] D. Pisinger. *Algorithms for knapsack problems*. Datalogisk Institut, Københavns Universitet, 1995.
- [583] C. Polychronopoulos and D. Kuck. Guided self-scheduling: A practical scheduling scheme for parallel computers. *IEEE Transactions on Computers*, C-36(12):1425–1439, December 1987.

- [584] F. Pop, A. Costan, C. Dobre, C. Stratan, and V. Cristea. Monitoring of complex applications execution in distributed dependable systems. In *Parallel and Distributed Computing, 2009. ISPDC '09. Eighth International Symposium on*, pages 241–244, July 2009.
- [585] Raluca Ada Popa, Jay Lorch, David Molnar, Helen J. Wang, and Li Zhuang. Enabling security in cloud storage slas with cloudproof. technical report, microsoft research, 2010.
- [586] Gerald J. Popek and Robert P. Goldberg. Formal requirements for virtualizable third generation architectures. *Commun. ACM*, 17:412–421, July 1974.
- [587] Gerald J. Popek and Robert P. Goldberg. Formal requirements for virtualizable third generation architectures. *Communications of the ACM*, 17(7):412–421, July 1974.
- [588] R. Prodan and S. Ostermann. A Survey and Taxonomy of Infrastructure as a Service & Web Hosting Cloud Providers. In *Proceedings of the Intl. Conference on Grid Computing*, Banff, Canada, Oct. 13–15, 2009.
- [589] FAWN Project. FAWN: A fast array of wimpy nodes. <http://www.cs.cmu.edu/~fawnproj/>.
- [590] Nimbus project. <http://www.nimbusproject.org/>.
- [591] J. Quemada, T. de Miguel, S. Pavon, G. Huecas, T. Robles, J. Salvachua, D.a.a. Ortiz, V. Sirvent, F. Escibano, and J. Sedano. Isabel: An Application for real time Collaboration with a flexible Floor Control. In *2005 International Conference on Collaborative Computing: Networking, Applications and Worksharing*, pages 1–9. IEEE, 2005.
- [592] Juan Quemada, Gabriel Huecas, Tomás de Miguel, Joaquín Salvachúa, Blanca Fernandez, Bernd Simon, Katherine Maillet, and Efie Lai-Cong. Educanext: a framework for sharing live educational resources with isabel. In *WWW (Alternate Track Papers & Posters)*, pages 11–18, 2004.
- [593] R. Akkiraju, J. Farrell, J. Miller, M. Nagarajan, M. Schmidt, A. Sheth, K. Verma. Web Service Semantics WSDL-S. A joint UGA-IBM Technical Note, version 1.0: <http://lsdis.cs.uga.edu/projects/METEOR-S/WSDL>, 2005.
- [594] Rackspace. <http://www.rackspacecloud.com/index.php>, Dec. 2010.
- [595] Sandro Rafaeli and David Hutchison. A survey of key management for secure group communication. *ACM Comput. Surv.*, 35:309–329, September 2003.

- [596] Barath Raghavan, Kashi Vishwanath, Sriram Ramabhadran, Kenneth Yocum, and Alex C. Snoeren. Cloud control with distributed rate limiting. *ACM SIGCOMM Computer Communication Review*, 37(4):337, oct 2007.
- [597] Arcot Rajasekar, Michael Wan, and Reagan Moore. Mysrb & srb: Components of a data grid. In *Proceedings of the 11th IEEE International Symposium on High Performance Distributed Computing*, HPDC '02, pages 301–, Washington, DC, USA, 2002. IEEE Computer Society.
- [598] S. Ran. A model for web services discovery with QoS. *ACM SIGecom Exchanges*, 4(1):1–10, 2003.
- [599] Omer Rana, Martijn Warnier, Thomas B. Quillinan, and Frances Brazier. Monitoring and reputation mechanisms for service level agreements. In *Proceedings of the 5th international workshop on Grid Economics and Business Models*, GECON '08, pages 125–139, Berlin, Heidelberg, 2008. Springer-Verlag.
- [600] EPA Technical Report. Report to congress on server and data center energy efficiency. http://www.energystar.gov/ia/partners/prod_development/downloads/EPA_Datacenter_Report_Congress_Final1.pdf, 2007.
- [601] RESERVOIR project. Project Website: <http://www.reservoir-fp7.eu>.
- [602] Cluster Resources. Moab access portal: <http://www.clusterresources.com/products/moab-cluster-suite/access-portal.php>, November 2010.
- [603] Cluster Resources. Moab adaptive hpc suite: <http://www.clusterresources.com/products/adaptive-hpc-suite.php>, November 2010.
- [604] Cluster Resources. Moab cluster manager: <http://www.clusterresources.com/products/moab-cluster-suite/cluster-manager.php>, November 2010.
- [605] Cluster Resources. Moab workload manager: <http://www.clusterresources.com/products/moab-cluster-suite/workload-manager.php>, November 2010.
- [606] B. P. Rimal, E. Choi, and I. Lumb. A Taxonomy and Survey of Cloud Computing Systems. In *Proceedings of the 5th International Conference on Networked Computing*, Seoul, Korea, Aug. 25–27 2009.
- [607] Bhaskar Prasad Rimal, Eunmi Choi, and Ian Lumb. A Taxonomy and Survey of Cloud Computing Systems. *Fifth International Joint Conference on INC, IMS and IDC*, pages 44–51, 2009.

- [608] Bhaskar Prasad Rimal, Eunmi Choi, and Ian Lumb. A taxonomy and survey of cloud computing systems. *Fifth International Joint Conference on INC, IMS and IDC*, 2009.
- [609] Bhaskar Prasad Rimal, Eunmi Choi, and Ian Lumb. A taxonomy and survey of cloud computing systems. *Networked Computing and Advanced Information Management, International Conference on*, 0:44–51, 2009.
- [610] Thomas Ristenpart, Eran Tromer, Hovav Shacham, and Stefan Savage. Hey, you, get off of my cloud: exploring information leakage in third-party compute clouds. In *CCS'09: Proceedings of the 16th ACM conference on Computer and communications security*, pages 199–212. ACM, 2009.
- [611] B. Rochwerger, D. Breitgand, E. Levy, A. Galis, K. Nagin, I. Llorente, R. Montero, Y. Wolfsthal, E. Elmroth, J. Caceres, M. Ben-Yehuda, W. Emmerich, and F. Galan. The RESERVOIR Model and Architecture for Open Federated Cloud Computing. *IBM Journal of Research and Development*, 53(4):4:1–4:11, July 2009.
- [612] Benny Rochwerger, David Breitgand, Eliezer Levy, Alex Galis, Kenneth Nagin, Ignacio M. Llorente, Ruben Montero, Yaron Wolfsthal, Erik Elmroth, Juan Caceres, Muli Ben-Yehuda, Wolfgang Emmerich, and Fermin Galan. The RESERVOIR Model and Architecture for Open Federated Cloud Computing. *IBM Systems Journal*, 2008.
- [613] Luis. Rodero-Merino, Luis M. Vaquero, Víctor Gil, Fermín Galán, Javier Fontán, Rubén Montero, and Ignacio M. Llorente. From infrastructure delivery to service management in clouds. *Future Generation Computer Systems*, 26:1226–1240, October 2010.
- [614] Pedro Rodríguez, Daniel Gallego, Javier Cerviño, Fernando Escribano, Juan Quemada, Salvachú, and Joaquin A. VaaS : Videoconference as a Service. In *CollaborateCom*, Washington, 2009.
- [615] D. Roman, U. Keller, H. Lausen, J. de Bruijn, R. Lara, M. Stollberg, A. Polleres, C. Feier, C. Bussler, and D. Fensel. Web service modeling ontology. *Applied Ontology*, 1(1):77–106, 2005.
- [616] F. Rosenberg, C. Platzer, and S. Dustdar. Bootstrapping performance and dependability attributes of web services. In *IEEE International Conference on Web Services (ICWS'06)*, 2006.
- [617] J. Rosenberg, H. Schulzrinne, G. Camarillo, A. Johnston, J. Peterson, and R. Spark. SIP: Session Initiation Protocol. RFC 3261 (Proposed Standard), June 2002.
- [618] Antony Rowstron and Peter Druschel. Storage management and caching in past, a large-scale, persistent peer-to-peer storage utility. *SIGOPS Oper. Syst. Rev.*, 35:188–201, October 2001.

- [619] Roy Bragg. Cloud computing: When computers really rule: <http://www.technewsworld.com/story/63954.html>, 2008.
- [620] RSoft Design. *Artifex v.4.4.2*. RSoft Design Group, Inc., San Jose, CA, 2003.
- [621] A. Ruiz-Cortés, O. Martin-Diaz, A. Duran, and M. Toro. Improving the automatic procurement of web services using constraint programming. *International Journal of Cooperative Information Systems*, 14(4):439–467, 2005.
- [622] J. Rumbaugh, I. Jacobson, and G. Booch. *The Unified Modeling Language Reference Manual*. Erehwon: Addison-Wesley, 1999.
- [623] P. Ruth, X. Jiang, D. Xu, and S. Goasguen. Virtual distributed environments in a shared infrastructure. *Computer*, 38(5):63 – 69, May 2005.
- [624] Matt Ryan, Sojan Markose, Xiaoqing (Frank) Liu, and Ying Cheng. Structured object-oriented co-analysis/co-design of hardware/software for the facts power system. In *COMPSAC '05: Proceedings of the 29th Annual International Computer Software and Applications Conference*, pages 396–402, Washington, DC, USA, 2005. IEEE Computer Society.
- [625] S. Battle, A. Bernstein, H. Boley, B. Grosz, M. Gruninger, R. Hull, M. Kifer, D. Martin, S. McIlraith, D. McGuinness, J. Su, S. Tabet. Semantic Web Services Language (SWSL): <http://www.w3.org/Submission/SWSF-SWSL/>, 2005.
- [626] T. L. Saaty. *Fundamentals of decision making and priority theory with the analytic hierarchy process*, volume 6. RWS Publications USA, 1994.
- [627] Jordi Sabater and Carles Sierra. Reputation and social network analysis in multi-agent systems. In *Proceedings of the first international joint conference on Autonomous agents and multiagent systems: part 1, AAMAS '02*, pages 475–482, New York, NY, USA, 2002. ACM.
- [628] F. D. Sacerdoti, M. J. Katz, M. L. Massie, and D. E. Culler. Wide area cluster monitoring with ganglia. In *Cluster Computing, 2003. Proceedings. 2003 IEEE International Conference on*, pages 289–298, December 2003.
- [629] A. Sahai, V. Machiraju, M. Sayal, A. Van Moorsel, and F. Casati. Automated SLA monitoring for web services. *Management Technologies for E-Commerce and E-Business Applications*, pages 28–41, 2002.
- [630] Usman Sait. The future of cloud computing. *Intel Executive Summary*, 2005.

- [631] Usman Sait. Welcome to www.cloudtutorial.com. *Intel Executive Summary*, 2005.
- [632] Salesforce. Appexchange: <http://sites.force.com/appexchange/home>, 17 November 2010.
- [633] SAP. Business by design. sap, 2010.
- [634] SAX. Simple API for XML. <http://sax.sourceforge.net/>.
- [635] Cristina Schmidt and Manish Parashar. Squid: Enabling search in dht-based systems. *J. Parallel Distrib. Comput.*, 68(7):962–975, 2008.
- [636] D. Schmidt and R. Guida. Hpc in the cloud - achieving ultra high performance in the cloud: <http://www.hpcinthecloud.com/features/101386074.html>, 24 August 2010.
- [637] W3C School. MathML3 Manual. 2011.
- [638] N. Seitz. Itu-t qos standards for ip-based networks. *Communications Magazine, IEEE*, 41(6):82 – 89, June 2003.
- [639] Peter Sempolinski and Douglas Thain. A Comparison and Critique of Eucalyptus, OpenNebula and Nimbus. *2nd IEEE International Conference on Cloud Computing Technology and Science*, December 2010.
- [640] Sequoia capital <http://www.sequoiacap.com/>.
- [641] Amazon Web Service. Amazon web services (aws), 2010.
- [642] A. ShaikhAli, O.F. Rana, R. Al-Ali, and D.W. Walker. Uddie: An extended registry for web services. In *Applications and the Internet Workshops, 2003. Proceedings. 2003 Symposium on*, pages 85–89. IEEE, 2003.
- [643] Jin Shao, Hao Wei, Qianxiang Wang, and Hong Mei. A runtime model based monitoring approach for cloud. In *Cloud Computing (CLOUD), 2010 IEEE 3rd International Conference on*, pages 313–320, July 2010.
- [644] R. Shearer, B. Motik, and I. Horrocks. Hermit: A highly-efficient owl reasoner. In *Proceedings of the 5th International Workshop on OWL: Experiences and Directions (OWLED 2008)*, 2008.
- [645] Justin Shi and Suntian Song. Apparatus and Method of Optimizing Database Clustering with Zero Transaction Loss. (Pending), 2007.
- [646] Justin Y. Shi. *Synergy v3.0 Manual*. <http://spartan.cis.temple.edu/synergy/>, 1995.
- [647] Justin Y. Shi. Fault tolerant self-optimizing multiprocessor system and method thereof. 2007.

- [648] Justin Y. Shi. Decoupling as a foundation for large scale parallel processing. In *Proceedings of 2009 High Performance Computing and Communications*, Seoul, Korea, 2009.
- [649] Y. Shi. Heterogeneous computing for graphics applications. In *National Conference on Graphics Applications*, April 1991.
- [650] Y. Shi. A distributed programming model and its applications to computation intensive applications for heterogeneous environments". In *International Space Year Conference on Earth and Space Information Systems*, pages 10–13, Pasadena, CA., February 1992.
- [651] Y. Shi. Program scalability analysis. In *International Conference on Distributed and Parallel Processing*, Georgetown University, Washington D.C., October 1997.
- [652] J. Shiers. The worldwide lhc computing grid (worldwide lcg). *Computer Physics Communications*, 177 (1-2), pages 219–223, 2007.
- [653] M. Shultz, G. Bronevetsky, R. Fernandes, D. M. K. Pingali, and P. Stodghill. Implementation and evaluation of a scalable application-level checkpoint-recovery scheme for mpi programs. In *Proceedings of Supercomputing 2004 Conference*, Pittsburgh, PA., November 2004.
- [654] Guttorm Sindre. Mal-activity diagrams for capturing attacks on business processes. In *REFSQ'07: Proceedings of the 13th international working conference on Requirements engineering*, pages 355–366, Berlin, Heidelberg, 2007. Springer-Verlag.
- [655] Guttorm Sindre and Andreas L. Opdahl. Eliciting security requirements with misuse cases. *Requir. Eng.*, 10(1):34–44, 2005.
- [656] Aameek Singh and Ling Liu. Sharoes: A data sharing platform for outsourced enterprise storage environments. In *ICDE*, pages 993–1002, 2008.
- [657] R. O. Sinnott, J. Jiang, J. Watt, and O. Ajayi. Shibboleth-based Access to and Usage of Grid Resources. In *Proceedings of the 7th International Conference on Grid Computing*, Barcelona, Spain, Sep. 28–29, 2006.
- [658] Radu Sion. Query execution assurance for outsourced databases. In *Proceedings of the 31st international conference on Very large data bases, VLDB '05*, pages 601–612. VLDB Endowment, 2005.
- [659] E. Sirin, B. Parsia, B. Grau, A. Kalyanpur, and Y. Katz. Pellet: A practical OWL-DL reasoner. *Web Semantics: Science, Services and Agents on the World Wide Web*, 5(2):51–53, June 2007.

- [660] K. Sivashanmugam, K. Verma, and A. Sheth. Discovery of Web Services in a Federated Registry Environment. In *Proceedings of the International Conference on Web Services (ICWS)*, San Diego, USA, June 6–9, 2004.
- [661] A. Skonnard. A developer’s guide to the microsoft .net service bus, white paper, pluralsight: <http://go.microsoft.com/fwlink/?LinkID=150834>, May 2009.
- [662] J. M. Smith. $M/G/c/K$ blocking probability models and system performance. *Perform. Eval.*, 52:237–267, May 2003.
- [663] Suntian Song. Method and apparatus for database fault tolerance with instant transaction replication using off-the-shelf database servers and low bandwidth networks. (#6,421,688), 2002.
- [664] B. Sotomayor, R. S. Montero, I. M. Llorente, and I. Foster. Virtual Infrastructure Management in Private and Hybrid Clouds. *Internet Computing, IEEE*, 13(5):14–22, Sept.-Oct. 2009.
- [665] B. Sotomayor, R. S. Montero, I. M. Llorente, and I. Foster. Virtual infrastructure management in private and hybrid clouds. *IEEE Internet Computing*, 13(5):14–22, 2009.
- [666] Borja Sotomayor, Ian Foster, Rubén S. Montero, and Ignacio M. Llorente. Virtual Infrastructure Management in Private and Hybrid Clouds. *IEEE Internet Computing*, 2009.
- [667] Borja Sotomayor, Rubén S. Montero, Ignacio M. Llorente, and Ian Foster. Virtual Infrastructure Management in Private and Hybrid Clouds. *IEEE Internet Computing*, 13(5):14–22, September 2009.
- [668] Matthew J. Sottile and Ronald G. Minnich. Supermon: A high-speed cluster monitoring system. *Cluster Computing, IEEE International Conference on*, 0:39, 2002.
- [669] L. Sotto, B. Treacy, and M. McLellan. Privacy & Data Security Risks in Cloud Computing. In *Electronic Commerce & Law Report*, Feb. 2010.
- [670] Scott Spetka, Haris Hadzimujic, Stephen Peek, and Christopher Flynn. High Productivity Languages for Parallel Programming Compared to MPI. *HPCMP Users Group Conference*, 0:413–417, 2008.
- [671] Amazon EC2 Spot Instances. <http://aws.amazon.com/ec2/spot-instances/>.
- [672] V. Springel. The cosmological simulation code gadget-2. *MNRAS, astro-ph/0505010*, 2005.
- [673] SPRUCE: Special PRiority and Urgent Computing Environment. <http://spruce.teragrid.org/>.

- [674] Shekhar Srikantaiah, Aman Kansal, and Feng Zhao. Energy aware consolidation for cloud computing. In *Proceedings of the 2008 Conference on Power Aware Computing and Systems*, HotPower'08, pages 10–10, Berkeley, CA, USA, 2008. USENIX Association.
- [675] N. Srinivas and K. Deb. Multiobjective optimization using non-dominated sorting in genetic algorithms. *Evolutionary computation*, 2(3):221–248, 1994.
- [676] Ion Stoica, Robert Morris, David Liben-Nowell, David R. Karger, M. Frans Kaashoek, Frank Dabek, and Hari Balakrishnan. Chord: A scalable peer-to-peer lookup protocol for internet applications. In *ACM SIGCOMM*, pages 149–160, 2001.
- [677] Michael Stonebraker. The case for shared nothing architecture. *Database Engineering*, 9(1), 1986.
- [678] Feijian Sun. *Automatic Program Parallelization Using Stateless Parallel Processing Architecture*. PhD thesis, Temple University, 2004.
- [679] Wei Sun, Xin Zhang, Chang J. Guo, Pei Sun, and Hui Su. Software as a Service: Configuration and Customization Perspectives. In *2008 IEEE Congress on Services Part II (services-2 2008)*, pages 18–25. IEEE, September 2008.
- [680] Yuanhui Sun, Zongshui Xiao, Dongmei Bao, and Jie Zhao. An architecture model of management and monitoring on cloud services resources. In *Advanced Computer Theory and Engineering (ICACTE), 2010 3rd International Conference on*, volume 3, pages V3–207 –V3–211, August 2010.
- [681] Sun Microsystems. *Introduction to Cloud Computing Architecture*, 1st edition, June 2009. White Paper.
- [682] Sonesh Surana, Brighten Godfrey, Karthik Lakshminarayanan, Richard Karp, and Ion Stoica. Load balancing in dynamic structured peer-to-peer systems. *Performance Evaluation*, 63(3):217–240, 2006.
- [683] Swedish seed accelerator <http://www.swedacc.com>.
- [684] Symantec. *Veritas Cluster Server (VCS)*. http://eval.symantec.com/mktginfo/products/Datasheets/High_Availability/cluster_server_datasheet.pdf, 2006.
- [685] M. Szeredi. Filesystem in userspace. <http://fuse.sourceforge.net>, February 2005.
- [686] E. Szmidt and J. Kacprzyk. Intuitionistic fuzzy sets in group decision making. *Notes on IFS*, 2(1):11–14, 1996.

- [687] Tom Szuba. *Safeguarding Your Technology*. Washington: National Center for Education Statistics, 1998.
- [688] B. Szymanski, Y. Shi, and N. Prywes. Synchronized distributed termination. *IEEE Transactions on Software Engineering*, SE11(10):1136–1140, 1985.
- [689] S.G.H. Tabatabaei, A.V. Dastjerdi, W.M.N.W. Kadir, S. Ibrahim, and E. Sarafian. Security conscious AI-planning-based composition of semantic web services. *International Journal of Web Information Systems*, 6(3):203–229, 2010.
- [690] H. Takagi. *Queueing Analysis*, volume 1: Vacation and Priority Systems. North-Holland, Amsterdam, The Netherlands, 1991.
- [691] Y. Takahashi. An approximation formula for the mean waiting time of an $M/G/c$ queue. *J. Operational Research Society*, 20:150–163, 1977.
- [692] Tingxi Tan and Cameron Kiddle. An Assessment of Eucalyptus Version 1.4. Technical report, Grid Research Centre, University of Calgary, Canada, 2009.
- [693] Andrew S. Tanenbaum and Maarten van Steen, editors. *Distributed Systems: Principles and Paradigms*. Prentice Hall, 2002.
- [694] Cloud Computing Interoperability Forum, Cloud taxonomy : <http://groups.google.com/group/cloudforum/web/ccif-cloud-taxonomy>.
- [695] Cloud Computing Use Case Discussion Group: Cloud Computing Use Cases, White Paper v0.1, Section 2: Definitions and Taxonomies: <http://groups.google.com/group/cloud-computing-use-cases>, 2009.
- [696] W. T. Luke Teacy, Jigar Patel, Nicholas R. Jennings, and Michael Luck. Coping with inaccurate reputation sources: experimental analysis of a probabilistic trust model. In *Proceedings of the fourth international joint conference on Autonomous agents and multiagent systems*, AAMAS '05, pages 997–1004, New York, NY, USA, 2005. ACM.
- [697] Akamai Technologies. Fast internet content delivery with freeflow. technical report, 2000.
- [698] Intel Virtualization Technology. <http://www.intel.com/technology/virtualization/technology.htm>.
- [699] TeraGrid. <http://www.teragrid.org/>.
- [700] Douglas Thain, Todd Tannenbaum, and Miron Livny. Condor and the Grid. In *Grid computing: making the global infrastructure a reality*, pages 299–350. John Wiley and Sons, 2003.

- [701] Douglas Thain, Todd Tannenbaum, and Miron Livny. Distributed computing in practice: the Condor experience. *Concurrency - Practice and Experience*, 17(2-4):323–356, 2005.
- [702] Douglas Thain, Todd Tannenbaum, and Miron Livny. Distributed computing in practice: The condor experience. *Concurrency and Computation: Practice and Experience*, 17:2–4, 2005.
- [703] Wolfgang Theilmann, Ramin Yahyapour, and Joe Butler. Multi-level sla management for service-oriented infrastructures. In *Proceedings of the 1st European Conference on Towards a Service-Based Internet*, ServiceWave '08, pages 324–335, 2008.
- [704] K Thompson. *UNIX implementation*, pages 26–41. Prentice-Hall, Inc., Upper Saddle River, NJ, USA, 1986.
- [705] H. C. Tijms. Heuristics for finite-buffer queues. *Probability in the Engineering and Informational Sciences*, 6:277–285, 1992.
- [706] H. C. Tijms, M. H. V. Hoorn, and A. Federgru. Approximations for the steady-state probabilities in the $M/G/c$ queue. *Advances in Applied Probability*, 13:186–206, 1981.
- [707] I. Toma, D. Foxvog, and M. C. Jaeger. Modeling QoS characteristics in WSMO. In *Proceedings of the 1st workshop on Middleware for Service Oriented Computing (MW4SOC 2006)*, pages 42–47. ACM, 2006.
- [708] I. Toma, D. Foxvog, and M. C. Jaeger. Modeling QoS characteristics in WSMO. In *Proceedings of the 1st workshop on Middleware for Service Oriented Computing (MW4SOC 2006)*, pages 42–47. ACM, 2006.
- [709] I. Toma, D. Roman, D. Fensel, B. Sapkota, and J. Gomez. A multi-criteria service ranking approach based on non-functional properties rules evaluation. *Service-Oriented Computing-ICSOC 2007*, pages 435–441, 2010.
- [710] TOP500 Supercomputers . Supercomputer’s application area share. <http://www.top500.org/stats/list/33/apparea> (2009).
- [711] J. Torres, D. Carrera, K. Hogan, R. Gavalda, V. Beltran, and N. Poggi. Reducing wasted resources to help achieve green data centers. In *Parallel and Distributed Processing, 2008. IPDPS 2008. IEEE International Symposium on*, pages 1 –8, April 2008.
- [712] V. Tomic, K. Patel, and B. Pagurek. WSOLUWeb Service Offerings Language. *Web Services, E-Business, and the Semantic Web*, pages 57–67, 2002.

- [713] V. X. Tran, H. Tsuji, and R. Masuda. A new QoS ontology and its QoS-based ranking algorithm for Web services. *Simulation Modelling Practice and Theory*, 17(8):1378–1398, 2009.
- [714] D. Tsarkov and I. Horrocks. Fact++ description logic reasoner: System description. In *Proc. of the Int. Joint Conf. on Automated Reasoning (IJCAR 2006)*, volume 4130 of *Lecture Notes in Artificial Intelligence*, pages 292–297. Springer, 2006.
- [715] D. Tsesmetzis, I. Roussaki, and E. Sykas. Modeling and simulation of QoS-aware Web service selection for provider profit maximization. *Simulation*, 83(1):93, 2007.
- [716] Alan Turing. On computable numbers, with an application to the entscheidungsproblem. *Proceedings of the London Mathematical Society*, Series 2(42):230–265, 1936.
- [717] Francesco Tusa, Maurizio Paone, Massimo Villari, and Antonio Puliafito. CLEVER: a CCloud Enabled Virtual EnviRonment. In *Computers and Communications (ISCC), 2010 IEEE Symposium on*, pages 477–482, June 2010.
- [718] Use Case Discussion Group. Cloud Computing Use Cases v2.0, 2009.
- [719] D.A. Van Veldhuizen. Multiobjective evolutionary algorithms: classifications, analyses, and new innovations. In *Evolutionary Computation*. Citeseer, 1999.
- [720] L. M. Vaquero, L. Rodero-Merino, J. Caceres, and M. Lindner. A break in the clouds: towards a cloud definition. *SIGCOMM Comput. Commun. Rev.*, 39:50–55, December 2008.
- [721] Luis M. Vaquero, Luis Rodero-Merino, Juan Caceres, and Maik Lindner. A Break in the Clouds : Towards a Cloud Definition. *Computer Communication Review*, 39(1):50–55, 2009.
- [722] Luis M. Vaquero, Luis Rodero-Merino, Juan Caceres, and Maik Lindner. A break in the clouds: towards a cloud definition. *SIGCOMM Comput. Commun. Rev.*, 39(1):50–55, 2009.
- [723] C. Vazquez, E. Huedo, R. S. Montero, and I. M. Llorente. Dynamic provision of computing resources from grid infrastructures and cloud providers. In *Grid and Pervasive Computing Conference, 2009. GPC '09. Workshops at the*, pages 113–120, May 2009.
- [724] Christian Vecchiola, Xingchen Chu, and Rajkumar Buyya. *High Speed and Large Scale Scientific Computing*, chapter Aneka: A Software Platform for .NET-based Cloud Computing. IOS Press, Amsterdam, Netherlands, 2009.

- [725] Christian Vecchiola, Xingchen Chu, and Rajkumar Buyya. *Aneka: A Software Platform for .NET-based Cloud Computing*. IOS Press, Amsterdam, Netherlands, 2010.
- [726] Christian Vecchiola, Xingchen Chu, Michael Mattess, and Rajkumar Buyya. *Cloud Computing: Principles and Paradigms*, chapter Aneka – Integration of Private and Public Clouds. John Wiley & Sons, August 2010.
- [727] Christian Vecchiola, Xingchen Chu, Michael Mattess, and Rajkumar Buyya. *Cloud Computing: Principles and Paradigms*, chapter Aneka Integration of Private and Public Clouds. Wiley Press, New York, USA, February 2011.
- [728] Kim H. Veltman. Syntactic and Semantic Interoperability: New Approaches to Knowledge and the Semantic Web. *New Review of Information Networking*, 7(1):159–183, 2001.
- [729] VMWare. VMWare vCloud.
- [730] VMware. VMware vsphere: <http://www.vmware.com/products/vsphere/mid-size-and-enterprise-business/features.html>, December 2010.
- [731] Werner Vogels. Eventually consistent. *Commun. ACM*, 52(1):40–44, 2009.
- [732] Mladen A Vouk. Cloud computing-issues, research and implementations. *Second International Symposium on Information Science and Engineering*, 2010.
- [733] J. Broberg, W. Voorsluys, and R. Buyya. *Introduction to Cloud Computing*, *Cloud Computing: Principles and Paradigms*. John Wiley and Sons (in print), 2011.
- [734] Luay A. Wahsheh and Jim Alves-Foss. Security policy development: Towards a life-cycle and logic-based verification model. *American Journal of Applied Sciences*, 5(9):1117–1126, 2008.
- [735] Phil Wainewright. Many degrees of multi-tenancy, 2008. *Technical Report*, available at <http://www.zdnet.com/blog/saas/many-degrees-of-multitenancy/533>
- [736] E. Walker. Benchmarking amazon ec2 for high-performance scientific computing: <http://www.usenix.org/publications/login/2008-10/openpdfs/walker.pdf>, 2008.
- [737] N. Wancheng, H. Lingjuan, L. Lianchen, and W. Cheng. Commodity-Market Based Services Selection in Dynamic Web Service Composition. In *Asia-Pacific Service Computing Conference, The 2nd IEEE*, pages 218–223. IEEE, 2007.

- [738] Cong Wang, Student Member, Sherman S Chow, Qian Wang, Kui Ren, and Wenjing Lou. Privacy-Preserving Public Auditing for Secure Cloud Storage. *INFOCOM, 2010 Proceedings IEEE*, pages 1–12, 2009.
- [739] Cong Wang, Qian Wang, Kui Ren, and Wenjing Lou. Ensuring data storage security in Cloud Computing. *2009 17th International Workshop on Quality of Service*, pages 1–9, July 2009.
- [740] G. Wang, A. Chen, C. Wang, C. Fung, and S. Uczekaj. Integrated quality of service (QoS) management in service-oriented enterprise architectures. In *Enterprise Distributed Object Computing Conference, 2004. EDOC 2004. Proceedings. Eighth IEEE International*, pages 21–32. IEEE, 2005.
- [741] Haiqin Wang, Guijun Wang, Changzhou Wang, A. Chen, and R. Santiago. Service level management in global enterprise services: from qos monitoring and diagnostics to adaptation, a case study. In *EDOC Conference Workshop, 2007. EDOC '07. Eleventh International IEEE*, pages 44 –51, Oct 2007.
- [742] L. Wang, G. V. Laszewski, A. Younge, X. He, M. Kunze, J. Tao, and C. Fu. Cloud computing: a perspective study. *New Generation Computing*, 28:137–146, 2010.
- [743] Lizhe Wang, Gregor Von Laszewski, Marcel Kunze, and Jie Tao. Cloud Computing: a Perspective Study. *Proceedings of the Grid Computing Environments (GCE) workshop*, November 2008.
- [744] Lizhe Wang, Gregor von Laszewski, Andrew Younge, Xi He, Marcel Kunze, Jie Tao, and Cheng Fu. Cloud computing: a perspective study. *New Generation Computing*, 28:137–146, 2010. 10.1007/s00354-008-0081-5.
- [745] P. Wang. QoS-aware web services selection with intuitionistic fuzzy set under consumer’s vague perception. *Expert Systems with Applications*, 36(3):4460–4466, 2009.
- [746] X. Wang, T. Vitvar, M. Kerrigan, and I. Toma. A QoS-aware selection model for semantic web services. *Service-Oriented Computing-ICSOC 2006*, pages 390–401, 2006.
- [747] X. Wang, K. Yue, J.Z. Huang, and A. Zhou. Service selection in dynamic demand-driven Web services. In *Web Services, 2004. Proceedings. IEEE International Conference on*, pages 376–383. IEEE, 2004.
- [748] Yan Wang and Kwei-Jay Lin. Reputation-oriented trustworthy computing in e-commerce environments. *IEEE Internet Computing*, 12:55–59, July 2008.

- [749] Yung-Terng Wang and Robert J. T. Morris. Load sharing in distributed systems. *IEEE Transactions on Computers*, C-34(3):204–217, March 1985.
- [750] M.S. Ware, J.B. Bowles, and C.M. Eastman. Using the common criteria to elicit security requirements with use cases. In *Proceedings of the IEEE SoutheastCon*, pages 273 –278. 10.1109/second.2006.1629363, 2006.
- [751] web.mit.edu. StarCluster. <http://web.mit.edu/stardev/cluster/index.html>.
- [752] Yi Wei and M.B. Blake. Service-oriented computing and cloud computing: Challenges and opportunities. *Internet Computing, IEEE*, 14(6):72 –75, Nov. 2010.
- [753] S. Weibel, J. Kunze, C. Lagoze, and M. Wolf. Dublin core metadata for resource discovery. *Internet Engineering Task Force RFC*, 2413, 1998.
- [754] M.H. Weik. Computer science and communications dictionary. Springer, 2000.
- [755] J. Weise. *Public Key Infrastructure Overview*. Sun Microsystems, August 2001.
- [756] Aaron Weiss. Computing in the clouds. *netWorker*, 11:16–25, December 2007.
- [757] Craig D. Weissman and Steve Bobrowski. The design of the force.com multitenant internet application development platform. In *Proceedings of the 35th SIGMOD international conference on Management of data*, SIGMOD ’09, pages 889–896, New York, NY, USA, 2009. ACM.
- [758] J. Whitehead and Y. A. Goland. WebDAV: A network protocol for remote collaborative authoring on the web. In *ECSCW 1999*, 1999.
- [759] Wikipedia. Wikipedia, free encyclopedia. *3rd International Conference on Grid and Pervasive Computing-gpc-workshops*, 2008.
- [760] Wikipedia. Wikipedia, free encyclopedia. *3rd International Conference on Grid and Pervasive Computing-gpc-workshops*, 2008.
- [761] Wikipedia. Wikipedia, free encyclopedia. 2009.
- [762] Wikipedia. Cloud computing. *Second International Symposium on Information Science and Engineering*, 2010.
- [763] D. Winer et al. Xml-rpc specification, 1999.
- [764] W.Kim. Cloud computing:today and tomorrow. *Journal of Object Technology*, 2009.

- [765] Rich Wolski, James S. Plank, John Brevik, and Todd Bryan. Analyzing market-based resource allocation strategies for the computational Grid. *The International Journal of High Performance Computing Applications*, 15(3):258–281, Fall 2001.
- [766] Rich Wolski, Neil T Spring, and Jim Hayes. The network weather service: a distributed resource performance forecasting service for meta-computing. *Future Generation Computer Systems*, 15(5-6):757 – 768, 1999.
- [767] Eugene Y. C. Wong, Alvin T. S. Chan, and Hong Va Leong. Xstream: A middleware for streaming XML contents over wireless environments. *IEEE Trans. Softw. Eng.*, 30:918–935, December 2004.
- [768] T. Wood, P. J. Shenoy, A. Venkataramani, and M. S. Yousif. Sandpiper: Black-box and gray-box resource management for virtual machines. *Computer Networks*, 53(17):2923–2938, 2009.
- [769] xCAT Extreme Cloud Administration Toolkit.
<http://xcat.sourceforge.net/>.
- [770] xen.org. Xen. <http://www.xen.org/>.
- [771] Fanfan Xiong. *Resource Efficient Parallel VLDB with Customizable Degree of Redundancy*. PhD thesis, Temple University, 2009.
- [772] K. Xiong and H. Perros. Service performance and analysis in cloud computing. In *Proceedings of the 2009 Congress on Services - I*, pages 693–700, Los Alamitos, CA, USA, 2009.
- [773] Li Xiong and Ling Liu. Peertrust: Supporting reputation-based trust for peer-to-peer electronic communities. *IEEE Trans. Knowl. Data Eng.*, 16(7):843–857, 2004.
- [774] Li Xiong, Ling Liu, and Mustaque Ahamad. Countering feedback sparsity and manipulation in reputation systems. In *Proceedings of the 2007 International Conference on Collaborative Computing: Networking, Applications and Worksharing*, pages 203–212, Washington, DC, USA, 2007. IEEE Computer Society.
- [775] B. Yang, F. Tan, Y. Dai, and S. Guo. Performance evaluation of cloud service considering fault recovery. In *Cloud Computing*, volume 5931 of *Lecture Notes in Computer Science*, pages 571–576. Springer Berlin Heidelberg, 2009.
- [776] Yijian Yang. *Fault Tolerance Protocol for Multiple Dependent Master Protection in a Stateless Parallel Processing Framework*. PhD thesis, Temple University, August 2007.

- [777] Yin Yang, Stavros Papadopoulos, Dimitris Papadias, and George Kollios. Authenticated indexing for outsourced spatial databases. *The VLDB Journal*, 18:631–648, June 2009.
- [778] D. D. Yao. Refining the diffusion approximation for the $M/G/m$ queue. *Operations Research*, 33:1266–1277, 1985.
- [779] Y Combinator seed accelerator <http://ycombinator.com>.
- [780] Chee Shin Yeo and Rajkumar Buyya. Pricing for utility-driven resource management and allocation in clusters. *Int. J. High Perform. Comput. Appl.*, 21(4):405–418, 2007.
- [781] Yieldex: advanced forecasting and delivery simulation algorithms to manage digital business for publishing companies. <http://www.yieldex.com>.
- [782] N. Yigitbasi, A. Iosup, D. Epema, and S. Ostermann. C-meter: A framework for performance analysis of computing clouds. In *CCGRID '09: Proceedings of the 2009 9th IEEE/ACM International Symposium on Cluster Computing and the Grid*, pages 472–477, Washington, DC, USA, 2009.
- [783] L. Youseff, M. Butrico, and D. Da Silva. Toward a Unified Ontology of Cloud Computing. In *Grid Computing Environments Workshop, 2008. GCE '08*, pages 1–10, Nov 2008.
- [784] H. Q. Yu and S. Reiff-Marganiec. A method for automated web service selection. In *Services-Part I, 2008. IEEE Congress on*, pages 513–520. IEEE, 2008.
- [785] H.Q. Yu and S. Reiff-Marganiec. Non-functional property based service selection: A survey and classification of approaches. In *Proc. of 2nd Non Functional Properties and Service Level Agreements in SOC Workshop (NFPSLASOCS08)*. Citeseer, 2008.
- [786] Shucheng Yu, Cong Wang, Kui Ren, and Wenjing Lou. Achieving Secure, Scalable, and Fine-grained Data Access Control in Cloud Computing. *2010 Proceedings IEEE INFOCOM*, pages 1–9, March 2010.
- [787] Y.Zhao, I.Raicu, and I.Foster. Scientific workflow systems for 21st century,new bottle or new wine. *IEEE Workshop on Scientific Workflows*, 2008.
- [788] Zabbix. Project Website: <http://www.zabbix.com>.
- [789] Giorgos Zacharia and Pattie Maes. Trust management through reputation mechanisms. *Applied Artificial Intelligence*, 14(9):881–907, 2000.
- [790] Michal Zalewski. *Browser Security Handbook*. Google, 2008.

- [791] Zencoder: video encoding on the cloud <http://www.zencoder.com>.
- [792] L. Zeng, B. Benatallah, A.H.H. Ngu, M. Dumas, J. Kalagnanam, and H. Chang. QoS-aware middleware for web services composition. *Software Engineering, IEEE Transactions on*, 30(5):311–327, 2004.
- [793] W. Zeng, Y. Zhao, and J. Zeng. Cloud service and service selection algorithm research. In *Proceedings of the first ACM/SIGEVO Summit on Genetic and Evolutionary Computation*, pages 1045–1048. ACM, 2009.
- [794] Zenoss. Zenoss Core - Open Source IT Management. <http://community.zenoss.org/docs/DOC-2614>.
- [795] Ya-Qin Zhang. The future of computing in the cloud - client. *The Economic Observer*, 2008.
- [796] Yingqian Zhang, Bin Sun, and Jia Liu. A markup language for parallel programming model on multi-core system. *SCALCOM-EMBEDDED COM 2009 International Conference*, pages 640–643, Sept. 2009.
- [797] Ben Y. Zhao, John D. Kubiatowicz, and Anthony D. Joseph. Tapestry: a fault-tolerant wide-area application infrastructure. *SIGCOMM Comput. Commun. Rev.*, 32:81–81, January 2002.
- [798] C. Zhou, L.T. Chia, and B.S. Lee. DAML-QoS ontology for web services. In *Web Services, 2004. Proceedings. IEEE International Conference on*, pages 472–479. IEEE, 2004.
- [799] Runfang Zhou and Kai Hwang. Powertrust: A robust and scalable reputation system for trusted peer-to-peer computing. *IEEE Trans. Parallel Distrib. Syst.*, 18:460–473, April 2007.
- [800] Dakai Zhu, Rami G. Melhem, and Bruce R. Childers. Scheduling with dynamic voltage/speed adjustment using slack reclamation in multiprocessor real-time systems. *IEEE Trans. Parallel Distrib. Syst.*, 14(7):686–700, 2003.
- [801] H.J. Zimmermann. Fuzzy programming and linear programming with several objective functions* 1. *Fuzzy sets and systems*, 1(1):45–55, 1978.
- [802] E. Zitzler and L. Thiele. Multiobjective evolutionary algorithms: A comparative case study and the strength pareto approach. *evolutionary computation, IEEE transactions on*, 3(4):257–271, 2002.
- [803] Zope. Project Website: <http://www.zope.org>.